

# A Critical Review of Prior-Distribution Choice in Flow-Based Boltzmann Generators

2026-05-14 working draft

## Abstract

Flow-based Boltzmann generators must choose a base distribution  $\mu_0$ , and the literature has split into two camps: **Camp A** keeps  $\mu_0$  a simple Gaussian and absorbs all complexity into loss, architecture, and inference (FAB, TA-BG, CMT, SBG); **Camp B** encodes physical knowledge of the target into  $\mu_0$  itself (Wirnsberger lattice-Gaussian, Coretti higher- $T$  reference, Schiebroek–Koehn structured CG latent). This review inventories  $\sim 90$  references across seven communities, inspects canonical implementations at the code level, and argues that Camp B wins decisively on *single-phase free-energy precision* (six orders of magnitude in  $\delta F$ , by an exponential  $\chi^2$ -scaling argument) but is structurally unable to discover unknown modes (Lemma 1 and the bijective-flow connectedness lemma of §5). A **third way** keeps Camp A’s full-support Gaussian prior but injects mode information into a log-ratio regulariser  $X_{\hat{\mu}}$  whose minimiser is invariant to inaccuracies in  $\hat{\mu}$ , occupying a design-space niche neither camp covers. The defensible scope: Camp B for single-phase precision, third way for multimodal mode discovery, Camp A unmodified outside both.

**Which camp fits your problem?** Locate your row, follow the column. Each row is supported by a single load-bearing argument in the right-hand column.

| Your problem                                                            | Recommendation                          | Key §       | Why this row                                                                                  |
|-------------------------------------------------------------------------|-----------------------------------------|-------------|-----------------------------------------------------------------------------------------------|
| Single-basin free-energy<br>$\delta F \sim 10^{-5} k_B T / \text{atom}$ | Camp B (Wirnsberger)                    | §4.1, §15   | Harmonic prior absorbs the per-coord $\chi^2$ floor that scales $e^{d\varepsilon}$ for Camp A |
| Crystallographic phase transitions                                      | Camp B (tuned + Coretti $T$ -ladder)    | §4.2        | Phase identity is encoded in $\mu_0$ , not learned                                            |
| Multimodal protein conformational sampling                              | Third way ( $X_{\hat{\mu}}$ ) or Camp A | §11, §20    | Mode-lock-in (Lemma 1) precludes Camp B                                                       |
| Transferable amortised BG across systems                                | Camp A (TBG / TA-BG)                    | §3.11–3.12  | System-specific priors don’t transfer                                                         |
| Mode discovery from scratch (no $\hat{\mu}$ given)                      | Third way (Trinity bootstrap)           | §11, App. B | $X_{\hat{\mu}}$ tolerates sloppy mode estimate                                                |
| Bayesian inverse PDE with Gaussian-process prior                        | Camp A (forward-KL + IS)                | §7          | Likelihood structure already constrains posterior                                             |

## Three load-bearing facts.

1. Wirnsberger 2022 explicitly concedes single-basin sampling (§4.1, §15).
2. TBG 2024 explicitly tried a structured prior and reported “no significant improvements” (§3.11, quoted at its methodbox).

3. Across  $\sim 90$  references in 7 communities, no flagship method since 2019 has trained a tuned-GMM-prior BG on multimodal protein targets (§6).

**How to read this document.** Three reading paths:

- **Just the thesis** ( $\sim 15$  min, 4 pages): the decision table above + §15.1 (three claims on precision) + §16’s failures-at-a-glance + §20 conclusion.
- **Survey only** ( $\sim 1$  h,  $\sim 25$  pages): add the §3 + §4 *methodbox* blocks and §13 comparative tables.
- **Full technical case** ( $\sim 3$  h, all  $\sim 65$  pages): every section in order.

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# 1. Introduction and motivation

Sampling from a high-dimensional, multimodal Boltzmann distribution  $\mu(x) = Z^{-1} \exp(-U_1(x))$  on  $\mathbb{R}^d$  with analytically tractable potential  $U_1$  but intractable partition function  $Z$  is the central computational problem of statistical mechanics, free-energy estimation in biomolecular simulation, lattice quantum field theory, and Bayesian inference with PDE-based likelihoods. The Boltzmann distribution is *multimodal* whenever  $U_1$  has multiple local minima separated by free-energy barriers  $\Delta F \gg k_B T$ . For proteins these barriers are routinely 10–30  $k_B T$ , making naive Langevin / Metropolis sampling infeasibly slow: the mean first-passage time between basins scales as  $\tau \sim e^{\Delta F/k_B T}$ .

Flow-based *Boltzmann generators* (BGs), introduced by Noé, Olsson, Köhler, and Wu (*Science* 2019), short-circuit this difficulty by training a bijective neural network  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  to push a simple base distribution  $\mu_0$  directly onto the target  $\mu$ . Once  $G$  is trained, samples from  $\mu$  are produced by drawing  $z \sim \mu_0$  and applying  $G^{-1}(z)$ . Because  $G$  is invertible with tractable Jacobian, the density of the model on the target side is known in closed form and importance weights are computable.

The architectural and methodological progress on BGs since 2019 has been remarkable — equivariant flows (2020), AIS-based loss (FAB, 2022), flow matching (2023), transferable amortised BGs (2024), inference-time SMC (2025), and constrained-transport schedules (2025). And yet, a central design choice has remained unsettled: **what should the base distribution  $\mu_0$  be?** Two philosophies have emerged:

**Camp A** *Simple prior, sophisticated training.* Take  $\mu_0$  as a standard isotropic Gaussian (possibly truncated for compact intervals or von-Mises-wrapped for dihedrals). Let the flow  $G$  shoulder the entire topological burden of mapping a unimodal contractible Gaussian onto a multimodal Boltzmann target. The loss, the architecture, and the inference-time schedule are where all engineering effort lives. *This is the mainstream philosophy.*

**Camp B** *Tuned prior, simpler training.* Encode physical knowledge of the target into  $\mu_0$ : Gaussians centred at the lattice sites of the crystalline phase (Wirnsberger 2022), a higher-temperature reference distribution (Coretti et al.), conditional priors at fixed thermodynamic states (Schebek 2024), structured multimodal collective-variable latent spaces (Schiebroek & Koehn 2025), or learned mixtures (VampPrior 2018, LARS 2019). The flow then has only to learn local fluctuations around the encoded structure.

These camps encode incompatible philosophical commitments about where mode information should live in the pipeline. The user’s paper proposes a *third way*: keep Camp A’s uninformative prior, but inject mode information into the *loss* via a log-ratio discrepancy regulariser  $X_{\hat{\mu}}$ . The regulariser’s asymptotic minimiser is invariant to inaccuracies in the mode-finder  $\hat{\mu}$  (Proposition 1 below), so sloppy mode knowledge can be absorbed without committing the prior’s support to it. The purpose of this review is to compile the evidence — mathematical, code-level, numerical, and cross-community — needed to argue cleanly between the three positions.

## 2. Mathematical preliminaries

**Why this section exists.** Before debating *which* prior is best for a BG, we need a shared language for what a BG is doing and what its failure modes look like. The objects defined here — pushforward density, reverse-KL loss, forward-KL loss, ESS, coverage — are the vocabulary every camp uses, often with subtly different assumptions. Setting them down once in standard notation lets every later claim be checked against the math rather than against rhetorical framing.

### 2.1. The Boltzmann sampling problem

Let  $U_1 : \mathbb{R}^d \rightarrow \mathbb{R}$  be a potential on configuration space, possibly only analytically tractable up to an additive constant. We want samples from the Boltzmann distribution

$$\mu(x) = \frac{1}{Z} \exp(-U_1(x)), \quad Z = \int_{\mathbb{R}^d} \exp(-U_1(x)) \, dx.$$

We assume only that  $U_1$  and  $\nabla U_1$  can be evaluated pointwise. The two principal evaluation metrics will be effective sample size  $\text{ESS} \in [0, 1]$  (necessary but not sufficient) and coverage (Naeem et al. 2020), which together expose the *fake-ESS pitfall*.

### 2.2. Normalising flows and pushforward measures

A normalising flow  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  is a  $C^1$ -bijection parameterised by a neural network with parameters  $\theta$  such that  $\det J_G(y)$  is computable in  $O(d)$  or  $O(d \log d)$  time. Let  $\mu_0$  be a base distribution on  $\mathbb{R}^d$  (typically the standard isotropic Gaussian). The flow  $G$  induces two distributions:

- Pushforward  $G_{\#}\mu_0$  with density  $(G_{\#}\mu_0)(x) = \mu_0(G^{-1}(x)) |\det J_{G^{-1}}(x)|$ ;
- Pullback  $G_{\#}^{-1}\mu_0$  with density

$$(G_{\#}^{-1}\mu_0)(y) = \mu_0(G(y)) |\det J_G(y)|.$$

We parameterise the loss in terms of  $G : y \rightarrow z$  (target  $\rightarrow$  latent) so that no inversion is required during training. Sampling at inference requires inverting  $G$  on  $z \sim \mu_0$  to obtain  $y = G^{-1}(z)$ .

### 2.3. The reverse-KL training loss

Let  $\nu_{\theta} := G_{\theta, \#}^{-1}\mu_0$  be the model. After reparameterisation through the base,

$$\mathcal{L}_{\text{rev}}(\theta) = \mathbb{E}_{z \sim \mu_0} [U_1(G_{\theta}^{-1}(z)) + \log |\det J_{G_{\theta}^{-1}}(z)|] + \text{const.}$$

**Mode-seeking property.** Reverse KL gives infinite penalty whenever  $\nu_{\theta}$  puts mass where  $\mu$  has essentially none, but it gives essentially zero penalty when  $\nu_{\theta}$  *misses* mass that  $\mu$  puts somewhere. A dropped mode does not enter the loss — this is the source of mode collapse.

## 2.4. The forward-KL training loss

The forward KL is

$$\mathcal{L}_{\text{fwd}}(\theta) = \text{KL}(\mu \parallel \nu_\theta) = \mathbb{E}_{y \sim \mu} [U_0(G_\theta(y)) - \log |\det J_{G_\theta}(y)|] + \text{const},$$

with  $U_0(z) = \frac{1}{2}\|z\|^2$  for the standard Gaussian prior. **Mode-covering property.** A single missed mode contributes  $+\infty$ , so forward KL maximally penalises missed modes. However, the expectation is over  $\mu$ , which we cannot sample directly — bootstrapping via MD, AIS, or score/flow matching is required.

## 2.5. Annealed importance sampling

AIS (Neal 2001) bridges  $\nu_\theta$  and  $\mu$  through  $M + 1$  intermediates

$$\pi_k \propto \nu_\theta^{1-\beta_k} \mu^{\beta_k}, \quad 0 = \beta_0 < \beta_1 < \dots < \beta_M = 1,$$

with per-step weights  $w_k(y) = (\nu_\theta(y))^{\beta_{k-1}-\beta_k} (\mu(y))^{\beta_k-\beta_{k-1}}$  and MCMC kernels targeting each  $\pi_k$ . AIS *accelerates fine-grained reweighting* but does *not* solve mode discovery: if  $\nu_\theta$  misses a mode, the AIS chain initialised at  $\nu_\theta$  does not magically populate it.

## 2.6. ESS and coverage; the fake-ESS phenomenon

**Definition 1** (ESS). For  $N$  samples  $\mathbf{y}$  with importance weights  $w_i = \mu(y_i)/\nu_\theta(y_i)$ ,

$$\text{ESS}(\mathbf{y}) = \frac{(\sum_i w_i)^2}{N \sum_i w_i^2} \in [0, 1],$$

attained at 1 iff all weights are equal.

**Definition 2** (Coverage; Naeem et al. 2020). For reference samples  $\mathbf{x} = (x_1, \dots, x_P)$  and candidate samples  $\mathbf{y}$ ,

$$\text{NND}_k(x_i; \mathbf{x}) = \inf\{r \geq 0 : \#\{j : 0 < |x_i - x_j| < r\} \geq k\},$$

and

$$\text{Coverage}_k(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1}[\exists j : |y_j - x_i| \leq \text{NND}_k(x_i; \mathbf{x})].$$

**Fake ESS.** If  $\nu_\theta$ 's support is strictly inside  $\mu$ 's, importance weights  $w_i = \mu(y_i)/\nu_\theta(y_i)$  remain computable and can be uniform on the visited region:  $\text{ESS} \rightarrow 1$  does *not* imply  $\text{supp}(\nu_\theta) = \text{supp}(\mu)$ . ESS is a necessary but not sufficient quality criterion; coverage is the sufficient complement.

With the shared vocabulary in place, the survey proper begins. We start with Camp A because it is both the historical and the empirical default of the BG field, and only against its full timeline do Camp B's commitments and the third way's design choices become legible.

## 3. Camp A — simple prior, sophisticated training

**Why this section exists.** “Camp A” has been the de facto default in the BG literature since Noé et al. 2019: keep  $\mu_0$  a standard Gaussian and pour all the engineering effort into the loss, the architecture, and the inference schedule. The thirteen papers below trace the canonical Camp A timeline from the founding 2019 paper to the 2025 state-of-the-art and document the exact loss formula, training procedure, and numerical headlines of each.

### 3.1. The original BG (Noé et al. 2019)

#### At a glance — Noé 2019 (founding BG)

**Target:** known-mode protein systems where short MD has already discovered every basin.

**Prior:** standard isotropic Gaussian (author’s own comment: “rather naive choice”).

**New idea:** hybrid loss reverse-KL + maximum-likelihood + reaction-coordinate; flow trained on each basin’s MD seeds simultaneously.

**Headline result:** first demonstration that a coupling-flow BG can reweight MD samples to compute free-energy profiles on biomolecules, when modes are known a priori.

**Verdict.** Camp A foundational paper; explicitly out-of-scope for mode discovery.

Noé, Olsson, Köhler, Wu, *Science* 365, eaaw1147 (2019); preprint [arXiv:1812.01729](https://arxiv.org/abs/1812.01729). Founding paper. Hybrid loss combining reverse-KL with a maximum-likelihood term over MD samples  $\{y_i^{\text{MD}}\}$  in each known metastable state:

$$\mathcal{L}_{\text{BG}}(\theta) = w_{\text{KL}} \mathcal{L}_{\text{rev}}(\theta) + w_{\text{ML}} \left( -\frac{1}{N_{\text{MD}}} \sum_i \log \nu_{\theta}(y_i^{\text{MD}}) \right) + w_{\text{RC}} \mathcal{L}_{\text{RC}}(\theta).$$

The framework explicitly requires that short MD has already discovered every metastable state of interest — mode discovery is not within scope. Latent prior: standard Gaussian (the canonical [noegroup/bgflow](#) notebook uses `bg.NormalDistribution(dim_ics, mean=mean)` with the author-supplied comment that this is a “rather naive choice”).

### 3.2. Stochastic normalising flows (Wu, Köhler, Noé 2020)

#### At a glance — SNFs

**Target:** multimodal targets where pure deterministic flows mode-collapse.

**Prior:** standard Gaussian.

**New idea:** interleave invertible coupling layers with stochastic MCMC kernels; Jarzynski work replaces the missing  $\log |\det J|$  for the stochastic layers.

**Headline result:** mode-mixing improves on 2D toy targets; biomolecular gains modest.

**Verdict.** Camp A architectural fix; the prior is left untouched.

[NeurIPS 2020](#). Interleave deterministic invertible layers with stochastic MCMC kernels. Path likelihood combines  $\log |\det J|$  terms from deterministic layers with Jarzynski work terms from stochastic ones. *Base distribution: standard Gaussian.*

### 3.3. Equivariant flows (Köhler, Klein, Noé 2020)

#### At a glance — Equivariant flows

**Target:** systems with permutation, translation, or rotation symmetry (LJ clusters, water).

**Prior:** mean-free centred Gaussian on the symmetry-invariant subspace (e.g.  $\{z : \sum z_i = 0\}$ ).

**New idea:**  $G$  built from  $G_{\text{sym}}$ -equivariant coupling layers; the prior’s structure is forced by the symmetry group, not by mode information.

**Headline result:** matches non-equivariant BGs with 5–10 $\times$  fewer parameters on LJ clusters.

**Verdict.** *Geometry, not mode tuning* — the prior structure is determined by the symmetry group, not by knowledge of  $\mu$ ’s modes.

ICML 2020, arXiv:2006.02425. If the target has symmetry group  $G_{\text{sym}}$ , build  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$   $G_{\text{sym}}$ -equivariant with  $G_{\text{sym}}$ -invariant base. The base is a centred mean-free Gaussian on  $\{z : \sum z_i = 0\}$  for translation-rotation symmetry — this is *symmetry-imposed* prior structure, not mode tuning.

### 3.4. Smooth normalising flows on tori and bounded intervals (Köhler–Krämer–Noé 2021; Rezende et al. 2020)

#### At a glance — Smooth NFs on tori

**Target:** molecular Boltzmann in internal coordinates (dihedrals on a torus, bonds/angles on compact intervals).

**Prior:** product of uniform-on-[0, 1] for dihedrals (Haar measure on the torus) and truncated Gaussian for bonded intervals.

**New idea:** coupling transformers built from a  $C^\infty$  smooth partition of unity times smooth invertible bumps, yielding a  $C^2$  density whose gradient  $\nabla_x \log \nu_\theta(x)$  is continuous and usable as a molecular-dynamics force.

**Headline result:** on alanine dipeptide internal-coordinate sampling, smooth-bump flows reach ESS  $\approx 60\%$  versus  $\approx 35\%$  for standard spline flows at matched parameter count.

**Setup.** For a molecule of  $n$  atoms, the Cartesian-to-internal map sends  $x \in \mathbb{R}^{3n}$  (modulo SE(3)) to a tuple  $(b, \alpha, \varphi)$  of  $n - 1$  bond lengths,  $n - 2$  bond angles, and  $n - 3$  dihedrals. The ambient manifold is  $\mathcal{M} = \prod_j I_j^{(b)} \times \prod_j I_j^{(\alpha)} \times \mathbb{T}^{d_{\text{dih}}}$  with  $I_j \subset \mathbb{R}$  bounded and  $\mathbb{T}$  the torus, not  $\mathbb{R}^d$ . A standard Gaussian-based flow  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  is geometrically inappropriate: it neither respects the periodicity of  $\varphi$  nor the boundedness of  $b, \alpha$ . Köhler–Krämer–Noé and Rezende et al. construct coupling layers whose elementary scalar transformer  $T : [0, 1] \rightarrow [0, 1]$  is intrinsically defined on the unit interval and (for the torus) periodic.

**Core math.** Write  $T : [0, 1] \rightarrow [0, 1]$  for the scalar transformer parameterised by conditioner outputs. Two requirements: smoothness ( $T \in C^k$ ,  $T'(y) > 0$ ) and periodicity ( $T(0) = 0$ ,  $T(1) = 1$ ,  $T'(0) = T'(1)$ ,  $T''(0) = T''(1)$ , ... up to the derivative order required by the loss).

*Smooth-bump construction.* Fix knots  $0 = y_0 < \dots < y_K = 1$  and a  $C^\infty$  bump  $\rho(t) = \exp(-1/(1 - t^2))$  supported on  $|t| < 1$ . With localised weights  $w_k(y) = \rho((y - c_k)/h_k)$  and the resulting partition of unity  $\pi_k(y) = w_k(y)/\sum_j w_j(y)$ , define on each interval  $[y_k, y_{k+1}]$  a smooth invertible bump  $\psi_k(y) = y + s_k(y)(y - y_k)(y_{k+1} - y)$  with  $s_k$  a conditioner-controlled shape function. The transformer

$$T(y) = \sum_{k=0}^{K-1} \pi_k(y) \psi_k(y)$$

is then  $C^\infty$ , strictly monotone, and (when  $\pi_k, \psi_k$  are defined modulo 1) periodic.

*Periodic rational-quadratic splines.* For the Rezende construction, the  $K + 1$  knot heights on the torus are constrained so that the first and last interior derivatives agree, and the spline segment crossing the seam  $y \equiv 0$  is parameterised by paired endpoint coefficients  $(\theta_K, \theta_0)$ .

*Forces.* For a coupling layer with one half held fixed and the other transformed by  $T$ , the Jacobian is diagonal on the transformed block and  $\log \nu_\theta(x) = \log \mu_0(G^{-1}(x)) + \sum_{\ell,j} \log T'_{\theta,\ell,j}$ . The  $C^2$  guarantee makes  $\nabla_x \log \nu_\theta(x)$  continuous, enabling a *force-matching* term

$$\mathcal{L}_{\text{force}}(\theta) = \frac{1}{N} \sum_i \|\nabla_x \log \nu_\theta(x_i) + \nabla_x U_1(x_i) - F_i^{\text{MD}}\|^2$$

on top of (or instead of) reverse/forward KL. With a  $C^1$ -only spline this term is discontinuous at the knots.

**Algorithm.**

1. Convert MD frames to internal coordinates  $(b, \alpha, \varphi)$ , rescaling each to  $[0, 1]$ .
2. Stack  $L$  coupling layers, each using the smooth-bump transformer on bond/angle channels and a periodic (rational-quadratic or smooth-bump) transformer on dihedral channels.
3. Train with reverse-KL, optionally augmented by forward-KL on MD seeds and by  $\mathcal{L}_{\text{force}}$ .
4. At inference, either sample  $z \sim \mu_0$  and push through  $G_\theta$  as an IS proposal, or integrate Langevin dynamics with potential  $-\log \nu_\theta$  (only possible because  $\nabla_x \log \nu_\theta$  is continuous).

**Numerical details.** Rezende et al. test on  $\mathbb{T}^2$  and  $S^2$ : on a 2D ring on  $\mathbb{T}^2$  they report KL to ground truth two orders of magnitude below an affine RealNVP baseline. Köhler et al. test on alanine dipeptide in 22 internal coordinates: ESS  $\approx 60\%$  for smooth-bump versus  $\approx 35\%$  for standard rational-quadratic at matched parameter count. Both papers use `normflows`; the smooth-bump layer ships in `bgflow` as `SmoothRingTransform`.

**Critical assessment.** The construction buys geometry-aware support and a flow whose density-gradient is a legitimate force. The base remains *untuned*: uniform-on-torus is the Haar measure (the unique symmetry-respecting choice), and the truncated Gaussian on bonds/angles is fixed by chemical bond stiffness, not by mode hints. In Camp-A vs. Camp-B terms this is *geometry, not mode tuning*.

What is *not* bought: a uniform marginal on each dihedral still puts the multimodal-transport burden on  $G_\theta$ . If the marginal of  $\mu$  on a dihedral has well-separated modes (e.g.  $\phi/\psi$  in alanine), the flow must again carry all multimodal transport, exactly the regime where reverse-KL collapses. Smooth flows on tori are a *correctness fix* (right topology, usable forces), not a mode-coverage fix.

### 3.5. FAB — Flow Annealed Importance Sampling Bootstrap (Midgley et al. 2022)

**At a glance — FAB**

**Target:** multimodal Boltzmann targets on  $\mathbb{R}^d$  with no MD data — mode-discovery, energy-only setting.

**Prior:** standard diagonal Gaussian  $\mu_0 = \mathcal{N}(0, I_d)$ , hardcoded in `fab-torch` as `nf.distributions.base.DiagGaussian(dim)`.

**New idea:** replace mode-seeking reverse-KL by the mass-covering  $\alpha = 2$  Rényi divergence; estimate its gradient from AIS chains anchored at the geometric-mean bridge  $\pi_{1/2} \propto \sqrt{\nu_\theta \mu}$  with a prioritised replay buffer.

**Headline result:** first flow-based BG to learn alanine dipeptide *without any MD samples*, at  $\sim 100\times$  fewer target evaluations than the Noé 2019 hybrid-ML pipeline; recovers all 40 components of a GMM-40 target where reverse-KL collapses to one or two modes.

**Setup.** Let  $\mu(x) \propto \exp(-U_1(x))$  on  $\mathbb{R}^d$  and  $\nu_\theta = (G_\theta)_\# \mu_0$  the flow’s pushforward. The aim is to find  $\theta^*$  such that  $\nu_{\theta^*}$  covers all modes, even those reverse-KL would miss. The bridge path

$$\pi_\beta(x) \propto \nu_\theta(x)^{1-\beta} \mu(x)^\beta, \quad \beta \in [0, 1],$$

interpolates between  $\nu_\theta$  and  $\mu$ ; AIS produces approximate samples from  $\pi_\beta$  given a sample from  $\pi_0 = \nu_\theta$ .

**The  $\alpha = 2$  divergence as variance of weights.** Let  $r(x) = \mu(x)/\nu_\theta(x)$ . Then

$$\begin{aligned}\text{Var}_{\nu_\theta}[r(X)] &= \mathbb{E}_{\nu_\theta}[r(X)^2] - (\mathbb{E}_{\nu_\theta}[r(X)])^2 \\ &= \int \frac{\mu(x)^2}{\nu_\theta(x)^2} \nu_\theta(x) dx - 1 = \int \frac{\mu(x)^2}{\nu_\theta(x)} dx - 1,\end{aligned}$$

since  $\mathbb{E}_{\nu_\theta}[r] = \int \mu dx = 1$ . Therefore

$$D_2(\mu \parallel \nu_\theta) = \log \int \frac{\mu(x)^2}{\nu_\theta(x)} dx = \log(1 + \text{Var}_{\nu_\theta}[r]).$$

Minimising  $D_2$  is equivalent to minimising importance-weight variance, which directly improves  $\text{ESS} = N/(1 + \text{Var}[r])$ . Note  $D_2$  requires  $\nu_\theta(x) > 0$  wherever  $\mu(x) > 0$ : a *full-support flow is mandatory*, explaining why FAB *must* keep a Gaussian base.

**The gradient identity.** Differentiating  $\mathcal{L}_{\text{FAB}}(\theta) = \int \mu^2/\nu_\theta dx$ ,

$$\nabla_\theta \mathcal{L}_{\text{FAB}}(\theta) = - \int \frac{\mu(x)^2}{\nu_\theta(x)} \nabla_\theta \log \nu_\theta(x) dx.$$

Writing  $\mu^2/\nu_\theta = \pi_{1/2}(x) \cdot (\pi_{1/2}(x)/\nu_\theta(x))$  with  $\pi_{1/2} \propto \sqrt{\nu_\theta \mu}$  the bridge, importance-sampling against  $\pi_{1/2}$  gives the practical estimator

$$\nabla_\theta \mathcal{L}_{\text{FAB}}(\theta) \approx - \sum_{i=1}^N w_i^{(\text{AIS})} \nabla_\theta \log \nu_\theta(y_i^{(\text{AIS})})$$

where  $y_i^{(\text{AIS})} \sim \pi_{1/2}$  are AIS samples anchored at  $\nu_\theta$ , with weights  $w_i^{(\text{AIS})} \propto r(y_i)$  accumulated along the chain.

**Prioritised replay buffer.** AIS is expensive, so past  $(y_i, w_i)$  are stored in a buffer  $\mathcal{B}$ . Minibatch priorities are

$$p_i \propto \frac{(w_i^{\text{old}})^2}{\nu_\theta(y_i)} = \frac{\mu(y_i)^2}{\nu_\theta(y_i) (\nu_{\theta_{\text{old}}}(y_i))^2},$$

with importance correction  $1/p_i$  applied for unbiasedness. The priority is exactly the  $\mathcal{L}_{\text{FAB}}$  integrand — high-priority samples are those where the  $D_2$  loss is largest.

### Algorithm.

1. Initialise: random flow + reverse-KL warm-up.
2. Sample  $y_0 \sim \nu_\theta$ ; run  $M$ -step HMC AIS through  $\beta_0 = 0 < \beta_1 < \dots < \beta_M = \frac{1}{2}$  targeting acceptance  $\approx 0.65$ . Accumulate  $w^{(\text{AIS})}$ .
3. Push  $(y_M, w^{(\text{AIS})})$  into buffer  $\mathcal{B}$ .
4. Sample a minibatch with priorities  $p_i \propto w_i^2/\nu_\theta(y_i)$ .
5. Adam step on  $\theta$  using  $\sum_i (w_i/p_i) \nabla_\theta \log \nu_\theta(y_i)$ .
6. Repeat until ESS plateaus.

## Numerical headlines.

- GMM-40 ( $d = 2$ ): all 40 modes recovered; reverse-KL recovers 1–3; forward-KL on  $10^5$  samples recovers  $\sim 25$ .
- Many-Well in  $d \leq 32$ : all  $2^{d/2}$  wells recovered; reverse-KL collapses for  $d \geq 8$ .
- Alanine dipeptide ( $d \approx 60$  internal coordinates): first BG learned without MD samples; ESS  $\approx 6\%$  at  $\sim 100\times$  fewer energy evaluations than Noé 2019.

Hyperparameters (fab-torch defaults):  $M = 4$  to 8; HMC inner  $L_{\text{HMC}} \in \{5, 10\}$  leapfrog; buffer size  $|\mathcal{B}| \in \{2^{14}, 2^{16}\}$ ; 12 RealNVP layers with 256-wide MLP conditioners.

**Critical assessment.** FAB’s  $D_2$  objective is genuinely mass-covering, and routing its gradient through AIS samples breaks the reverse-KL feedback loop that ruins vanilla flow training; the dramatic mode discovery on a 40-component 2D GMM is the canonical demonstration. The catch is that the gradient’s quality *is* the AIS chain’s quality. If a target mode is separated from  $\nu_\theta$ ’s bulk by a barrier that HMC cannot cross within  $L_{\text{HMC}}$  leapfrog steps at intermediate temperatures, no AIS sample lands in that mode, and the gradient is computed as though the mode did not exist; rugged molecular landscapes such as chignolin’s folded and unfolded basins routinely violate this. Two further frailties compound. The prioritised replay buffer recomputes its priorities at the current  $\theta$  but the AIS samples were drawn at  $\theta_{\text{old}}$ , so when  $\theta$  moves faster than the buffer turns over, the gradient is biased; and the base distribution is hardcoded as `DiagGaussian(dim)` inside `make_normflow_realnvp`, with no path for the user to substitute a structured prior. The net effect is the canonical Camp A position: all engineering investment goes into the loss and the AIS machinery; the prior remains  $\mathcal{N}(0, I_d)$  by default.

### 3.6. Adaptive MC + flow (Gabri , Rotskoff, Vanden-Eijnden 2022)

Gabri , Rotskoff, Vanden-Eijnden, *PNAS* 119(10), e2109420119 (2022); preprint [arXiv:2107.08001](https://arxiv.org/abs/2107.08001). The flowMC library that follows (Wong, Gabri , Foreman-Mackey, *JOSS* 2023; [kazewong/flowMC](https://github.com/kazewong/flowMC)) packages the recipe and demonstrates dramatic wall-clock wins on gravitational-wave parameter estimation and primordial-feature cosmology benchmarks.

#### At a glance — Adaptive MCMC + flow

**Target.** A possibly multimodal  $\mu \propto e^{-U_1}$  on  $\mathbb{R}^d$  with tractable energy  $U_1$ , in moderate dimension  $d \sim 10\text{--}50$ .

**Prior / base of the flow.** Standard isotropic Gaussian; nothing problem-specific is encoded there.

**New idea.** Don’t pre-train a flow and hope it generalises: *train it online on the chain history* so that, as the chains explore, the flow becomes a progressively better global Metropolis–Hastings proposal that complements local HMC/MALA moves.

**Headline.** flowMC reports 25–50 $\times$  speed-ups over nested sampling on gravitational-wave and cosmology problems; the framework remains exact (every flow proposal is wrapped in an MH accept/reject step).

**Setup.** A bank of  $K$  parallel chains  $\{y_t^{(k)}\}_{k=1}^K$  is evolved by a local kernel (HMC, MALA, or unadjusted Langevin), accumulating a history  $\mathcal{D}_n = \{y_j^{(k)}\}_{k,j}$  up to round  $n$ . A real-NVP-style flow

$\nu_\theta$  with standard-Gaussian base is fitted to  $\mathcal{D}_n$  by maximum likelihood and used as an independent-MH global proposal. The expectation is that local moves explore each basin while flow proposals jump between basins; the two kernels are composed in a standard mixture or cycle and the chain is still exact for  $\mu$ .

**Core math.** Two ingredients. (i) The flow is fitted to history by forward-KL / maximum likelihood, i.e. a Monte Carlo estimate of  $\text{KL}(\mu_{\mathcal{D}_n} \parallel \nu_\theta)$ :

$$\mathcal{L}_{\text{adapt}}(\theta; \mathcal{D}_n) = -\frac{1}{|\mathcal{D}_n|} \sum_{y \in \mathcal{D}_n} \log \nu_\theta(y).$$

(ii) Given a current state  $y$ , the flow proposes  $y' \sim \nu_\theta$  and the move is accepted with the independent-MH ratio

$$\alpha(y, y') = \min \left\{ 1, \frac{\mu(y') \nu_\theta(y)}{\mu(y) \nu_\theta(y')} \right\}.$$

Because  $\nu_\theta$  is exactly evaluable (the flow is bijective and  $\log \nu_\theta$  is available via change-of-variables),  $\alpha$  is computable in closed form and the resulting kernel leaves  $\mu$  invariant by construction. The key dynamical fact is that the local kernel is what *discovers* new modes; the flow’s role is to *share* mass between modes the chains have already visited at least once.

**Algorithm.** The training loop alternates exploration and refit:

1. Initialise  $K$  parallel chains  $\{y_0^{(k)}\}$  from a coarse over-dispersed distribution; initialise  $\nu_\theta$  with a few warm-up local-MCMC steps.
2. For  $t = 1, 2, \dots$ :
  - Take  $\Delta t$  local-kernel steps on each chain (HMC, MALA, or Langevin), appending samples to  $\mathcal{D}_n$ .
  - Interleave one or more independent-MH proposals from  $\nu_\theta$  per chain, accepting with ratio  $\alpha$  above.
3. Every  $T_{\text{retrain}}$  outer steps, re-fit the flow by minimising  $\mathcal{L}_{\text{adapt}}(\theta; \mathcal{D}_n)$  for a few hundred Adam steps on the accumulated history (with optional thinning).
4. Repeat until target diagnostics (effective sample size,  $\hat{R}$ ) stabilise; the final  $\mathcal{D}_n$  is the posterior sample.

**Numerical details.** The PNAS paper studies a 2D double-well and an  $\sim 10$ -dimensional Bayesian inverse problem. flowMC extends the approach to gravitational-wave parameter estimation (binary black hole and binary neutron-star sources,  $d \sim 10$ –15 astrophysical parameters) and CMB primordial-feature inference. Reported headlines: 25–50 $\times$  faster than nested-sampling baselines on these problems; high acceptance for the flow kernel when modes are reasonably close. The library uses JAX, supports HMC and MALA local kernels, real-NVP and rational-quadratic-spline flows, and exposes a small configuration surface (chain count, retrain frequency, flow depth). Practical guidance from the authors is to use  $K \sim 10^2$  chains and refit the flow every  $10^2$ – $10^3$  local steps.

**Critical assessment.** The Gabri  -Rotskoff-Vanden-Eijnden recipe is exact (every flow proposal is wrapped in a Metropolis–Hastings accept step, so the chain targets  $\mu$  regardless of how badly  $\nu_\theta$  fits) and remarkably modular: the same machinery serves gravitational-wave posteriors and Boltzmann targets, both being of the form  $\mu \propto e^{-U_1}$ . Training the flow on chain history is also a benign forward-KL regime where flows are well-behaved, since training never sees an undiscovered mode. The limitations are equally clear. Mode discovery is delegated entirely to the local kernel, and if HMC misses a basin for  $10^6$  steps then  $\nu_\theta$  has zero density there and the global proposal cannot retrieve it; the independent-Metropolis acceptance rate of a flow proposal degrades exponentially in the effective dimension and the mode separation, which is why published demonstrations sit in moderate dimension  $d \sim 10\text{--}50$ ; and bursts of low acceptance during retraining inflate autocorrelation and produce exactly the fake-ESS pathology analysed in §2.6. What distinguishes this method is the candour of its stance on the prior-knowledge problem: the prior of  $\nu_\theta$  stays Gaussian, but operational mode knowledge enters through chain initialisation and the local kernel rather than through any explicit prior tuning. This is precisely the externalisation pattern the  $X_{\hat{\mu}}$  construction formalises in §9.

### 3.7. Resampled base distributions / LARS (Stimper et al. 2022)

Stimper, Sch  lkopf, Hern  ndez-Lobato, AISTATS 2022; [arXiv:2110.15828](https://arxiv.org/abs/2110.15828). Released as part of the [normflows](#) library, the same package later used as the flow backbone of FAB and TA-BG.

#### At a glance — Resampled base / LARS

**Target.** Boltzmann-style  $\mu \propto e^{-U_1}$  on  $\mathbb{R}^d$  whose topology is hard for a single-Gaussian-base flow (multimodal, multi-component, complicated support).

**Prior / base.** A learned-rejection base on top of a Gaussian:  $\tilde{p}_\phi(z) \propto \pi(z) a_\phi(z)$  with  $\pi = \mathcal{N}(0, I)$  and a learned acceptance function  $a_\phi : \mathbb{R}^d \rightarrow [0, 1]$ .

**New idea.** Move some of the geometric flexibility from the flow into the *base* itself by learning a Gaussian-times-acceptance density — a mild step toward Camp B that keeps the prior data-driven rather than hand-tuned.

**Headline.** Mode-faithful 2D synthetic densities at fixed flow depth; modest but consistent gains over a plain Gaussian base on small Boltzmann benchmarks; never adopted as default by mainstream BG papers.

**Setup.** The construction — “Learned Accept/Reject Sampling” (LARS) in the AISTATS title — builds an enriched base distribution  $\tilde{p}_\phi$  that the flow then transports onto  $\mu$ . The acceptance function  $a_\phi$  is parameterised by a small neural network with output squashed to  $[0, 1]$  (sigmoid). Sampling from  $\tilde{p}_\phi$  is by rejection: draw  $z \sim \pi$ , accept with probability  $a_\phi(z)$ , otherwise resample. Density evaluation requires the normalising constant  $Z_\phi = \int \pi(z) a_\phi(z) dz$ , estimated by an independent Monte Carlo average over  $\pi$ .

**Core math.** The resampled-base density and its log are

$$\tilde{p}_\phi(z) = \frac{\pi(z) a_\phi(z)}{Z_\phi}, \quad Z_\phi = \int \pi(z) a_\phi(z) dz,$$

$$\log \tilde{p}_\phi(z) = \log \pi(z) + \log a_\phi(z) - \log Z_\phi.$$

Composing with a flow  $G_\theta$  gives the pushforward model density

$$\nu_{\theta, \phi}(x) = \tilde{p}_\phi(G_\theta(x)) \left| \det J_{G_\theta}(x) \right|.$$

Training is by reverse-KL (or forward-KL on samples, whichever the downstream loss requires); both  $\theta$  and  $\phi$  are updated jointly:

$$\mathcal{L}_{\text{rev}}(\theta, \phi) = \mathbb{E}_{z \sim \tilde{p}_\phi} [U_1(G_\theta^{-1}(z)) - \log |\det J_{G_\theta^{-1}}(z)|] + \mathbb{E}_{z \sim \tilde{p}_\phi} [\log \tilde{p}_\phi(z)] + \text{const.}$$

Two technical subtleties matter in practice. (i) The normalising constant  $Z_\phi$  is a noisy MC estimate; the paper uses a running exponential-moving average to reduce gradient variance. (ii) Sampling-time rejection cost is  $\mathbb{E}[1/a_\phi(z)]$  rejections per accepted sample under  $\pi$ ; the loss implicitly penalises pathologically peaky  $a_\phi$  by inflating  $-\log Z_\phi$ .

**Algorithm.** A representative training round:

1. Initialise flow parameters  $\theta$  and acceptance-network parameters  $\phi$ ; set  $a_\phi \equiv 1/2$  (uniform acceptance) so that  $\tilde{p}_\phi = \pi$  at initialisation.
2. At each step, draw a minibatch from  $\pi$  and run rejection with  $a_\phi$  to obtain samples  $\{z_i\} \sim \tilde{p}_\phi$ ; in parallel estimate  $Z_\phi$  by an independent MC average over  $\pi$ .
3. Push the accepted  $z_i$  through  $G_\theta^{-1}$  to obtain  $x_i$ ; evaluate  $U_1(x_i)$  and the log-Jacobian and form the reverse-KL gradient.
4. Backpropagate through both  $\theta$  and  $\phi$ ; the dependence of the sampling distribution on  $\phi$  is handled with a likelihood-ratio (REINFORCE) correction inside the normflows implementation.
5. Update the EMA of  $\log Z_\phi$ ; repeat until the joint loss plateaus.

**Numerical details.** The AISTATS paper evaluates the construction on 2D synthetic densities (rings, mixtures, spirals), on image generation as a flow-only model, and on small Boltzmann-style benchmarks. Reported gains over Gaussian base are modest but consistent at fixed flow depth, and the resampled-base flow can fit topology-mismatched 2D targets that defeat plain real-NVP. The construction is available as `nf.distributions.base.ResampledGaussian` in normflows; downstream BG pipelines such as FAB and TA-BG depend on normflows but instantiate `DiagGaussian` rather than the resampled variant.

**Critical assessment.** Conceptually LARS sits exactly on the Camp A versus Camp B fault line: the base is no longer a fixed Gaussian, yet the modification is data-driven through the learned acceptance function  $a_\phi$  rather than physics-driven through a hand-placed mixture. In low dimension the construction is appealing because the acceptance map can carve out the right topology on 2D toy targets, but several factors have kept it out of mainstream BG practice. First, the variance of the partition-function estimator  $Z_\phi$  grows with  $d$ , so the loss becomes noisy precisely where multimodal protein targets live; in practice the EMA estimate drifts slowly enough to corrupt early-training gradients. Second, the gain over a deeper flow with a plain Gaussian base is marginal once the flow has enough capacity, since bijective coupling-layer flows can absorb most of what  $a_\phi$  offers by routing more capacity into the early couplings. Third, the construction does not solve the support-mismatch problem that motivates Camp B in the first place:  $\tilde{p}_\phi$  still has the support of the underlying Gaussian, so any target mode requires a flow trajectory to carry mass across a contractible base. The field’s empirical verdict has been that the same engineering budget is better spent on flow depth, richer couplings, or AIS-based reweighting; the `ResampledGaussian` class in normflows is essentially never the default in production BG notebooks. Stimper et al. deserve credit for taking the base distribution seriously — the construction remains a useful benchmark for arguments about *where* expressiveness should live (in the base, in the bijection, or in a separate proposal density) — but the empirical case for it at protein scale has not been made.

### 3.8. Equivariant flow matching (Klein, Krämer, Noé 2023)

Klein, Krämer, Noé, NeurIPS 2023; [arXiv:2306.15030](https://arxiv.org/abs/2306.15030). Built on top of the flow-matching framework of Lipman, Chen, Ben-Hamu, Nickel, Le (ICLR 2023). The first flow-based Boltzmann generator that operates on Cartesian coordinates of a molecule without an internal-coordinate featurisation.

#### At a glance — Equivariant flow matching

**Target.** Boltzmann distribution of a molecule with permutation-invariant atom labels and full  $\text{SE}(3) \times S_n$  symmetry, e.g. alanine dipeptide in Cartesian coordinates.

**Prior / base.** Mean-free Gaussian on the linear subspace  $\{z \in \mathbb{R}^{3n} : \sum_i z_i = 0\}$  — the maximally entropic  $\text{SE}(3) \times S_n$ -invariant Gaussian.

**New idea.** Train the *velocity field* of a probability-flow ODE by regressing onto closed-form conditional velocities (flow matching), and constrain the velocity field to be exactly  $\text{SE}(3) \times S_n$ -equivariant via an EGNN. The flow itself never needs to be parameterised or inverted during training.

**Headline.** First flow-based BG to handle a peptide entirely in Cartesian coordinates; equivariance gives substantial sample efficiency over non-equivariant flow matching; free-energy differences match MD reference.

**Setup.** Let  $\mu_0$  be the mean-free Gaussian and  $\mu_1 = \mu$  the Boltzmann target. Flow matching constructs a probability path  $\{\rho_t\}_{t \in [0,1]}$  with  $\rho_0 = \mu_0$ ,  $\rho_1 = \mu_1$  and a corresponding velocity field  $u_t$  such that  $\partial_t \rho_t + \nabla \cdot (\rho_t u_t) = 0$ . The probability path is built by mixing per-sample conditional paths  $\rho_t(\cdot|y_1)$  that interpolate from a wide Gaussian at  $t = 0$  to a  $\delta_{y_1}$  at  $t = 1$ . A neural velocity  $v_\theta(x, t)$  is then regressed onto the conditional velocity  $u_t(\cdot|y_1)$ , which is available in closed form. At inference the ODE  $\dot{y} = v_\theta(y, t)$  is integrated from  $y_0 \sim \mu_0$  to  $y_1 \sim \mu_1$ .

**Core math.** The Gaussian conditional path used in the paper is

$$\rho_t(x|y_1) = \mathcal{N}(x; ty_1, (1-t)^2 I),$$

which interpolates between  $\rho_0(\cdot|y_1) = \mathcal{N}(0, I)$  and  $\rho_1(\cdot|y_1) = \delta_{y_1}$ . The associated conditional velocity is

$$u_t(x|y_1) = \frac{y_1 - x}{1-t}.$$

The flow-matching loss is the conditional-velocity regression

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{y_1 \sim \mu_1} \mathbb{E}_{x \sim \rho_t(\cdot|y_1)} \left[ \|v_\theta(x, t) - u_t(x|y_1)\|^2 \right].$$

At the optimum, the ODE  $\dot{y} = v_\theta(y, t)$  transports  $\rho_0 = \mu_0$  to  $\rho_1 = \mu_1$ , and the model density at  $y_1$  is obtained by the continuous-flow change-of-variables

$$\log \nu_\theta(y_1) = \log \mu_0(y_0) - \int_0^1 \nabla \cdot v_\theta(y_t, t) dt,$$

with the divergence term computed via Hutchinson’s trace estimator at  $\mathcal{O}(d)$  per ODE step. Equivariance is imposed by parameterising  $v_\theta$  with an  $\text{E}(3)$ -equivariant graph neural network (EGNN) acting on the fully-connected atom graph, with  $S_n$  equivariance inherited from the unordered set structure.

**Algorithm.** A representative training round:

1. Pre-process MD samples: subtract the centre of mass so that  $y_1 \in \{x : \sum_i x_i = 0\}$ ; this places the data in the support of the mean-free base.
2. Sample  $t \sim U[0, 1]$ , draw a target sample  $y_1$  from the MD pool, and sample  $x \sim \rho_t(\cdot|y_1) = \mathcal{N}(ty_1, (1-t)^2 I)$  by the reparameterisation  $x = ty_1 + (1-t)\varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, I)$ .
3. Compute the closed-form conditional velocity  $u_t(x|y_1) = (y_1 - x)/(1-t)$ .
4. Forward-pass the EGNN  $v_\theta(x, t)$  and form the squared-error loss  $\|v_\theta(x, t) - u_t(x|y_1)\|^2$ ; back-propagate.
5. At inference, draw  $y_0 \sim \mu_0$  (mean-free Gaussian) and integrate the ODE with a standard adaptive solver (Dopri5 or RK45) to obtain  $y_1 \sim \nu_\theta$ ; if importance weights are desired, accumulate the Hutchinson trace along the trajectory to obtain  $\log \nu_\theta(y_1)$ .

**Numerical details.** Reported experiments include alanine dipeptide and small peptides in Cartesian coordinates with the molecule’s full Cartesian graph. Equivariance gives substantial sample-efficiency gains over a non-equivariant flow-matching baseline at matched parameter count; reported free-energy differences along the canonical  $\phi/\psi$  Ramachandran transitions agree with MD reference within statistical error. Inference cost is dominated by the ODE solve and the Hutchinson trace; a typical sample costs on the order of  $10^2$  network evaluations.

**Critical assessment.** Equivariant flow matching is the cleanest example of Camp A’s design philosophy in action: the base distribution is the maximally symmetric Gaussian implied by the target’s exact symmetry group, and all engineering goes into the loss (closed-form conditional velocities turn training into a regression problem, sidestepping adjoint-sensitivity backprop) and the architecture (the EGNN imposes symmetry exactly rather than penalising violations). The shift from parameterising  $G$  to parameterising its infinitesimal generator  $v_\theta$  has the further pleasant consequence that the model is continuous in  $t$ : one can stop the ODE early for partial annealing, or chain it with a discrete coupling-layer flow as residual correction. Three costs are worth flagging. Tractable log-density at inference requires an ODE solve plus a stochastic-trace estimator, which makes importance-weighting corrections expensive — this is why downstream pipelines needing exact  $\log \nu_\theta$  for FAB-style  $\alpha = 2$  training still prefer coupling-layer flows. Mode discovery is not addressed: training samples come from MD, so the method inherits the metastable-state coverage of the MD reference, and the conditional path  $\rho_t(x|y_1) = \mathcal{N}(ty_1, (1-t)^2 I)$  is sharply concentrated on a tube around the segment  $0 \rightarrow y_1$ , so any mode unseen in the MD pool contributes nothing to the loss. Finally the equivariance constraint, valuable as it is for shrinking the learning problem, is incompatible with a Camp B labelled-atom GMM prior unless the mixture is symmetrised over  $|S_n| = n!$  images. The contribution is therefore correctly read as a better Camp A backbone for systems with known  $\text{SE}(3) \times S_n$  symmetry, not as a step toward multimodal discovery; the 2024 TBG follow-up carries this backbone into multi-system amortisation.

### 3.9. iDEM — Iterated Denoising Energy Matching (Akhound-Sadegh et al. 2024)

#### At a glance — iDEM

**Target.**  $\mu \propto \exp(-U_1)$  on  $\mathbb{R}^d$ ,  $U_1$  differentiable, no target samples available. Benchmarks: GMM-40 (2D), DW-4 (8D), LJ-13 (39D), LJ-55 (165D).

**Prior.** Standard isotropic Gaussian  $\mathcal{N}(0, I)$ , fixed as the terminal distribution of a variance-preserving (VP) forward SDE; not tuned, not learned.

**New idea.** Bypass the absence of target data by training a diffusion score network against a Monte Carlo estimator of the noised-marginal score, built from  $U_1$  alone; iterate between sampling with the current score and refitting it on the resulting replay buffer.

**Headline.** First energy-only neural sampler to scale to LJ-55 (165-dimensional), with reported 2–5 $\times$  wall-clock speedups over FAB on shared benchmarks and competitive sample-quality / ESS.

Akhound-Sadegh et al., ICML 2024, [arXiv:2402.06121](https://arxiv.org/abs/2402.06121); code: [jarriidrb/DEM](https://github.com/jarriidrb/DEM). iDEM is the canonical entry of *score-based* samplers into the Boltzmann-generator literature and is the central reference for the diffusion lineage that continues through iEFM, EWFM, and Sequential BG.

**Setup.** Let  $X_0 \sim \mu_1$  where  $\mu_1(x) \propto \exp(-U_1(x))$  is the target on  $\mathbb{R}^d$ . Define a forward variance-preserving SDE driven by a positive noise schedule  $\beta : [0, 1] \rightarrow \mathbb{R}_{>0}$ ,

$$dX_t = -\frac{1}{2}\beta(t) X_t dt + \sqrt{\beta(t)} dW_t, \quad t \in [0, 1], \quad (1)$$

arranged so that  $X_1 \approx \mathcal{N}(0, I)$ . The transition kernel admits a closed form,

$$q_{t|0}(x_t | x_0) = \mathcal{N}(x_t; \alpha_t x_0, \sigma_t^2 I), \quad \alpha_t = \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right), \quad \sigma_t^2 = 1 - \alpha_t^2, \quad (2)$$

so the noised marginal at time  $t$  is  $p_t(x_t) = \int q_{t|0}(x_t | x_0) \mu_1(x_0) dx_0$ . Sampling at inference time will run the reverse-time SDE  $dX_t = [-\frac{1}{2}\beta(t)X_t - \beta(t)\nabla_x \log p_t(X_t)] dt + \sqrt{\beta(t)} d\bar{W}_t$  from  $t = 1$  down to  $t = 0$ , which requires the time-dependent score  $s(x, t) := \nabla_x \log p_t(x)$ .

**Core math contribution.** The standard score-matching identity gives two equivalent expressions for  $s(x, t)$ . Tweedie’s formula yields

$$\nabla_x \log p_t(x) = \mathbb{E}_{x_0 \sim q_{0|t}(\cdot | x)}[\nabla_x \log q_{t|0}(x | x_0)] = \mathbb{E}_{x_0 | x} \left[ \frac{\alpha_t x_0 - x}{\sigma_t^2} \right], \quad (3)$$

where the posterior  $q_{0|t}(x_0 | x)$  satisfies

$$q_{0|t}(x_0 | x) \propto q_{t|0}(x | x_0) \mu_1(x_0) \propto q_{t|0}(x | x_0) \exp(-U_1(x_0)). \quad (4)$$

The key obstruction is that one cannot draw  $x_0 \sim q_{0|t}(\cdot | x)$  directly without already solving the very sampling problem one is trying to solve. iDEM’s contribution is to replace the posterior expectation in (3) by an importance-sampled estimator using  $q_{t|0}(x | \cdot)$ , viewed as a function of  $x_0$  at fixed  $x$ , as the proposal. Concretely, draw

$$x_0^{(k)} \sim \mathcal{N}\left(\frac{x}{\alpha_t}, \frac{\sigma_t^2}{\alpha_t^2} I\right), \quad k = 1, \dots, K, \quad (5)$$

which is exactly the Gaussian whose density is proportional to  $q_{t|0}(x | x_0)$  in  $x_0$ . The self-normalised importance weights are

$$w_k \propto \exp(-U_1(x_0^{(k)})), \quad \tilde{w}_k = w_k / \sum_j w_j, \quad (6)$$

because the Gaussian factor cancels between target and proposal. The self-normalised score estimator is

$$\widehat{s}(x, t) = \sum_{k=1}^K \tilde{w}_k \frac{\alpha_t x_0^{(k)} - x}{\sigma_t^2}. \quad (7)$$

The iDEM training objective is then the conditional score-matching loss against this stochastic target,

$$\mathcal{L}_{\text{iDEM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1]} \mathbb{E}_{x_t \sim \mathcal{B}} \left[ \| s_\theta(x_t, t) - \widehat{s}(x_t, t) \|^2 \right], \quad (8)$$

where  $\mathcal{B}$  is a replay buffer of  $x_t$  values constructed by running the reverse SDE under the *current* parameters  $\theta$  and then noising the resulting endpoint samples back to a random time  $t$ .

A short calculation justifies (7). With proposal  $r(x_0 | x) \propto q_{t|0}(x | x_0)$  (Gaussian in  $x_0$ ) and target posterior  $q_{0|t} \propto q_{t|0} \mu_1$ , the unnormalised importance ratio in  $x_0$  is  $\mu_1(x_0)/Z_{r,x,t}$ , where  $Z_{r,x,t}$  does not depend on  $x_0$ . Self-normalisation absorbs  $Z_{r,x,t}$  and the partition function of  $\mu_1$ , leaving the formula in (6).

### Algorithm.

1. Initialise  $\theta$ ; initialise the replay buffer  $\mathcal{B}$  with samples drawn from  $\mathcal{N}(0, I)$  noised to random  $t \in [0, 1]$ .
2. (*Outer loop, sampling phase.*) Using the current  $s_\theta(\cdot, t)$ , integrate the reverse-time SDE on  $[1, 0]$  with an Euler–Maruyama discretisation to produce  $M$  candidate endpoint samples  $\{x_0^{(m)}\}_{m=1}^M$ .
3. For each  $x_0^{(m)}$ , draw a random  $t^{(m)} \sim \mathcal{U}[0, 1]$  and noise it via (2) to obtain  $x_t^{(m)}$ ; append the pair  $(x_t^{(m)}, t^{(m)})$  to  $\mathcal{B}$  (with optional priority weighting on low energy).
4. (*Inner loop, fitting phase.*) For  $J$  gradient steps, sample minibatches  $(x_t, t) \sim \mathcal{B}$ , draw  $K$  proposal samples from (5), evaluate  $U_1$  to form the weights (6), and compute  $\widehat{s}(x_t, t)$  via (7). Take an Adam step on (8).
5. Return to step 2 until a time budget is exhausted. At inference, integrate the reverse SDE under the final  $s_\theta$  to obtain samples.

**Numerical details.** The released benchmarks are GMM-40 (2D mixture of 40 Gaussians as a stress test for mode coverage), DW-4 (4 particles in  $\mathbb{R}^2$ , double-well coupling,  $d = 8$ ), LJ-13 ( $d = 39$ ) and LJ-55 ( $d = 165$ ). For Lennard–Jones targets the energy is clipped to mitigate the  $r^{-12}$  singularity. Reported MC budget:  $K \in \{500, 1000\}$  proposal draws per estimator call. Buffer size on the order of  $10^4$ – $10^5$ . Time discretisation: 1000 Euler–Maruyama steps for inference,  $\sim 100$ – $500$  for training sample generation. Optimiser: Adam, learning rate  $\sim 10^{-3}$ , exponential-moving-average parameters used for sampling. Architecture: an  $\text{SE}(3) \times S_N$ -equivariant EGNN for the particle-system benchmarks, a standard MLP score net for GMM-40. Wall-clock comparison against FAB on LJ-13 and DW-4 shows a  $2$ – $5\times$  training-time speedup at matched sample quality. The headline LJ-55 run, which FAB cannot complete, takes on the order of a day on a single A100.

**Critical assessment.** iDEM is the first energy-only sampler that scales to LJ-55 without requiring any MD samples, and the high-noise regime of its forward SDE delivers mode discovery by smoothing energy barriers rather than by any prior tuning. Four costs sharpen its place in the prior-choice debate. First, iDEM requires a cheap differentiable  $U_1$ : the inner IS step demands  $K$  energy evaluations per minibatch element, which makes the method inapplicable to PDE-likelihood problems where each  $U_1$

evaluation requires a forward solve. Second, iDEM has no tractable model log-density — inference is via the reverse SDE, and converting to a probability-flow ODE would buy  $\log \nu_\theta$  only at the cost of a stochastic trace estimator at every integration step and a recommitment of the equivariant architecture. Without  $\log \nu_\theta$  the standard importance-correction  $w = \exp(-U_1)/\nu_\theta$  is unavailable, and ESS is reported indirectly through rejection sampling against an independent reference; this is a meaningful gap against the FAB tradition. Third, the base distribution  $\mu_0 = \mathcal{N}(0, I)$  is structurally fixed by the variance-preserving SDE:  $\mu_0$  is the SDE’s terminal distribution, not a free knob. Encoding mode information would require either reweighting the SDE so that  $X_1 \sim \sum_k \pi_k \mathcal{N}(c_k, \Sigma_k)$  (with the score formula updated accordingly) or running a non-Gaussian reference process, and iDEM does neither. Mode discovery here comes from noise smoothing, not from any prior structure — the canonical Camp A by default pattern. Fourth, the bilevel structure (sample with the current model, train against the energy-Monte-Carlo target derived from those samples) inherits the convergence caveats of self-distillation: early drift can bias the replay buffer toward a wrong mode set, after which the energy-MC target is locally consistent with the wrong basins. The authors mitigate this with a wide noise schedule and aggressive buffer turnover, but no formal guarantee is provided, and we register this as the primary risk surface for diffusion-based BGs going forward.

### 3.10. iEFM and EWFM — energy-supervised flow matching

#### At a glance — iEFM / EWFM

**Target.** Same Boltzmann targets as iDEM (GMM-40, DW-4, LJ-13).  $U_1$  differentiable and cheap.

**Prior.** Standard Gaussian  $\mu_0 = \mathcal{N}(0, I)$ , used as the flow-matching source distribution. EWFM keeps the same source but adds an explicit energy reweighting in the integrand.

**New idea.** Replace iDEM’s score-matching objective by a velocity-matching (CNF / flow-matching) objective with an analogous energy-based Monte Carlo target. The resulting flow admits a probability-flow ODE and hence a tractable model log-density.

**Headline.** Matches or improves on iDEM’s GMM / DW-4 / LJ-13 sample quality with ODE rather than SDE inference, restoring the importance correction  $w = \exp(-U_1)/\nu_\theta$  that iDEM gives up.

Woo et al., iEFM 2024, arXiv:2408.16249 and Hahn et al., EWFM 2025, arXiv:2509.03726. Both papers are direct flow-matching analogues of iDEM and adopt the same prior choice; we treat them together as one design point.

**Setup.** Flow matching (Lipman et al. 2023) trains a velocity field  $v_\theta : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$  so that the ODE  $dx_t = v_\theta(x_t, t) dt$  pushes  $\mu_0$  to  $\mu_1$ . The optimal-transport coupling fixes the path

$$x_t = (1 - t)x_0 + tx_1, \quad x_0 \sim \mu_0, \quad x_1 \sim \mu_1, \quad (9)$$

with conditional velocity  $u_t(x | x_0, x_1) = x_1 - x_0$ . In the *supervised* flow-matching setting one has access to  $\mu_1$  samples  $x_1$ ; the standard objective is then  $\mathbb{E}_{t, x_0, x_1} \|v_\theta((1 - t)x_0 + tx_1, t) - (x_1 - x_0)\|^2$ . For Boltzmann sampling no such  $x_1$  samples exist, and the iEFM contribution is to construct an MC estimator of the marginal velocity  $u_t(x) := \mathbb{E}_{(x_0, x_1)|x} [x_1 - x_0]$  using  $U_1$  alone.

**Core math contribution.** The marginal density along the OT interpolation factorises through  $(x_0, x_1)$  and the linear constraint  $x = (1 - t)x_0 + tx_1$ . Eliminating  $x_0 = (x - tx_1)/(1 - t)$ , the

conditional law of  $x_1$  given  $(x, t)$  is

$$p_{1|t}(x_1 | x) \propto \mu_1(x_1) \mu_0\left(\frac{x-tx_1}{1-t}\right), \quad (10)$$

and one obtains

$$u_t(x) = \mathbb{E}_{x_1|x}\left[x_1 - \frac{x-tx_1}{1-t}\right] = \frac{1}{1-t}\left(\mathbb{E}_{x_1|x}[x_1] - x\right). \quad (11)$$

The conditional expectation in (11) is intractable but can be approximated by self-normalised importance sampling against the Gaussian factor  $\mu_0((x-t\cdot)/(1-t))$ . With proposal  $r(x_1 | x) \propto \mu_0((x-tx_1)/(1-t))$  (a Gaussian in  $x_1$  centred at  $x/t$  with covariance  $((1-t)/t)^2 I$  when  $\mu_0 = \mathcal{N}(0, I)$ ) and importance weights  $\tilde{w}_k \propto \exp(-U_1(x_1^{(k)}))$  as in iDEM, the estimator is

$$\hat{u}_t(x) = \frac{1}{1-t}\left(\sum_{k=1}^K \tilde{w}_k x_1^{(k)} - x\right). \quad (12)$$

The iEFM loss is

$$\mathcal{L}_{\text{iEFM}}(\theta) = \mathbb{E}_{t, x_t \sim \rho_t^{\text{model}}}\left[\|v_\theta(x_t, t) - \hat{u}_t(x_t)\|^2\right], \quad (13)$$

where  $\rho_t^{\text{model}}$  is the marginal of  $x_t = (1-t)x_0 + t\bar{x}_1$  with  $\bar{x}_1$  drawn from the current flow’s output — the same bilevel structure as iDEM.

EWFM extends this by inserting an explicit energy weight inside the expectation,  $\hat{u}_t^{\text{EWFM}}(x) = \sum_k \tilde{w}_k^\gamma (x_1^{(k)} - x_0^{(k)})$  with  $\gamma > 1$ , sharpening the implicit posterior toward low-energy regions and biasing the trained velocity toward mode coverage. The price is increased variance: large  $\gamma$  collapses the effective sample size of the estimator.

#### Algorithm (iEFM).

1. Initialise  $v_\theta$  and an empty buffer  $\mathcal{B}$ . Populate  $\mathcal{B}$  with  $\mu_0$  samples paired with random  $t$ .
2. Integrate the ODE  $dx = v_\theta(x, t) dt$  from  $t = 0$  to  $t = 1$  under the current  $\theta$  to produce  $\bar{x}_1$ ; reverse-interpolate by drawing  $t$  and  $x_0 \sim \mu_0$  to obtain  $x_t = (1-t)x_0 + t\bar{x}_1$ ; append.
3. Take inner-loop gradient steps on (13): per minibatch  $(x_t, t)$ , draw  $K$  Gaussian proposals for  $x_1$ , weight by  $\exp(-U_1)$ , evaluate  $\hat{u}_t(x_t)$  from (12), backpropagate.
4. Repeat steps 2–3. At inference, draw  $x_0 \sim \mu_0$  and integrate the ODE forward.

**Numerical details.** Benchmarks: GMM-40, DW-4, LJ-13. Architectures:  $\text{SE}(3) \times S_N$ -equivariant EGNN for particle systems. Reported  $K \in \{200, 500\}$ . ODE integrator: Dormand–Prince (`dopri5`) for inference, 50–100 Euler steps for training-time forward passes. EWFM reports modest improvements in mode-coverage metrics on GMM-40 over iEFM at matched compute, with  $\gamma \in [1, 3]$  as the sweet spot.

**Critical assessment.** The defining advantage over iDEM is that the probability-flow ODE permits  $\log \nu_\theta$  via the instantaneous change of variables,

$$\log \nu_\theta(x_1) = \log \mu_0(x_0) - \int_0^1 \text{tr}(\nabla v_\theta(x_t, t)) dt,$$

where  $x_0$  is recovered by backward ODE integration. This makes the importance correction  $w = \exp(-U_1)/\nu_\theta$  tractable and restores compatibility with FAB-style  $\alpha = 2$  training or post-hoc

reweighting. EWFm’s energy-weighted variant is a worthwhile design point but inherits the variance amplification of any reweighted MC estimator; the paper does not formally analyse the bias–variance trade-off introduced by  $\gamma$ . Both methods share iDEM’s bilevel concern: the buffer is generated by the current model, and early-training drift biases the IS estimator’s effective support. Neither paper varies the prior; both fix  $\mu_0 = \mathcal{N}(0, I)$  at the OT source. The prior-choice debate is, again, untouched.

### 3.11. TBG — Transferable Boltzmann Generators (Klein, Krämer, Noé 2024)

#### At a glance — TBG

**Target.** Equilibrium distributions of small peptides under a classical force field (AMBER), parametrised by chemical identity. Training set: 200 dipeptides; held-out test set: 100 unseen dipeptides; flagship single target: alanine dipeptide.

**Prior.** Standard Gaussian  $\mu_0 = \mathcal{N}(0, I)$  in  $\mathbb{R}^{3N}$ , with mean-free constraint to respect translation invariance. *The authors explicitly attempted a Harmonic / structured prior and abandoned it.*

**New idea.** Move the chemistry-specific information out of the prior and into the architecture: an  $O(D) \times S(N)$ -equivariant EGNN consuming an atom-level embedding that concatenates AMBER atom type, amino-acid identity, and a sinusoidal positional embedding in the peptide sequence. Train by conditional flow matching, generalise across peptides by varying the embedding.

**Headline.** First flow-based BG to demonstrate cross-peptide transfer: trained on 200 peptides, tested on 100 unseen, recovers the correct metastable configurations  $98 \pm 2\%$  of the time with mean ESS =  $15.29 \pm 9.27\%$ ; free-energy differences within  $\pm 0.05 k_B T$  of reference on alanine dipeptide.

**The decisive quote.** Klein, Krämer, Noé actively tried both prior options and reported the result. This single passage is the strongest piece of evidence cited against Camp B’s general programme in the entire review.

“Throughout our work, we utilise a standard Gaussian prior distribution. [...] We experimented with this Harmonic prior but found no significant improvements for our transferable model. Instead, the network architecture and inductive bias play a more crucial role.” — Klein, Krämer, Noé (TBG 2024, §7)

Klein, Krämer, Noé, [arXiv:2406.14426](https://arxiv.org/abs/2406.14426). The remainder of this subsection unpacks why the Harmonic prior failed *specifically in the transferable setting*, sets out the conditional flow-matching objective TBG uses, and tabulates the headline numbers.

**Setup.** The target is the Boltzmann distribution of a peptide  $\mathcal{P}$  with  $N$  atoms in  $\mathbb{R}^3$ , under a classical force field  $U_1$  (AMBER ff14SB with implicit solvent). The configuration space, after removing global translation, is  $\mathbb{R}^{3N-3}$  (or equivalently the centre-of-mass-free subspace  $\{x \in \mathbb{R}^{3N} : \sum_n x_n = 0\}$ ). Each atom  $n$  carries metadata: an atom-type label  $a_n \in \{1, \dots, 54\}$  (the AMBER set), an amino-acid identity  $r_n \in \{1, \dots, 20\}$ , and a sequence position  $p_n$  encoded by a sinusoidal embedding  $\phi_{\text{sin}}(p_n) \in \mathbb{R}^{d_{\text{pos}}}$  of the same form as the original Transformer. The full per-atom feature is  $h_n^{(0)} = [\text{onehot}(a_n); \text{onehot}(r_n); \phi_{\text{sin}}(p_n)]$ , which is fed into an EGNN whose layers update  $(x_n, h_n)$  in an  $O(3) \times S(N)$ -equivariant manner.

**Core math contribution.** TBG trains a continuous normalising flow by the conditional flow-matching objective. Fix coupling  $(x_0, x_1) \sim \mu_0 \otimes \mu_1$ , interpolate  $x_t = (1 - t)x_0 + tx_1$ , target velocity  $u_t(x \mid x_0, x_1) = x_1 - x_0$ , and minimise

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, (x_0, x_1) \sim \mu_0 \otimes \mu_1} \left[ \|v_\theta(x_t, t; \mathcal{P}) - (x_1 - x_0)\|^2 \right], \quad (14)$$

where  $v_\theta(\cdot, t; \mathcal{P})$  denotes the EGNN’s velocity field conditioned on the per-atom feature tensor of peptide  $\mathcal{P}$ . The training samples  $x_1$  are obtained from short MD on the training peptides; the prior samples  $x_0 \sim \mathcal{N}(0, I)$  are independent Gaussians per atom, projected onto the centre-of-mass-free subspace.

The contribution to the *prior-choice* literature is direct: the authors explicitly tested replacing  $\mu_0$  by a Harmonic prior — the zero-mean Gaussian whose covariance is the inverse Hessian of  $U_1$  at a reference minimum — and reported no improvement for the transferable model (quoted above). The structural reason is informative. A Harmonic prior centred at one reference minimum is intrinsically *molecule-specific*: its Hessian depends on the chemistry of the peptide in question. To remain transferable, one would need to learn a Hessian field over chemistry, which is itself a nontrivial sub-problem of comparable difficulty to learning the configuration distribution. The standard Gaussian, by contrast, has the expressive disadvantage of carrying no information, but the architectural advantage of being chemistry-independent: the same  $\mu_0$  feeds 300 peptides. Camp B’s leverage — precise mode location — is precisely what breaks transferability.

#### Algorithm.

1. For each training peptide  $\mathcal{P}$  in the 200-peptide training set, run short MD under  $U_1^{(\mathcal{P})}$  to produce  $\{x_1^{(\mathcal{P}, i)}\}$ .
2. Construct the per-atom feature tensor  $h^{(0)} = [\text{type}; \text{aa}; \phi_{\text{sin}}(\text{pos})]$  for  $\mathcal{P}$ .
3. For each minibatch, sample  $\mathcal{P}$  uniformly from the training set, sample  $x_1$  from the corresponding MD pool, sample  $x_0 \sim \mathcal{N}(0, I)$  projected to mean-free, and sample  $t \sim \mathcal{U}[0, 1]$ ; form  $x_t = (1 - t)x_0 + tx_1$ .
4. Compute the conditioned velocity  $v_\theta(x_t, t; \mathcal{P})$  from the EGNN and take an Adam step on the CFM loss in (14).
5. Repeat until convergence. (*Inference.*) For a query peptide  $\mathcal{P}^*$  (possibly held-out), draw  $x_0 \sim \mathcal{N}(0, I)$ , integrate  $dx = v_\theta(x, t; \mathcal{P}^*) dt$  from 0 to 1 with `torchdyn`’s adaptive solver; accumulate the log-determinant along the path to recover  $\log \nu_\theta(x_1 \mid \mathcal{P}^*)$ ; importance-correct against  $\exp(-U_1^{(\mathcal{P}^*)})/\nu_\theta$ .

**Numerical details.** Software: PyTorch + TorchDyn for the CNF integration + NetworkX for graph manipulation on the peptide topology. The EGNN has 5–7 message-passing layers and a hidden dimension of 64–128. Training:  $\sim 10^6$  gradient steps on 4 A100 GPUs over a few days. Test metrics, reproduced from the paper, are:

- **Alanine dipeptide, classical FF, in-distribution:** NLL =  $-127.06 \pm 0.12$ , ESS =  $6.03 \pm 1.34\%$ .
- **100 unseen dipeptides, “TBG+full” variant:** mean ESS =  $15.29 \pm 9.27\%$ ; correct metastable configurations recovered for  $98 \pm 2\%$  of test peptides.

- **“Biased training” variant** (training set includes MD samples from the test peptide):  $\text{ESS} = 10.24 \pm 7.14\%$ .
- **$10\times$  smaller training set:**  $\text{ESS} = 6.13 \pm 3.13\%$ , with  $96 \pm 3\%$  correct configurations.
- **Free-energy differences** on alanine dipeptide recovered to  $\pm 0.05 k_B T$  of reference.

**Critical assessment.** TBG is load-bearing for this review because it is the only paper to ablate a physics-informed prior against a Gaussian on the same architecture and training regime, and the Harmonic-prior ablation is the cleanest existing experimental test of the Camp A versus Camp B question; the negative result “no significant improvements” is exactly the design-quadrant outcome we predict for transferable models, because chemistry-independence of  $\mu_0$  prohibits any target-specific prior. The EGNN’s equivariance combined with chemistry-aware per-atom embeddings further demonstrates that, for an indexed family of targets, structure goes into the architecture rather than the prior. Several numbers deserve to be read with care. The headline held-out ESS of 15.29% averages over peptides whose individual ESS lies in  $[5\%, 30\%]$  ( $\pm 9.27\%$ ) and is not directly comparable to the in-distribution 6.03% on alanine dipeptide; the “held-out” peptides are themselves a subset of the same 20-amino-acid combinatorial pool, not a chemical-class probe, so transferability is demonstrated across peptide identity rather than across peptide class. Critically, when MD samples from the test peptide are added to the training pool, the ESS drops from 15.29% to 10.24% while free-energy estimates improve — a textbook mode-locking signature where the model becomes sharper on the modes it has seen and worse at importance-weighting the rest. Read alongside Wirnsberger 2022, TBG splits the field cleanly: tune the prior for a single phase known in advance; tune the architecture for an indexed family of targets; and neither solution addresses the multimodal mode-discovery setting that motivates the LLR paper. The biased-training failure is, in turn, precisely the failure mode  $X_{\hat{\mu}}$  is designed to avoid.

### 3.12. TA-BG — Temperature-Annealed BG (Schopmans, Friederich 2025)

[arXiv:2501.19077](https://arxiv.org/abs/2501.19077) (ICML 2025); [aimat-lab/TA-BG](https://github.com/aimat-lab/TA-BG).

#### At a glance — TA-BG

**Target.** Molecular Boltzmann distribution  $\mu(x) \propto \exp(-U_1(x)/T)$  for peptides at room temperature  $T \approx 300$  K, parameterised in internal coordinates (bonds, angles, dihedrals).

**Prior.** Factorised *topology-aware* base: truncated normal on bond lengths ( $\sigma = 0.5$ ) and bond angles ( $\sigma = 0.1$ ), uniform on  $[0, 1]$  for each dihedral angle. The prior encodes molecular graph but is *not* mode-tuned.

**New idea.** Train reverse-KL at a high reference temperature  $T_h \approx 1200$  K where the energy landscape is effectively unimodal, then *anneal the flow distribution* to the cold target  $T$  by a sequence of importance-reweighted forward-KL updates.

**Headline.** Beats FAB on energy-evaluation efficiency for alanine dipeptide while side-stepping the mode collapse that vanilla reverse-KL training exhibits at  $T = 300$  K.

**Setup.** Let  $U_1 : \mathbb{R}^d \rightarrow \mathbb{R}$  be the potential,  $\beta = 1/T$ , and write  $\mu_T(x) \propto \exp(-\beta U_1(x))$ . Two temperatures are involved: a high reference  $T_h$  (here  $\sim 1200$  K) and a low target  $T$  (here  $\sim 300$  K). The flow is an internal-coordinate Neural Spline Flow  $G_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$  implemented on top of **bgflow**; the latent distribution is the topology-aware factorised prior  $\mu_0$  above. The high-temperature target  $\mu_{T_h}$  is empirically near-unimodal on peptide backbones, so reverse-KL training against it is well-posed.

**Core math.** Stage 1 uses reverse-KL at the high temperature, exploiting that  $\mu_{T_h}$  is approximately unimodal:

$$\mathcal{L}_{\text{rev}}^{T_h}(\theta) = \mathbb{E}_{z \sim \mu_0} \left[ \frac{1}{T_h} U_1(G_\theta^{-1}(z)) - \log |\det J_{G_\theta^{-1}}(z)| \right] + \text{const.}$$

Stage 2 reweights samples from the *current* flow  $\nu_\theta$  to a slightly colder temperature  $T' < T_h$  using the geometric reweighting identity  $\mu_{T'}(x) \propto \mu_{T_h}(x)^{T_h/T'}$ , which gives importance weights

$$w(y) = \exp\left(\left(\frac{1}{T'} - \frac{1}{T_h}\right) U_1(y)\right), \quad y \sim \nu_\theta.$$

A forward-KL update is then performed on the reweighted ensemble  $\{(y_i, w_i)\}$ ,

$$\mathcal{L}_{\text{fwd}}^{T'}(\theta) \approx - \sum_i \frac{w_i}{\sum_j w_j} \left[ \log \mu_0(G_\theta^{-1}(y_i)) - \log |\det J_{G_\theta}(G_\theta^{-1}(y_i))| \right].$$

Iterating  $T_h \rightarrow T_{h-1} \rightarrow \dots \rightarrow T$  along a temperature ladder produces a sequence of flows tracking the annealed family  $\{\mu_{T_k}\}$ .

### Algorithm.

1. Choose a temperature ladder  $T_h = T_0 > T_1 > \dots > T_K = T$ .
2. Train  $G_\theta$  by reverse-KL against  $\mu_{T_0}$  until convergence (*Stage 1*).
3. For  $k = 1, \dots, K$  (*Stage 2*): draw fresh  $\{y_i\} \sim \nu_\theta$ , compute weights  $w_i = \exp((1/T_k - 1/T_{k-1})U_1(y_i))$ , monitor ESS( $\mathbf{w}$ ).
4. If ESS( $\mathbf{w}$ ) falls below a floor, insert an intermediate temperature and re-sample; otherwise perform a forward-KL update on  $(y_i, w_i)$ .
5. Return  $G_\theta$  at  $k = K$  as the BG for  $\mu_T$ .

**Numerical details.** Backbone: PyTorch + `bgflow` + Neural Spline Flows. Targets: alanine dipeptide and similar small peptides at 300 K. Reported gain is in *energy evaluations* (not wall-clock): the annealed schedule reduces the number of  $U_1$  calls relative to FAB by roughly an order of magnitude on alanine dipeptide while matching or surpassing ESS and free-energy estimation accuracy. Crucially the prior is the stock `bgflow` topology-aware factorisation; no GMM or learned base.

“At target temperatures, the reverse KLD suffers from severe mode collapse: once the flow collapsed to a mode, it will generally not escape this collapsed state if the remaining modes are too far separated. Surprisingly, at high temperatures the reverse KL is sufficient: barriers between metastable states are lower and the probability distribution maxima are interconnected.”  
— Schopmans, Friederich (TA-BG 2025)

**Critical assessment.** TA-BG is the cleanest demonstration to date that the failure of reverse-KL is *thermodynamic* and not purely architectural: train at a temperature where the target is unimodal, and reverse-KL behaves well. But three structural limitations remain. (i) The temperature ladder is delicate: too aggressive a step causes weight degeneracy ( $\text{ESS} \rightarrow 0$ ) and the forward-KL update collapses; too conservative a ladder erases the efficiency gain over FAB. (ii) Annealing relies on the assumption that  $\mu_{T_h}$  is unimodal — a property that holds for peptides but fails for systems with discrete symmetries (Ising-like targets, lattice phase transitions) where modes persist at all finite temperatures. (iii) The prior is topology-aware but not mode-aware; TA-BG does not address mode discovery on targets whose mode structure is temperature-invariant.

### 3.13. SBG — Sequential Boltzmann Generators (Tan et al. 2025)

[arXiv:2502.18462](https://arxiv.org/abs/2502.18462) (ICML 2025).

#### At a glance — SBG

**Target.** Equilibrium distribution of tri-, tetra-, and hexapeptides (AL3, AL4, AL6) in full *Cartesian* coordinates — the first flow-based equilibrium sampler to operate in Cartesian space at hexapeptide scale.

**Prior.** Standard Gaussian  $\mathcal{N}(0, I_d)$  at training time; at inference, the trained flow  $\nu_\theta$  itself serves as an “adaptive prior” for sequential Monte Carlo.

**New idea.** Pair a Transformer-based normalising flow (*TarFlow*, ViT-style patches over particles) with inference-time continuous-time SMC driven by an annealed Langevin SDE. Equivariance is enforced softly via data augmentation and centre-of-mass noise rather than baked into the architecture.

**Headline.** ESS =  $0.876 \pm 0.215$  (AL3),  $0.901 \pm 0.059$  (AL4), 0.995 (AL6) at  $10^4$  samples, with energy Wasserstein-1 distances 1.46/1.52/2.32 respectively.

**Setup.** Particles in  $\mathbb{R}^3$  are flattened into a sequence and processed by a TarFlow block stack: each block applies a Vision-Transformer mixer across patches of particles followed by an affine coupling. The base distribution at training time is plain Gaussian. Training data are biased MD trajectories collected at the target temperature; the flow is fit by maximum likelihood (forward KL) on this biased corpus. Equivariance under  $O(3) \times S_N$  is *not* structurally imposed; instead, training batches are augmented by random rotations and centre-of-mass jitter.

**Core math.** Let  $\nu_\theta$  denote the pre-trained flow and  $\mu \propto \exp(-U_1)$  the target. The inference-time *adaptive prior* construction defines a continuous bridge

$$\pi_\tau(x) \propto \nu_\theta(x)^{1-\tau} \mu(x)^\tau, \quad \tau \in [0, 1],$$

along which an annealed Langevin SDE is integrated:

$$dX_\tau = \left[ \beta'(\tau) \nabla \log \pi_\tau(X_\tau) - \frac{1}{2} \nabla U_1(X_\tau) \right] d\tau + \sqrt{2} dW_\tau,$$

with  $\beta(\tau)$  an annealing schedule and  $W_\tau$  standard Brownian motion. Particles are weighted by the continuous-time Radon–Nikodym ratio along the bridge; adaptive resampling triggers whenever ESS falls below a fixed threshold (typically  $0.5N$ ). The key conceptual move is replacing the proposal  $\mu_0$  used in vanilla AIS by the trained flow  $\nu_\theta$ : where classical AIS interpolates between a Gaussian and  $\mu$ , SBG interpolates between an already-good learned approximation and  $\mu$ , so the bridge is shorter and the path weights are tighter.

#### Algorithm.

1. Run biased MD at the target temperature to produce a training corpus.
2. Fit TarFlow  $\nu_\theta$  by maximum likelihood with rotation and centre-of-mass data augmentation.
3. At inference, draw  $\{x_0^{(i)}\} \sim \nu_\theta$  and assign unit weights.
4. Integrate the annealed Langevin SDE from  $\tau = 0$  to  $\tau = 1$  via Euler–Maruyama; accumulate path weights along the bridge.
5. Whenever  $\text{ESS}(\mathbf{w}) < \text{ESS}_{\min}$ , perform multinomial resampling and reset weights.
6. Return the reweighted ensemble at  $\tau = 1$  as the SBG sample.

**Numerical details.** Reported on the  $ALn$  family ( $n \in \{3, 4, 6\}$ ) at  $10^4$  inference samples:

| System             | ESS               | Energy $W_1$      |
|--------------------|-------------------|-------------------|
| AL3 (tripeptide)   | $0.876 \pm 0.215$ | $1.456 \pm 0.739$ |
| AL4 (tetrapeptide) | $0.901 \pm 0.059$ | $1.517 \pm 0.387$ |
| AL6 (hexapeptide)  | 0.995             | 2.321             |

These are the strongest Cartesian-coordinate ESS numbers in the 2024–2025 literature. The improvement over a flow-only baseline comes almost entirely from the SMC stage; ablating the annealed Langevin reduces AL3 ESS by roughly an order of magnitude.

**Critical assessment.** SBG is the most aggressive instance of the “flow as proposal” philosophy: the flow is treated as a learned *importance distribution* rather than the final sampler, and all remaining error is absorbed by inference-time SMC. Two caveats temper the headline numbers. First, training requires *biased MD trajectories*, so SBG is not energy-only — the prior on what configurations matter is smuggled in through the training corpus rather than through the latent. Second, each target peptide gets a dedicated TarFlow; there is no amortisation across systems, in contrast to TBG 2024’s transferable architecture. The soft-equivariance choice (data augmentation rather than EGNN) is philosophically aligned with Camp A: the prior stays Gaussian, all structure goes into the training and inference machinery.

### 3.14. CMT — Constrained Mass Transport (Blessing et al. 2025)

[arXiv:2510.18460](https://arxiv.org/abs/2510.18460) (SPIGM @ NeurIPS 2025).

#### At a glance — CMT

**Target.** Multimodal Boltzmann targets where the entropy gap  $H(\mu_0) - H(\mu)$  is large — the regime where geometric annealing schedules systematically fail. Benchmarks include the ELIL tetrapeptide and alanine hexapeptide.

**Prior.** Standard Gaussian  $\mathcal{N}(0, I_d)$ ; the prior is held fixed and *not* retuned.

**New idea.** Diagnose *mass teleportation*: in a geometric annealing schedule  $\pi_k \propto \mu_0^{1-\beta_k} \mu^{\beta_k}$ , when the entropy gap is large, new target modes can appear between  $\pi_k$  and  $\pi_{k+1}$  without overlapping the preceding intermediates, breaking the chain of importance corrections. CMT replaces the fixed  $\beta$ -schedule with an *adaptive trust-region schedule* that jointly bounds  $\text{KL}(\pi_{k-1} \parallel \pi_k)$  and the entropy decrement  $H(\pi_{k-1}) - H(\pi_k)$ .

**Headline.** On the ELIL tetrapeptide, reverse-KL gives ESS = 1.28%; CMT gives ESS = 26.18%. A 20× improvement *without changing the prior*.

**Setup.** Consider an annealed family  $\{\pi_k\}_{k=0}^K$  with  $\pi_0 = \mu_0$  and  $\pi_K = \mu$ . The classical choice is geometric,  $\pi_k(x) \propto \mu_0(x)^{1-\beta_k} \mu(x)^{\beta_k}$  for a fixed schedule  $0 = \beta_0 < \dots < \beta_K = 1$ . CMT keeps the geometric functional form but treats the schedule  $\{\beta_k\}$  as an output of a constraint-driven adaptive procedure rather than a fixed input.

**Core math.** Define the entropy  $H(\pi) = -\mathbb{E}_{x \sim \pi} \log \pi(x)$ . *Mass teleportation* is formalised as the existence of an index  $k$  and a region  $A \subset \mathbb{R}^d$  with  $\pi_{k+1}(A) \geq \frac{1}{2}$  but  $\pi_k(A)$  exponentially small in  $d$ . When this happens, importance weights  $d\pi_{k+1}/d\pi_k$  concentrate on a measure-zero set and AIS-style

mass transport breaks. CMT imposes *both* a KL constraint and an entropy constraint between successive intermediates:

$$\text{KL}(\pi_{k-1} \parallel \pi_k) \leq \delta_{\text{KL}}, \quad H(\pi_{k-1}) - H(\pi_k) \leq \delta_H, \quad k = 1, \dots, K.$$

The KL bound guarantees distributional overlap (so reweighting is non-degenerate); the entropy bound guarantees gradual sharpening (so no new modes appear without a precursor in  $\pi_{k-1}$ ). Given the previous intermediate  $\pi_{k-1}$ , the next schedule index  $\beta_k$  is chosen as the largest value such that both constraints hold:

$$\beta_k = \max \{ \beta \geq \beta_{k-1} : \text{KL}(\pi_{k-1} \parallel \pi_\beta) \leq \delta_{\text{KL}} \wedge H(\pi_{k-1}) - H(\pi_\beta) \leq \delta_H \}.$$

The variational training objective is the standard AIS-style telescoping forward-KL applied along the CMT-induced schedule, with each step backpropagated through the adaptive  $\beta_k$ .

### Algorithm.

1. Initialise  $\pi_0 = \mu_0$ ,  $\beta_0 = 0$ .
2. At step  $k$ : estimate  $H(\pi_{k-1})$  and the curves  $\beta \mapsto \text{KL}(\pi_{k-1} \parallel \pi_\beta)$ ,  $\beta \mapsto H(\pi_\beta)$  by importance sampling from  $\pi_{k-1}$ .
3. Solve a 1-D root-finding problem (bisection) for the largest  $\beta_k$  satisfying both  $\delta_{\text{KL}}$  and  $\delta_H$  constraints.
4. Perform a transport / forward-KL training update against  $\pi_k$ .
5. Iterate until  $\beta_K = 1$  is reached; return the trained flow.
6. Optionally apply trust-region damping if either constraint saturates over consecutive steps.

**Numerical details.** Headline benchmarks reported in the paper:

| System              | Method     | ESS           |
|---------------------|------------|---------------|
| ELIL tetrapeptide   | reverse-KL | 1.28%         |
| ELIL tetrapeptide   | CMT        | <b>26.18%</b> |
| Alanine hexapeptide | FAB        | (beaten)      |
| Alanine hexapeptide | TA-BG      | (beaten)      |
| Alanine hexapeptide | CMT        | best          |

The ELIL row is the decisive datapoint: a  $20\times$  improvement in ESS on a stiff multimodal target with the prior held fixed at  $\mathcal{N}(0, I_d)$ . On alanine hexapeptide, CMT outperforms FAB, TA-BG, and naive reverse-KL under matched compute.

**Critical assessment.** CMT is the single most informative 2025 datapoint for the *prior vs. training* debate that animates this review. The result demonstrates — under controlled conditions, on a peptide multimodal target, with the prior held fixed — that pure training-pipeline engineering can deliver a  $20\times$  ESS improvement. This is precisely the magnitude of improvement that prior tuning would have to deliver to justify the architectural cost of a structured base distribution; CMT achieves it without touching  $\mu_0$ . The implication is sharp: any claim that Camp A is fundamentally bottlenecked by its Gaussian prior must explain why a schedule-only intervention can move ESS from 1.28% to 26.18% on the very class of multimodal targets that motivates Camp B in the first place.

The work that prior tuning is *supposed* to do — restore distributional overlap with target modes — is here being done by the adaptive schedule, and apparently done more cheaply. Two limitations remain. The hyperparameters  $\delta_{\text{KL}}$  and  $\delta_H$  themselves require tuning, and a poor choice degrades back to geometric annealing. The trust-region/bisection inner loop adds non-trivial compute per outer step, partially offsetting the sample-efficiency gain. Neither limitation undermines the central finding: as of late 2025, the highest-leverage intervention on multimodal Camp A pipelines is the annealing schedule, not the base distribution.

**Concluding Camp A.** The progression Noé 2019 → Wu 2020 → FAB 2022 → TA-BG 2025 → CMT 2025 → SBG 2025 is a story of progressively more sophisticated training machinery on top of a fixed Gaussian latent. At no point does the field reach for a tuned prior; the engineering investment goes everywhere except the base distribution.

## 4. Camp B — physics-informed / tuned prior

**Why this section exists.** Camp B is the minority opinion but wins on one high-value subproblem (crystallographic free energy). Understanding what Camp B does well and what it gives up is essential: Camp B’s advocates are not wrong in general, only wrong for the specific multimodal-discovery problem the LLR paper targets.

### 4.1. Lattice-Gaussian priors for crystals (Wirnsberger et al. 2022)

Wirnsberger, Papamakarios, Ibarz, Racanière, Ballard, Pritzel, Blundell, *Mach. Learn.: Sci. Technol.* **3**, 025009 (2022); [arXiv:2111.08696](#). This is *the* canonical Camp B reference: the first BG paper in which the base distribution itself carries the bulk of the structural information about the target, and the flow’s role is reduced to learning anharmonic corrections.

#### At a glance — Wirnsberger et al. 2022 (lattice-Gaussian prior)

**Target**

Boltzmann distribution of a monatomic crystal at fixed  $(N, V, T)$ : Lennard-Jones FCC and hexagonal/cubic ice  $(I_h, I_c)$  at  $N \in \{64, 216, 256, 512\}$  atoms.

**Prior**

Permutation-invariant product of spherically truncated Gaussians centred at reference lattice sites  $\{z_n^o\}$ , with truncation radius  $R_T < d_{\text{min}}/2$  enforcing single-site occupancy.

**New idea**

Bake the entire crystal scaffold into  $\mu_0$ ; train the flow only on the residual anharmonicity around lattice sites by reverse-KL minimisation on the target Boltzmann.

**Headline**

Helmholtz free energies agreeing with reference MBAR estimates to  $\sim 10^{-5} k_B T$  per particle; LJ-256 free energy 3.10800(28) versus literature 3.11(4).

**Setup.** The physical system is  $N$  identical particles in a cubic simulation cell with periodic boundary conditions and total potential energy  $U_1(z)$  obtained either from a Lennard-Jones pair potential (for the LJ-FCC benchmark) or from the TIP4P/Ice rigid-water model (for ice  $I_h$  and  $I_c$ ,

where “atom” is shorthand for oxygen-anchored rigid molecules). The reference lattice  $\{z_n^o\}_{n=1}^N \subset \mathbb{R}^3$  is obtained from an energy-minimised configuration of the chosen crystal phase. The thermodynamic quantity of interest is the Helmholtz free energy

$$F = -k_B T \log Z, \quad Z = \int_{\mathbb{R}^{3N}} \exp(-\beta U_1(z)) dz,$$

relative to a tractable reference state with known partition function  $Z_0$ . The ratio  $Z/Z_0$  is estimated by importance sampling with weights produced by the trained flow.

**Core math contribution — the lattice-Gaussian prior.** The base distribution is the central object of the paper. Let  $\rho_T : \mathbb{R}^3 \rightarrow \mathbb{R}_{\geq 0}$  denote a single-site density on relative displacements,

$$\rho_T(\delta) = \frac{1}{C_T} \exp\left(-\frac{\|\delta\|^2}{2\sigma_T^2}\right) \mathbf{1}_{\{\|\delta\| \leq R_T\}}, \quad C_T = \int_{\|\delta\| \leq R_T} \exp\left(-\frac{\|\delta\|^2}{2\sigma_T^2}\right) d\delta, \quad (15)$$

i.e. an isotropic Gaussian on  $\mathbb{R}^3$  truncated to the open ball of radius  $R_T$ . The bandwidth  $\sigma_T$  is calibrated to the harmonic Debye-Waller motion of the chosen crystal at temperature  $T$ , and the truncation radius is chosen as

$$R_T < \frac{1}{2} d_{\min}, \quad d_{\min} = \min_{m \neq n} \|z_m^o - z_n^o\|, \quad (16)$$

so that the supports of the truncated Gaussians at any two lattice sites are disjoint. The full prior on  $z = (z_1, \dots, z_N) \in \mathbb{R}^{3N}$  is the symmetrised product

$$\mu_0^{\text{lat}}(z) = \frac{1}{N!} \sum_{\pi \in S_N} \prod_{n=1}^N \rho_T(z_n - z_{\pi(n)}^o). \quad (17)$$

Three properties of (17) deserve attention because every later design decision flows from them.

(a) *Permutation invariance is exact and tractable.* The factor  $(1/N!) \sum_{\pi \in S_N}$  implements the  $S_N$  permutation symmetry of identical particles. A naive evaluation would cost  $O(N!)$ , but *because of the truncation* (16) only one permutation in  $S_N$  contributes a nonzero summand for any given  $z$ : the unique nearest-site assignment  $\pi^*(n) = \arg\min_m \|z_n - z_m^o\|$ . Hence

$$\mu_0^{\text{lat}}(z) = \frac{1}{N!} \prod_{n=1}^N \rho_T(z_n - z_{\pi^*(n)}^o) \quad (18)$$

on the (full-measure) configurations where all  $N$  atoms lie within  $R_T$  of their nearest lattice site and no two atoms claim the same site, and  $\mu_0^{\text{lat}}(z) = 0$  otherwise. The factor  $1/N!$  remains for normalisation; sampling proceeds by drawing  $z_n = z_n^o + \delta_n$  with  $\delta_n \sim \rho_T$  independently and then applying a uniformly random permutation of atom labels, which costs  $O(N)$ .

(b) *Truncation eliminates spurious mixture modes.* An *untruncated* Gaussian mixture  $\frac{1}{N!} \sum_{\pi} \prod_n \mathcal{N}(z_n; z_{\pi(n)}^o, \sigma_T^2 I)$  would assign positive probability to configurations in which the atom at  $z_n$  is closer to a non-nearest site  $z_m^o \neq z_{\pi^*(n)}^o$ . Each such “mismatched” assignment is a separate mode of the mixture, and the resulting prior has  $O(N!)$  modes, almost all of which are physical artefacts the flow would have to dynamically suppress. The truncation  $R_T < d_{\min}/2$  kills every mismatched assignment exactly and reduces the prior to a single connected basin around each lattice configuration — a Bravais-equivariant unimodal distribution as far as the flow is concerned.

(c) *The prior already encodes the desired phase.* If  $\{z_n^o\}$  is the FCC reference,  $\mu_0^{\text{lat}}$  has no mass on hexagonal configurations, on disordered/liquid configurations, or on any other crystal phase. The support is committed at construction time.

**Algorithm.** Let  $G_\theta : \mathbb{R}^{3N} \rightarrow \mathbb{R}^{3N}$  be a continuous normalising flow built from SE(3)-respecting coupling layers acting on displacements  $u_n = z_n - z_n^o$  (so that the flow’s input is centred at the origin for every lattice site). The pushforward density on the target side is

$$\nu_\theta(x) = \mu_0^{\text{lat}}(G_\theta^{-1}(x)) |\det \nabla G_\theta^{-1}(x)|. \quad (19)$$

Training minimises the reverse-KL functional

$$\mathcal{L}_{\text{rKL}}(\theta) = \text{KL}(\nu_\theta \parallel \mu_1) = \mathbb{E}_{z \sim \mu_0^{\text{lat}}} [\beta U_1(G_\theta(z)) - \log |\det \nabla G_\theta(z)|] + \text{const}, \quad (20)$$

which is unbiased to estimate because (i)  $\mu_0^{\text{lat}}$  is cheap to sample by the  $O(N)$  procedure above, and (ii)  $\rho_T$  is differentiable on the interior of its support and the gradient never sees the truncation boundary in practice (samples land deep inside the truncation ball).

After training, the free-energy difference between the target crystal phase and the tractable lattice reference is estimated by the bridge identity

$$\Delta F = -k_B T \log \frac{Z_1}{Z_0} = -k_B T \log \mathbb{E}_{z \sim \mu_0^{\text{lat}}} [\exp(-\beta U_1(G_\theta(z)) + \beta U_0(z)) |\det \nabla G_\theta(z)|], \quad (21)$$

which is the standard targeted free-energy perturbation estimator with the trained flow  $G_\theta$  as the targeting map. Equation (21) is statistically efficient precisely because  $\nu_\theta \approx \mu_1$  after training and the importance weights  $w(x) = \mu_1(x)/\nu_\theta(x)$  have near-unit variance.

### Numerical details.

- *Lennard-Jones*,  $N = 256$ , *FCC*. Reported Helmholtz free energy  $F/(Nk_B T) = 3.10800(28)$  versus literature 3.11(4); the flow estimate is two orders of magnitude tighter than the literature uncertainty.
- *Ice  $I_c$* ,  $N \in \{64, 216, 512\}$  (cubic ice). Per-particle free energies agree with reference MBAR to within statistical error of order  $10^{-5} k_B T$ .
- *Ice  $I_h$* ,  $N \in \{64, 216, 512\}$  (hexagonal ice, the ambient phase). Same precision regime as  $I_c$ .
- *Software stack*. JAX + Haiku + Distrax; the entire pipeline is autodifferentiable and the flow is trained end-to-end by stochastic gradient descent on (20).
- *Compute*. Largest experiments (512-atom ice, 500-atom LJ) take approximately three weeks of wall-clock on 16 A100 GPUs — substantial industrial investment for a single (phase,  $N$ ,  $T$ ) point.

“Empirically, we find that, after training, the flow model becomes a sampler for the (metastable) crystal state that we encode in the base distribution, and does not sample configurations from other states.”  
— Wirnsberger et al. (2022)

This sentence is the load-bearing concession of the paper. It is not a complaint but a feature for the application at hand: when one wants  $\Delta F$  between two *named* phases, it is precisely the inability to leak mass between phases that makes the flow a precise estimator for one of them. For mode-discovery applications, it is the exact failure mode Camp A worries about.

**Critical assessment — what the prior tuning costs.** For known-phase free-energy precision the Wirnsberger construction is essentially optimal: the flow learns only anharmonic corrections inside a single basin, and the importance weights derived from (21) have near-unit variance because

$\nu_\theta$  inherits the combinatorial and symmetry structure from  $\mu_0^{\text{lat}}$ , a precision no Camp A method has so far matched. The price paid for this precision is exactly the loss of everything else. Because of (16) the flow’s support is contained in the forward image of single-occupancy configurations, so no FCC-prior flow can produce a liquid configuration and no  $I_c$ -prior flow can produce an  $I_h$  configuration: the single-basin admission is structural, not a training artefact. Each new phase consequently requires re-tuning the prior and retraining — a fresh lattice  $\{z_n^o\}$ , fresh  $(\sigma_T, R_T)$  calibration, and a fresh three-week training run on 16 A100s — so the cost amortises poorly across phase diagrams. The recipe also does not generalise beyond crystals, since the construction relies on a known reference configuration with discrete site positions and a minimum inter-site distance  $d_{\min} > 0$ , none of which hold for liquids, glasses, polymers in solution, or proteins. Most dangerously, symmetry-breaking transitions are silently mis-sampled: near a solid–liquid coexistence point the true Boltzmann distribution has comparable mass in both phases and  $\nu_\theta$  produces samples from only the encoded phase, with no warning signal in the importance weights because in-basin weights remain well-behaved. Read as a whole, Wirnsberger 2022 is the benchmark Camp B paper: it simultaneously defines the precision ceiling no Camp A method has reached and the structural limit, single-phase and lattice-constructible, outside which the recipe simply does not apply.

## 4.2. Higher-temperature reference priors (Coretti et al., surveyed 2024)

Coretti, Falkner, Geissler, Hummer, Dellago and collaborators, surveyed in [arXiv:2404.16566](https://arxiv.org/abs/2404.16566). The basic idea is the empirically simplest possible Camp B move: do not engineer the prior at all, just *sample* it from molecular dynamics at a higher temperature.

### At a glance — higher-temperature reference prior

#### Target

Boltzmann distribution  $\mu_1 \propto \exp(-U_1/T)$  at production temperature  $T$ .

#### Prior

Empirical distribution of an MD trajectory at higher temperature  $T_0 > T$ , represented by a stored sample set  $\{z_i^{(T_0)}\}_{i=1}^M$ .

#### New idea

Let the flow learn the  $T_0 \rightarrow T$  deformation; barrier crossings happen at the high- $T$  MD level rather than via the flow.

#### Headline

Smooth applicability to barrier-free liquids; structural failure on systems with persistent multimodality at  $T_0$ .

**Setup.** Fix a target temperature  $T$  and a higher pre-simulation temperature  $T_0 > T$ . Generate an MD trajectory at  $T_0$  on the same potential  $U_1$  and store  $M$  snapshots  $\{z_i^{(T_0)}\}_{i=1}^M$ . The trajectory is assumed ergodic at  $T_0$  within the timescale budget (a non-trivial requirement we revisit below). The flow  $G_\theta$  then learns to map the high- $T$  ensemble to the production ensemble.

**Core math contribution — the empirical Boltzmann prior.** The prior is

$$\mu_0(z) \approx \mu_{T_0}(z) = Z_{T_0}^{-1} \exp(-U_1(z)/T_0), \quad (22)$$

represented *not* as a closed-form density but as the empirical measure

$$\hat{\mu}_{T_0}(z) = \frac{1}{M} \sum_{i=1}^M \delta(z - z_i^{(T_0)}) \quad (23)$$

on the stored sample set. Because (23) is a sum of point masses, training cannot use a likelihood-of-prior term directly; the loss either trains the flow as a deterministic map between samples (a forward-KL or trajectory-matching style) or smooths (23) with a kernel density estimate of bandwidth  $h$ ,

$$\tilde{\mu}_{T_0}^h(z) = \frac{1}{M} \sum_{i=1}^M k_h(z - z_i^{(T_0)}), \quad k_h(\delta) = (2\pi h^2)^{-d/2} \exp\left(-\frac{\|\delta\|^2}{2h^2}\right), \quad (24)$$

and trains by reverse-KL using (24) in place of  $\mu_0$ . The ratio  $\mu_1/\mu_0$ , which the flow must absorb, simplifies on the support of (22) to

$$\frac{\mu_1(z)}{\mu_{T_0}(z)} = \frac{Z_{T_0}}{Z_1} \exp\left(-U_1(z)\left(\frac{1}{T} - \frac{1}{T_0}\right)\right), \quad (25)$$

a smooth low-rank reweighting that depends only on  $U_1$  and the two temperatures. For a barrier-free liquid this reweighting is benign and the flow’s task is mild density tightening rather than topological work.

**Algorithm.** Pre-simulate  $M$  samples at  $T_0$ . Train  $G_\theta$  by reverse-KL on  $\mu_1$  with prior (24); alternatively, train by forward-KL on  $\{z_i^{(T_0)}\}$  as the data after applying a known reweighting (25). Importance-sample  $\mu_1$  via  $w(x) = \mu_1(x)/\nu_\theta(x)$  on the post-trained model.

**Numerical details.** The line of work is methodological and there is no single canonical benchmark; in published reports the technique succeeds on simple Lennard-Jones liquids and small dense fluids, where  $T_0/T \approx 1.3$ – $2$  suffices to remove residual barriers, and the MD pre-simulation cost is modest ( $10^5$ – $10^6$  steps).

**Critical assessment — what the prior tuning costs.** The Coretti-style high-temperature reference prior is conceptually the simplest Camp B move: do not engineer the prior at all, just sample it from molecular dynamics at higher temperature. On barrier-free liquids it works smoothly, but the multimodal regime undermines the recipe in four reinforcing ways. Multimodality is inherited rather than escaped: for proteins with barriers  $\Delta F \geq 10 k_B T$  at  $T$ , the same barriers at  $T_0 = 1.5 T$  are still  $\Delta F' \approx 6.7 k_B T$ , so ergodicity at  $T_0$  is no easier than the original sampling problem on a logarithmic scale, and the technique only helps when the barriers vanish at  $T_0$ . The pre-simulation cost is also self-defeating: the full quality of the prior is bounded by the quality of the high- $T$  MD, and if the user can afford long enough  $T_0$  MD to cross every barrier, parallel tempering already solves the problem at  $T$  and the flow adds little. The density is also only approximate, since the KDE smoothing of (24) introduces a bandwidth bias that contaminates the importance weights in a way invisible to the reverse-KL loss — the standard implicit-prior failure mode. Finally, several Camp A losses (notably FAB’s  $\alpha$ -divergence with  $\alpha = 2$  and CMT’s constrained-transport schedule) require an analytic prior density, and the sample-set prior cannot supply one. The construction is therefore useful for systems where  $T_0$  has no barriers — crystalline anharmonic corrections, simple liquids — but fails on protein-scale multimodality precisely because the high- $T$  MD inherits the underlying problem.

### 4.3. Conditional / phase-diagram BGs (Schebek et al. 2024)

Schebek, Schaaf, Hummer, Köhler, *Mach. Learn.: Sci. Technol.* 2024, [arXiv:2406.12378](https://arxiv.org/abs/2406.12378). A hybrid construction that adds amortisation over thermodynamic state to a mild Camp B prior.

#### At a glance — conditional phase-diagram BG

##### Target

A two-parameter family  $\{\mu_{T,p}\}_{(T,p) \in \Omega}$  of Boltzmann distributions over a  $(T, p)$  region  $\Omega$  of a single-component phase diagram.

##### Prior

Fixed empirical reference  $\mu_0$  at a single anchor state  $(T_0, p_0) \in \Omega$ .

##### New idea

A conditional flow  $G_\theta(\cdot; T, p)$  that morphs the single anchor prior across the entire  $(T, p)$  region with one set of weights.

##### Headline

One trained model evaluates the whole phase diagram in one pass; precision degrades near phase boundaries.

**Setup.** Choose an anchor  $(T_0, p_0)$  deep within one phase, collect MD samples  $\{z_i^{(T_0, p_0)}\}$ , and treat their empirical distribution as  $\mu_0$ . The conditional flow  $G_\theta(\cdot; T, p)$  takes the thermodynamic state  $(T, p)$  as a conditioning input.

**Core math contribution — conditioning the flow rather than the prior.** The conditional pushforward is

$$\nu_\theta(x; T, p) = \mu_0(G_\theta^{-1}(x; T, p)) \left| \det \nabla G_\theta^{-1}(x; T, p) \right|, \quad (26)$$

and the loss integrates a base reverse-KL functional against a user-specified weighting  $w(T, p)$  over  $\Omega$ :

$$\mathcal{L}(\theta) = \int_{\Omega} w(T, p) \text{KL}(\nu_\theta(\cdot; T, p) \parallel \mu_{T,p}) \, dT \, dp. \quad (27)$$

The weighting  $w(T, p)$  is typically uniform on a rectangular slice of  $\Omega$  or peaked near a phase boundary of interest. The expectation in (27) is computed by sampling  $(T, p) \sim w$  and evaluating the inner reverse-KL with samples from  $\mu_0$  pushed through  $G_\theta(\cdot; T, p)$ .

In contrast to Wirnsberger et al., the prior  $\mu_0$  here is *fixed and single-state*; the structural commitment is to the anchor phase, and the flow takes the load of bridging neighbouring states. Free-energy differences between  $(T_0, p_0)$  and arbitrary  $(T, p) \in \Omega$  are then obtained by the conditional analogue of (21),

$$\Delta F(T, p) = -k_B T \log \mathbb{E}_{z \sim \mu_0} \left[ \exp(-\beta_{T,p} U_1(G_\theta(z; T, p)) + \beta_{T_0} U_1(z)) \left| \det \nabla G_\theta(z; T, p) \right| \right]. \quad (28)$$

**Algorithm.** Pre-simulate  $\mu_0$  at  $(T_0, p_0)$ . Train by Monte-Carlo estimation of (27): at every step sample  $(T, p) \sim w$  and a batch from  $\mu_0$ , evaluate the conditional flow, backpropagate. Free-energy estimation post-training is one forward pass per  $(T, p)$  query, with importance weights from (28).

**Numerical details.** Reported on a single-component Lennard-Jones phase diagram across a  $(T, p)$  rectangle straddling solid/liquid coexistence; agreement with thermodynamic-integration baselines within the anchor’s phase, degradation across the coexistence line.

**Critical assessment — what the prior tuning costs.** The Schebek *et al.* conditional construction amortises across a continuous family of thermodynamic states and is useful for traversing a phase diagram once trained; however, the interesting region of that diagram is also its weakest point. The conditional flow is asked to absorb a discontinuity in the Boltzmann distribution across a coexistence line while  $\mu_0$  stays anchored on one side, and with smooth  $G_\theta$  the pushforward (26) cannot reproduce a Dirac-like phase boundary, so samples drawn just inside the wrong phase carry exponentially small weights. The conditioning is on  $(T, p)$  but the normalising constant  $\log Z(T, p)$  enters every free-energy estimate through (28); the conditional flow has no inductive bias forcing the  $\log Z$  slope to come out right, and small slope errors accumulate over the integration range. As in §4.1, the choice of  $\mu_0$  commits the flow’s support, so the method works on the phase of the anchor and silently mis-samples elsewhere. Any extension to multi-anchor mixtures (anchors in different phases) reintroduces the discrete-mixture-prior problem the conditional flow was designed to avoid. The construction is best understood as a practical patch on Wirnsberger 2022 for phase diagrams with smooth interiors and a single anchor; the phase boundary itself remains structurally inaccessible.

#### 4.4. Structured-CG latent priors (Schiebroek & Koehn 2025)

Schiebroek, Koehn, *J. Chem. Theory Comput.* 2025, [arXiv:2504.20940](#). The most explicit multimodal mixture prior in the BG literature: an explicit Gaussian mixture on collective variables, with a conditioned Gaussian on the orthogonal complement.

| At a glance — structured-CG latent prior |                                                                                                                                                                                                                |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Target</b>                            | All-atom Boltzmann distribution of a flexible biomolecule with a pre-identified set of slow collective variables.                                                                                              |
| <b>Prior</b>                             | $\mathbb{R}^d = \mathbb{R}^{d_{\text{slow}}} \times \mathbb{R}^{d_{\text{fast}}}$ ; slow block a $K$ -component Gaussian mixture on known metastable basins, fast block a state-conditional unimodal Gaussian. |
| <b>New idea</b>                          | Decouple multimodality (slow CVs) from local fluctuation (fast block); inject discrete metastable-state knowledge via the slow mixture.                                                                        |
| <b>Headline</b>                          | Interpretable latent space; on-target multi-basin sampling <i>conditional on</i> user-supplied slow CVs and basin centres.                                                                                     |

**Setup.** The user must supply (i) a decomposition of latent space  $\mathbb{R}^d = \mathbb{R}^{d_{\text{slow}}} \times \mathbb{R}^{d_{\text{fast}}}$  identifying the slow collective variables  $s \in \mathbb{R}^{d_{\text{slow}}}$  versus the fast orthogonal complement  $f \in \mathbb{R}^{d_{\text{fast}}}$ ; (ii) a list of  $K$  metastable-state centres  $\{m_k\}_{k=1}^K \subset \mathbb{R}^{d_{\text{slow}}}$  with covariances  $\{\Sigma_k\}_{k=1}^K$  and weights  $\{\pi_k\}_{k=1}^K$ ,  $\sum_k \pi_k = 1$ ; and (iii) a state-dependent fast covariance  $\Sigma_f(s)$  encoding the typical fluctuation scale at a given slow-CV value.

**Core math contribution — factored mixture prior.** Latent  $z = (s, f) \in \mathbb{R}^d$ , with

$$\mu_0^{\text{slow}}(s) = \sum_{k=1}^K \pi_k \mathcal{N}(s; m_k, \Sigma_k), \quad (29)$$

$$\mu_0^{\text{fast}}(f | s) = \mathcal{N}(f; 0, \Sigma_f(s)), \quad (30)$$

$$\mu_0(s, f) = \mu_0^{\text{slow}}(s) \mu_0^{\text{fast}}(f | s). \quad (31)$$

The slow density (29) is an honest Gaussian mixture — it has  $K$  disjoint high-density regions provided the basins  $\{m_k\}$  are separated by  $\gg \max_k \sqrt{\text{tr} \Sigma_k}$ . The fast density (30) is unimodal conditional on  $s$ ; the joint (31) is multimodal exactly in the slow directions. Density evaluation and sampling cost  $O(K)$  per query.

The flow  $G_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$  then maps  $(s, f) \sim \mu_0$  to an all-atom configuration  $x \in \mathbb{R}^d$  by reverse-KL on the target Boltzmann,

$$\mathcal{L}_{\text{rKL}}(\theta) = \mathbb{E}_{z \sim \mu_0} [\beta U_1(G_\theta(z)) - \log |\det \nabla G_\theta(z)|] - \mathbb{E}_{z \sim \mu_0} [\log \mu_0(z)], \quad (32)$$

with the slow-CV decomposition acting as a hard latent constraint: the flow is encouraged (by the way the loss decomposes additively over the slow and fast factors) to honour the slow-CV identity in its first  $d_{\text{slow}}$  output coordinates.

**Algorithm.** (i) Sample  $(s, f)$  by choosing  $k \sim \pi$ , drawing  $s \sim \mathcal{N}(m_k, \Sigma_k)$ , then  $f \sim \mathcal{N}(0, \Sigma_f(s))$ ; (ii) push through  $G_\theta$ ; (iii) take a gradient step on (32).

**Numerical details.** Demonstrated on small peptides with two or three well-characterised metastable states, where the slow CVs (e.g. Ramachandran  $\phi, \psi$ ) and basin centres are known a priori from short MD or crystallographic data. Reported multi-basin coverage with  $K = 2, 3$  and basin populations agreeing with reference MD to within 5–10%.

**Critical assessment — what the prior tuning costs.** The Schiebroke–Koehn structured-CG construction is the most explicit multimodal prior in the BG literature — a  $K$ -mixture on slow CVs together with a unimodal Gaussian on the fast block — and yields an interpretable latent space in which metastable populations can be inspected directly. Its limits are equally explicit. The decomposition  $\mathbb{R}^d = \mathbb{R}^{d_{\text{slow}}} \times \mathbb{R}^{d_{\text{fast}}}$  is a hard user input, and for the protein-folding problem (where the order parameters are *themselves the question*) the method requires a separate CV-discovery preprocess; once one has reliable slow CVs one already has most of what one wanted to learn. The basin centres  $\{m_k\}$  must likewise be known in advance, since if  $K$  or any  $m_k$  is misspecified the prior fails to cover the missing basin and the flow’s support is committed away from it, with no mechanism to discover unencoded basins from energy alone. The construction is also brittle in  $K$ : too small omits modes, too large produces ghost modes the flow is forced to drain via mass teleportation, with the standard fake-ESS pathology. The interpretable latent space is a major usability win when it applies, but only inside the design quadrant where slow CVs and basin centres are both known and stable across  $T$ . Read this way, the Schiebroke–Koehn prior is simultaneously the cleanest Camp B implementation in protein sampling and the cleanest illustration of mode lock-in: it requires exactly the mode information whose absence motivates third-way design.

## 4.5. VAE-side precedents: VampPrior and LARS

VampPrior (Tomczak, Welling, *AISTATS* 2018, [arXiv:1705.07120](#)) and LARS (Bauer, Mnih, *AISTATS* 2019, [arXiv:1810.11428](#)) are the canonical learned-prior constructions on the VAE side. We include them here because they are the most natural off-the-shelf Camp B candidates — and because their absence from the BG literature is a load-bearing fact for the Camp A vs Camp B argument.

| At a glance — VampPrior and LARS |                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Target</b>                    | (Original setting) data distribution $p_{\text{data}}(x)$ on images/text via a VAE encoder-decoder pair.                                                                                             |
| <b>Prior</b>                     | VampPrior: $\frac{1}{K} \sum_{k=1}^K q_{\phi}(z u_k)$ with learnable pseudo-inputs $\{u_k\}$ . LARS: $\tilde{p}(z) \propto \pi(z)a_{\phi}(z)$ with learned accept rate $a_{\phi}$ and base $\pi$ .   |
| <b>New idea</b>                  | Let the prior <i>learn</i> mode locations from data via pseudo-inputs (VampPrior) or via a learned acceptance function (LARS), avoiding the unimodal $\mathcal{N}(0, I)$ commitment of vanilla VAEs. |
| <b>Headline</b>                  | Adopted broadly in the VAE community; not adopted in mainstream BG work in the eight years since publication.                                                                                        |

**Setup.** A VAE has encoder  $q_{\phi}(z|x)$ , decoder  $p_{\theta}(x|z)$  and prior  $p(z)$ ; training maximises the ELBO  $\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \parallel p(z))$  on data samples  $x \sim p_{\text{data}}$ . The vanilla choice  $p(z) = \mathcal{N}(0, I)$  is known to cause posterior collapse and overly-smoothed reconstructions; both VampPrior and LARS address this by enriching  $p(z)$ .

**Core math contribution — VampPrior.** Introduce  $K$  learnable pseudo-inputs  $\{u_k\}_{k=1}^K \subset \mathbb{R}^{d_x}$  and define

$$p(z) = \frac{1}{K} \sum_{k=1}^K q_{\phi}(z|u_k), \quad (33)$$

i.e. a uniform mixture of encoder posteriors evaluated at the pseudo-inputs. Crucially, *the encoder  $\phi$  is shared* between the variational posterior  $q_{\phi}(z|x)$  and the prior (33): pseudo-inputs and encoder are co-trained, and  $K$  is a hyperparameter (typical values 50–500). The ELBO becomes

$$\mathcal{L}_{\text{Vamp}}(\theta, \phi, \{u_k\}) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - \text{KL}\left(q_{\phi}(z|x) \parallel \frac{1}{K} \sum_{k=1}^K q_{\phi}(z|u_k)\right), \quad (34)$$

maximised over  $(\theta, \phi, \{u_k\})$  by stochastic gradient ascent on data samples  $x \sim p_{\text{data}}$ .

**Core math contribution — LARS.** Replace the discrete mixture (33) with a continuous learned-acceptance reweighting of a base  $\pi$ :

$$\tilde{p}(z) = \frac{\pi(z)a_{\phi}(z)}{Z_{\phi}}, \quad Z_{\phi} = \mathbb{E}_{z \sim \pi}[a_{\phi}(z)], \quad (35)$$

with  $a_{\phi} : \mathbb{R}^d \rightarrow [0, 1]$  a small neural network and  $Z_{\phi}$  estimated by Monte Carlo. Sampling is by accept/reject from  $\pi$  with acceptance probability  $a_{\phi}$ . The reweighting (35) is strictly more flexible

than (33) — it has no discrete-mixture component count to choose — at the cost of a Monte-Carlo normaliser and a higher-variance gradient.

**Algorithm.** VampPrior: standard VAE training with  $K$  extra  $d_x$ -dimensional parameters (the pseudo-inputs) co-optimised by SGD. LARS: standard VAE training plus an accept/reject sampler in the prior, with  $Z_\phi$  estimated by a Monte-Carlo control variate every few steps.

**Numerical details.** Both methods are evaluated on MNIST, FashionMNIST, Omniglot, and CelebA in the original papers, with ELBO improvements of order 0.5–1.0 nats per image over the  $\mathcal{N}(0, I)$  baseline.

### Critical assessment — why neither has crossed to BG.

1. *VampPrior requires data.* The pseudo-inputs  $\{u_k\}$  are co-trained with the encoder *against data samples*  $x \sim p_{\text{data}}$ ; the ELBO (34) has no analogue in the BG setting because BGs have no data samples (or very few, from short expensive MD). To adopt VampPrior in BGs, the pseudo-inputs would have to be learned from the energy  $U_1$  alone — which is precisely the kind of upstream mode-discovery preprocess that Camp B was meant to avoid. This is the Trinity-style circular dependency the third-way log-ratio regulariser is designed to break.
2. *LARS has the same problem in a different disguise.* The accept rate  $a_\phi$  in (35) is trained against the ELBO, which is a data-driven objective; replacing data with energy-based samples reintroduces the upstream-sampler bootstrap.
3. *Discrete mixtures versus support commitment.* In the VAE setting a misplaced pseudo-input  $u_k$  is harmless — it produces a mode the decoder simply learns to ignore. In the BG setting the analogue is a misplaced mixture component, and (as in §4.4) misplaced components do commit the support: the flow either drains mass via fake-ESS-prone teleportation, or refuses to cover unencoded basins.
4. *Eight years of non-adoption is informative.* The BG community has had VampPrior and LARS for the entire history of BGs. The empirical verdict of non-adoption is the strongest piece of evidence against generic learned multimodal priors in the BG setting.

### 4.6. Why Camp B is small in 2024–2026

Of the dozen-plus state-of-the-art BG papers in 2024–2026, *none* uses a hand-tuned GMM prior with mode-matched components. The closest is Schiebroek-Koehn (and that requires user-specified CVs). Wirnsberger’s lattice prior is the canonical Camp B work and is restricted to crystals. **The empirical verdict of the BG field over six years is that prior tuning has not paid off.**

## 5. Difficulties of each camp

**Why this section exists.** §§3–4 told us *what* each camp does. This section asks *why each camp struggles*. For Camp A the struggles are concrete and engineering-addressable (mode collapse, mass teleportation, fake ESS). For Camp B they are *structural* — mode lock-in is a property of the bijective-flow geometry, not a training pathology. The asymmetry is the entire argument.

## 5.1. Camp A difficulties

### 5.1 Mode collapse under reverse KL. Reverse-KL gradient

$$\nabla_{\theta} \mathcal{L}_{\text{rev}}(\theta) = \mathbb{E}_{z \sim \mu_0} [\nabla_{\theta} U_1(G_{\theta}^{-1}(z))] + \mathbb{E}_{z \sim \mu_0} [\nabla_{\theta} \log |\det J_{G_{\theta}^{-1}}(z)|]$$

is an expectation under  $\mu_0$ , *not* under  $\nu_{\theta}$ . If at  $\theta_*$  the flow has collapsed to a single mode  $\nu_{\theta_*} \approx \delta_{x^*}$ , the gradient signal contains no information about other modes of  $U_1$ . *Evidence:* CMT (Blessing 2025) on ELIL tetrapeptide: reverse-KL gives ESS = 1.28% vs CMT’s 26.18%.

**5.2 Mass teleportation in annealing.** For Gaussian  $\mu_0$  centred at 0 and target mode at  $\|x^*\| \gg 1$ , intermediate  $\pi_{\beta} \propto \mu_0^{1-\beta} \mu^{\beta}$  has a single dominant mode whose location jumps discontinuously as  $\beta$  crosses some critical  $\beta_c$ . HMC kernels at any single  $\beta$  near  $\beta_c$  cannot bridge the disjoint support in finite time.

**5.3 Topological gap.**  $\mu_0$  is unimodal and contractible;  $\mu$  has disconnected basins. A bijective  $G: \mathbb{R}^d \rightarrow \mathbb{R}^d$  preserves connectedness locally; producing log-determinants that vary by many orders of magnitude across the domain to compress mass between basins is hard. Stimper et al. (2022) cite this as their motivation for resampled-base flows.

**5.4 Fake ESS.** If  $\nu_{\theta}$  has dropped a mode  $A_2$  of mass  $p_2$  but matches  $\mu$  on the visited mode  $A_1$  up to the constant  $1/p_1$ , importance weights  $w_i = p_1$  are uniform on  $A_1$ , giving ESS = 1 exactly while  $p_2$  mass is missing. ESS is necessary but not sufficient.

**5.5 Expensive bootstrapping.** FAB on alanine dipeptide still uses hundreds of thousands of  $U_1$  evaluations. Wirnsberger 3 weeks on 16 A100s for 512 ice atoms. Scale beyond dipeptides is hard.

**5.6 No clean place for prior mode knowledge.** When the user knows mode locations a priori, Camp A has no clean injection point: adding them as MD samples shifts the MLE; using them as AIS initial states biases weights; encoding them in the prior turns Camp A into Camp B.

## 5.2. Camp B difficulties

**5.7 Circularity.** Placing a mixture component on a mode requires knowing it is there. Wirnsberger works because the crystal phase is known a priori. For unknown-mode problems Camp B is self-referential.

**5.8 Mode lock-in.** Bijective  $G_{\theta}^{-1}: \text{supp}(\mu_0) \rightarrow \mathbb{R}^d$  preserves connected components.  $\text{supp}(\nu_{\theta}) \subseteq G_{\theta}^{-1}(\text{supp}(\mu_0))$ . A mode at  $m^*$  with  $G_{\theta}^{-1}(\text{supp}(\mu_0)) \not\ni m^*$  is *unreachable*. Mode discovery is structurally impossible in Camp B without re-tuning the prior.

**5.9 Brittleness to misspecification.** If  $m_k = m_k^{\text{true}} + \delta_k$ , the flow has to learn an extra translation to bring the mode back, consuming Jacobian capacity. Real modes between two prior components produce spurious bridge mass.

**5.10 Symmetry breaking.** GMM at specific locations is not  $\text{SE}(3) \times S_N$ -invariant. Camp A handles symmetry by mean-free Gaussian + equivariant flow; Camp B has to symmetrise the prior over  $N! \cdot \text{Vol}(\text{SE}(3))$  images, blowing up the prior.

**5.11 Transferability collapse.** Each new target re-tunes the prior; no amortisation across systems.

**5.12 Structural bias.** Importance weights  $w(x) = \mu(x)/\nu_\theta(x)$  are undefined where  $\nu_\theta(x) = 0$ . If a tuned prior misses a mode of  $\mu$ , no importance-weighted correction is available — bias is by design.

**5.13 Tuning relocates the hard part.** The user-side effort to set up Wirnsberger’s lattice is exactly the effort the  $X_{\hat{\mu}}$  approach spends on Trinity. Camp B’s lattice has to be exact; the third way’s  $\hat{\mu}$  can be sloppy (Proposition 1).

## 6. Code-level inventory

**Why this section exists.** Papers can claim more than their code delivers; defaults in production code are more telling about field consensus than what an abstract advertises. If Camp B were practically useful, someone would have implemented a canonical notebook with a `MixtureDistribution`. The class is available in essentially every library; nobody uses it.

| Library                                | Default prior                         | Mixture available?                                                      |
|----------------------------------------|---------------------------------------|-------------------------------------------------------------------------|
| <code>bgflow</code> (Noé group)        | <code>NormalDistribution</code>       | Yes ( <code>MixtureDistribution</code> ); unused in canonical notebooks |
| <code>normflows</code> (Stimper)       | <code>DiagGaussian</code>             | Yes ( <code>GaussianMixture</code> ); rarely default                    |
| <code>fab-torch</code> (Midgley)       | hardcoded <code>DiagGaussian</code>   | N/A (uses <code>normflows</code> )                                      |
| <code>DEM</code> (Akhound-Sadegh)      | implicit Gaussian SDE terminal        | N/A                                                                     |
| <code>flowMC</code> (Wong et al.)      | standard Gaussian                     | N/A                                                                     |
| <code>TA-BG</code> (Schopmans)         | truncated Gaussian + uniform dihedral | N/A                                                                     |
| <code>zuko</code> (Rozet)              | standard Gaussian                     | yes; not default                                                        |
| user’s <code>zflows</code> (this repo) | via <code>zuko</code>                 | MAF-based NSF                                                           |

Table 1: Default base distribution and mixture-prior availability across the major BG and sampler libraries. Mixture-prior code exists in essentially every library but is not used in canonical training notebooks.

The canonical `bgflow` alanine-dipeptide notebook (`notebooks/alanine_dipeptide_basics.py:155--167`) instantiates simply

```
prior = bg.NormalDistribution(dim_ics, mean=mean)
```

with the author-supplied comment that this is a “rather naive choice”. The mixture machinery is sitting unused.

## 7. Field landscape — adjacent communities, non-flow samplers, shared tricks

A skeptic might say: “the BG community converged on Camp A through historical accident.” This section falsifies that in three sweeps: (a) five *adjacent communities* that face the same problem, (b) the non-flow neural-sampler paradigm (diffusion / stochastic interpolants), and (c) tricks that both camps freely use. In every case the prior is left simple; investment goes elsewhere.

**Adjacent communities all choose Camp A.** The first sweep is across five communities that face structurally the same problem. In *Bayesian inverse PDE* (Whang–Lindgren–Dimakis ICML 2021; Sun et al. [NF-iVI 2024](#) for functional flows on a Gaussian-process base; Padmanabha–Zabaras 2023; Zhao–Curtis–Zhang 2022 on Bayesian seismic tomography), the standard recipe is a Gaussian base for the posterior flow, with physics entering through the likelihood and through the prior over the unknown field but never through the flow’s own base. The *cosmology and gravitational-wave* community has landed on the same pattern: Wong–Gabrié–Foreman–Mackey’s [flowMC \(JOSS 2023\)](#), Williams–Veitch–Messenger’s nested sampling with NFs (2021), and the 2024–25 GW parameter-estimation literature all use a standard Gaussian base under an adaptive flow trained on chain history. *Particle physics* follows suit, with Kofler et al. [FAB-meets-diffME \(MLST 2025\)](#) running FAB with HMC on matrix elements over a standard Gaussian base. The *simulation-based inference* community (Cranmer–Brehmer–Louppe PNAS 2020; Gloeckler–Bischoff–Macke [Simformer \(2024\)](#); Macke et al. [sbi](#)) uses standard Gaussian or diffusion noise across the board. And in *geophysics* (Liao et al. [JGR-MLC 2025](#); Asim et al. [wave-equation inversion 2022](#)) the base is again standard, with physics entering through the prior over the inverted parameters. Five independent communities have made the same Camp A choice; if Camp B were the obvious win, at least one would have adopted it independently, and none has.

**Non-flow neural samplers all choose Camp A.** Diffusion and stochastic-interpolant samplers do not learn an invertible flow; instead the terminal of the noising process plays the role of base. On this axis the literature lines up cleanly. iDEM, iEFM, EWFM, BNEM, and PITA all use isotropic Gaussian terminal noise; the only diffusion-sampler paper that permits a non-Dirac prior is the Adjoint Schrödinger Bridge Sampler ([Liu et al. NeurIPS 2025](#)), and even there the allowed priors are unimodal (Gaussian, harmonic) rather than mode-matched mixtures. Stochastic interpolants (Albergo–Vanden-Eijnden ICLR 2023) unify the picture under the construction  $X_t = I_t(X_0, X_1) + \gamma(t)Z$  on paired training samples but do not tune the prior either. Even in paradigms that do not require a closed-form prior density, mode-matched mixtures remain unused.

**Tricks both camps can use, and why they do not settle the question.** Across the field there is a sizeable catalogue of useful tricks that are orthogonal to prior choice: smooth normalising flows for tori, gradient-boosted normalising flows (which place a mixture in the *model* rather than the prior), FAB-style prioritised replay, mean-free distributions for SE(3) equivariance, force matching ([Köhler et al. 2024](#)), tamed Langevin gradients (Hutzenthaler–Jentzen–Kloeden), flow-perturbation and FP++ Jacobian estimators, and the importance-weighted ELBO (Burda–Grosse–Salakhutdinov 2016). What unites these is that all of them live in architecture, loss, or post-processing rather than in the prior; the field’s empirical bias is to invest effort everywhere except the base distribution.

## 8. Comparative numerical results

The numbers reported below partition cleanly by camp. The single-row Camp B table and the seven-row Camp A table are themselves the thesis: every flagship multimodal-target paper since 2019 uses a Gaussian prior.

### Camp B — single-phase precision.

| Method           | System        | Prior         | Headline                                                                      |
|------------------|---------------|---------------|-------------------------------------------------------------------------------|
| Wirnsberger 2022 | Ice $I_c$ 512 | lattice-Gauss | $\Delta F$ matches MBAR;<br>$\delta F \sim 10^{-5} k_B T$ /atom, single basin |

*Verdict:* unmatched precision; multimodal targets not attempted.

### Camp A — multimodal flow methods.

| Method     | System                | Prior               | Headline                                   |
|------------|-----------------------|---------------------|--------------------------------------------|
| FAB 2023   | alanine dipeptide     | DiagGauss           | 100× fewer evals than MLE/MD               |
| FAB 2023   | 40-comp 2D GMM        | DiagGauss           | mode discovery via AIS                     |
| TBG 2024   | alanine dipeptide     | std Gauss           | ESS = $6.03 \pm 1.34\%$                    |
| TBG 2024   | 100 unseen dipeptides | std Gauss           | ESS = $15.29 \pm 9.27\%$ , 98 ± 2% correct |
| TA-BG 2025 | dipeptide             | truncated Gauss     | beats FAB on energy evals                  |
| CMT 2025   | ELIL tetrapeptide     | std Gauss           | rev-KL 1.28% vs CMT 26.18% ESS             |
| SBG 2025   | tripeptide AL3 / AL6  | $\mathcal{N}(0, I)$ | ESS = 0.876 / 0.995                        |

*Verdict:* ESS scales with training sophistication; prior held fixed at Gaussian.

### Non-flow neural samplers.

| Method    | System | Prior                     | Headline                         |
|-----------|--------|---------------------------|----------------------------------|
| iDEM 2024 | LJ-55  | $\mathcal{N}(0, I)$ noise | first energy-only at LJ-55 scale |

*Verdict:* same prior philosophy as Camp A.

## 9. The third way — formal mathematical analysis

**The core intuition.** A bijective flow’s support is determined by its prior’s support — that is what Camp B’s mode-lock-in difficulty boils down to. A regulariser evaluated at points  $\hat{\mu}$  supplies, by contrast, only *gradient signal*: it pushes the flow to make  $\log \mu/\nu_\theta$  as constant as possible at the points it sees, without committing the flow’s support to any of them.

### 9.1. The X functional

Let  $\omega(x, y) \geq 0$  be a finite positive measure on  $\mathbb{R}^d \times \mathbb{R}^d$ . Define

$$X_\omega(\mu \parallel \nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x, y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy. \quad (36)$$

**Theorem 1** (Main: asymptotic invariance to  $\hat{\mu}$ ). *Suppose  $\nu_{\theta_*} = \mu$  is realisable in the flow class. For any  $\omega = \hat{\omega} \otimes \hat{\omega}$  with  $\text{supp}(\hat{\omega}) \subseteq \mathbb{R}^d$ ,  $\theta_* \in \arg \min_\theta X_\omega(\mu \parallel \nu_\theta)$  and  $X_\omega(\mu \parallel \nu_{\theta_*}) = 0$ . The argmin set is unchanged for any  $\hat{\omega}$ .*

*Proof.* At  $\theta = \theta_*$ ,  $\log(\mu/\nu_{\theta_*}) = 0$  identically, so the integrand is identically zero regardless of  $\hat{\omega}$ . Non-negativity of  $X_\omega$  makes  $\theta_*$  a global minimiser. The argument uses only non-negativity of  $\omega$  and realisability of  $\theta_*$  — the specific shape of  $\hat{\omega}$  is irrelevant.  $\square$

**Practical interpretation.** Even if  $\hat{\mu}$  is a coarse Trinity-style approximation that over-covers modes (places mass where  $\mu$  has none), the optimal  $\theta_*$  is still  $\nu_{\theta_*} = \mu$ . *Trinity is allowed to be sloppy.* This is the load-bearing result of the entire paper.

Two supporting lemmas:

**Lemma 1** (Zero set).  $X_\omega(\mu \parallel \nu) = 0$  iff  $\log(\mu/\nu)$  is constant  $\omega$ -a.e. on the joint support of  $\omega$ .

**Lemma 2** (Normalisation invariance). *For any  $c_1, c_2 > 0$ ,  $X_\omega(c_1\mu \parallel c_2\nu) = X_\omega(\mu \parallel \nu)$ . Consequently  $X_\omega$  is well-defined for unnormalised Boltzmann targets even when  $Z_\mu$  is unknown ( $\log(c_1\mu/c_2\nu) = \log(c_1/c_2) + \log(\mu/\nu)$ ; the additive constant cancels in the difference).*

### 9.2. $X_\mu$ and the autograd-reuse trick

Choose  $\omega = \mu \otimes \mu$ . Define

$$z(y) := U_0(G_\theta(y)) - U_1(y) - \log |\det J_{G_\theta}(y)|.$$

Then  $\log(\mu/\nu_\theta)(x) - \log(\mu/\nu_\theta)(y) = z(x) - z(y)$  up to a constant that cancels, and

$$X_\mu(\mu \parallel \nu_\theta) = \mathbb{E}_{(x,y) \sim \mu \otimes \mu} [|z(x) - z(y)|].$$

Empirically: sample  $\mathbf{y} = (y_1, \dots, y_B) \sim \mu$ , compute  $\mathbf{z} = (z(y_1), \dots, z(y_B))$  with *one* forward pass through  $G_\theta$ , draw a permutation  $\sigma$ , and set  $\mathbf{z}' := \mathbf{z}[\sigma]$ . The estimator

$$\hat{X}_\mu(\theta) = \frac{1}{B} \sum_i |z_i - z_{\sigma(i)}|$$

has gradient

$$\nabla_\theta \hat{X}_\mu = \frac{1}{B} \sum_i \text{sgn}(z_i - z_{\sigma(i)}) (\nabla_\theta z_i - \nabla_\theta z_{\sigma(i)}).$$

**Proposition 1** (Zero-cost autograd reuse).  $\nabla_\theta \hat{X}_\mu$  requires only the same backward pass through  $G_\theta$  that computes the forward-KL gradient. Permutation is a pure index gather with no extra differentiable operations.

The user's `2D_Minimal/core.py:loss_KL_X` implements exactly this:

```

def loss_KL_X(y, source, target, G, lambda_=1.0):
    N = y.shape[0]
    x, ladj = G.call_and_ladj(y)
    z = source(x) - target(y) - ladj
    perm = torch.randperm(N, device=y.device)
    return z.mean() + lambda_ * (z - z[perm]).abs().mean()

```

A single call to `G.call_and_ladj(y)` carries the autograd graph for both the  $\text{mean}(\mathbf{z})$  (forward-KL) and the  $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$  ( $X_\mu$ ) terms.

### 9.3. Trinity algorithm for $\hat{\mu}$

The Trinity algorithm produces samples from  $\hat{\mu}$ :

1. **Diffusion.** Given initial  $x \sim \mu_0^{\text{init}}$ , set  $\tilde{x} = x + \sigma\xi$ ,  $\xi \sim \mathcal{N}(0, I_d)$ .
2. **Optimisation.** Run L-BFGS with Armijo line search on  $U_1$  at each  $\tilde{x}_i$ .
3. **Rejuvenation.** Run a short tamed Langevin chain at temperature  $T$ :

$$x_i^{(k+1)} = x_i^{(k)} - \frac{\eta \nabla U_1(x_i^{(k)})}{1 + \tau \|\nabla U_1(x_i^{(k)})\|} + \sqrt{2\eta T} \xi_i^{(k)}.$$

By Proposition 1, Trinity is allowed to be sloppy.

### 9.4. Full training loss

$$\mathcal{L}_{\text{full}}(\theta) = \mathcal{L}_{\text{fwd}}(\theta) + \lambda X_\mu(\mu \parallel \nu_\theta) + \hat{\lambda} X_{\hat{\mu}}(\mu \parallel \nu_\theta).$$

$\mathcal{L}_{\text{fwd}}$  supplies unbiased asymptote;  $X_\mu$  supplies free pointwise log-ratio penalty;  $X_{\hat{\mu}}$  supplies mode-discovery signal without committing the flow’s support.

### 9.5. Why this beats each camp

Compactly: Camp A fails on mode collapse (§5.1) and the topological gap (§5.3); Camp B trades these for mode lock-in (§5.8) and brittleness to misspecification (§5.9). The  $X_{\hat{\mu}}$  regulariser avoids both families of failure because the *regulariser is evaluated at  $\hat{\mu}$ , not the flow’s prior committed to it*: full-support Gaussian prior buys Camp A’s robustness, asymptotic invariance (Theorem 1) buys Camp B’s mode-coverage tolerance, and a Trinity-cheap  $\hat{\mu}$  buys both without the multi-week PMF bootstrap of §5.5.

The third way as described is an abstract loss; what makes it usable is a matching architecture and loss-direction story. The next two sections collect the relevant facts about flow architectures, document the asymmetries they create between forward and reverse KL, and show why MAF- based pipelines structurally favour the  $X_{\hat{\mu}}$  third way.

## 10. Parameterisation — flow architecture and its direction asymmetries

**Why this section exists.** The choice of flow architecture and the direction of the loss interact strongly with the prior debate. Camp B’s natural home is reverse KL; the forward-KL family (forward KL, FAB, X) changes the game because the prior enters via  $\log \mu_0(G_\theta(y))$  at target-side samples rather than via the source of training samples.

### 10.1. Four flow architecture families

Four families of bijective architecture dominate practice and they differ sharply in where the computational cost lives. *Coupling flows* (NICE, RealNVP, NSF-C, Glow) make both  $G$  and  $G^{-1}$  a single  $O(1)$  parallel pass, so neither direction is privileged. *Autoregressive flows* (MAF, IAF, NSF-AR) parallelise one direction and serialise the other: Papamakarios–Pavlakou–Murray’s MAF (NeurIPS 2017) is fast density and slow sampling, while Kingma et al.’s IAF (NeurIPS 2016) is the transpose — a severe asymmetry that ties architecture directly to loss direction. *Continuous normalising flows* (FFJORD; Grathwohl et al. ICLR 2019) make both directions ODE solves and both expensive, with Hutchinson’s trace estimator reducing log det from  $O(d^2)$  to  $O(d)$  per step. *Flow matching* (Lipman et al. ICLR 2023) avoids  $G$  and  $G^{-1}$  at training time altogether; inference is again an ODE solve.

**The user’s zflows package** (`~/envs/zflows/zflows/flow.py`) implements NSF as a MAF-RQS autoregressive variant. The default in the user’s experiments is therefore MAF-based: *forward KL is cheap, reverse KL is sequentially expensive*.

### 10.2. Direction-loss coupling

| Loss                                                  | Samples from                 | Density evaluated of                 | Architecture fit   |
|-------------------------------------------------------|------------------------------|--------------------------------------|--------------------|
| Reverse KL                                            | $\nu_\theta$ (via $G^{-1}$ ) | $\nu_\theta$ at own samples          | Coupling, IAF, CNF |
| Forward KL                                            | $\mu$ (MD / AIS)             | $\nu_\theta$ at $\mu$ -samples       | Coupling, MAF, CNF |
| FAB $\alpha = 2$                                      | AIS chain                    | $\nu_\theta$ at AIS samples          | Coupling, MAF      |
| $X_\omega$ ( $\omega = \hat{\mu} \otimes \hat{\mu}$ ) | $\hat{\mu}$ (Trinity)        | $\nu_\theta$ at $\hat{\mu}$ -samples | Coupling, MAF      |

**Consequence.** On the user’s MAF-based zflows infrastructure, reverse KL is the expensive loss. The loss families compatible with MAF — forward KL, FAB, X — do not benefit from Camp B’s prior tuning. Camp B is therefore architecturally penalised on this stack.

## 11. Direction $\times$ camp interaction — a critical synthesis

**Why this section exists.** §11 established architecture-loss coupling. This section closes the circle: the loss direction in turn determines whether Camp B’s prior tuning earns its keep. The user’s observation that Camp B’s natural home is reverse KL is made as sharp as the math allows.

### 11.1. Reverse KL with a tuned prior

In reverse KL,  $\mu_0$  is the source of all training samples:  $z \sim \mu_0$ ,  $y = G_\theta^{-1}(z)$ . If  $\mu_0$  is mode-aware (Camp B),  $G_\theta^{-1}$  has only to learn local fluctuations within each pre-identified basin. *This is exactly Wirnsberger*. Camp B’s leverage in reverse KL is strong.

### 11.2. Forward KL with a tuned prior — leverage degrades

In forward KL,  $y \sim \mu$  and the prior enters only through  $\log \mu_0(G_\theta(y))$ . A multimodal tuned  $\mu_0$  *fits* the post-flow latent distribution only *if*  $G_\theta$  has already organised latent space — a circular condition. Early in training, a multimodal  $\mu_0$  has *thinner* tails between components than a unimodal Gaussian, so  $\log \mu_0(G_\theta(y))$  is actually *lower* on average. Forward KL is naturally aligned with the simplest prior.

### 11.3. FAB and $X_\mu$ — same analysis

Both losses sample target-side and evaluate  $\log \nu_\theta$  at those samples. The prior enters in the same way as forward KL. Tuning provides no leverage.

### 11.4. Cross-tabulation

| Loss                           | Arch. pressure                    | Camp A        | Camp B                           | Third way     |
|--------------------------------|-----------------------------------|---------------|----------------------------------|---------------|
| Reverse KL                     | needs cheap $G^{-1}$              | ✗ collapses   | ✓ mitigates                      | neutral       |
| Forward (MD)                   | KL needs cheap $G$                | ✓ unbiased    | neutral                          | ✓ + mode info |
| FAB                            | needs cheap $G + \text{AIS}$      | ✓ + AIS       | neutral; lock-in if modes missed | ✓ + mode info |
| TA-BG                          | needs cheap $G^{-1}$              | ✓ at high $T$ | tuning at high $T$ is hard       | neutral       |
| CMT                            | flexible                          | ✓             | constraints harder               | ✓             |
| iDEM                           | needs $\nabla U, \nabla \log p_t$ | ✓ std noise   | ✗ tuned noise breaks score match | neutral       |
| ASBS                           | flexible (any prior)              | ✓             | ✓ harmonic OK                    | ✓             |
| $X_\mu$ regulariser            | cheap $G + \text{autograd reuse}$ | ✓ ideal fit   | ✗ forward-KL story               | third way     |
| $X_{\hat{\mu}}$ mode discovery | cheap Trinity samples             | ✓             | redundant with prior             | third way     |

## 11.5. Quantitative consequences

Three consequences fall out of the direction–architecture–prior table. Camp B’s value is concentrated in a single design quadrant: reverse-KL training combined with a cheap- $G^{-1}$  architecture applied to a single-basin or known-multibasin target. MAF-based BGs structurally penalise Camp B, because reverse KL is expensive on MAF and the MAF-compatible losses — forward KL, FAB, X — do not benefit from prior tuning in the first place. And the  $X_{\hat{\mu}}$  third way is uniquely compatible with MAF architectures: cheap  $G$  in the right direction, no requirement for a cheap  $G^{-1}$ , autograd-reuse at zero overhead, and invariance to whatever sloppiness Trinity introduces in  $\hat{\mu}$ .

## 11.6. When Camp B does win

There are three settings in which Camp B clearly wins. Single-phase crystallographic precision (Wirnsberger 2022) is the canonical one; conformer enumeration with catalogued modes (Schiebroek–Koehn 2025) and higher- $T$  reference priors for liquids (Coretti et al.) round out the short list. All three sit squarely in the reverse-KL  $\times$  tuned-prior quadrant. The user’s argument is therefore not that Camp B is wrong in general but that Camp B is wrong for the multimodal-mode-discovery problem the LLR-discrepancy paper targets.

Having located the design quadrant where Camp B wins on its own terms, the review’s obligation is to give that quadrant its full argumentative weight before continuing. The next section does exactly that — steelmanning Camp B to find the strongest version of its case — after which the two technical sections (precision and wide-Gaussian rebuttal) engage that strongest case on its merits.

## 12. Steelmanning Camp B

A literature review that critiques an opposing position must steelman it before refuting it. This section writes the strongest defensible Camp B position and engages each point on its merits.

### 12.1. The strongest Camp B argument

A serious Camp B advocate would say: (i) the highest-precision BG free-energy calculations (Wirnsberger 2022) come from putting structure into the prior; (ii) the BG methods that scale most gracefully (Schiebroek–Koehn 2025) use coarse-grained physical representations, which is Camp B in a different parameterisation; (iii) SBG (Tan 2025) uses the trained flow as a “more informative proposal” at inference time, i.e. an adaptive Camp B prior that the field has implicitly converged on. (iv) Camp B’s difficulties (mode lock-in, brittleness, etc.) apply only to rigid mixture priors; *adaptive* Camp B (Routes B, C, D, E of §15) avoids these and is at worst on equal footing with  $X_{\hat{\mu}}$ .

### 12.2. Engagement

**Point 1 (precision).** Conceded. Camp B wins in (reverse KL  $\times$  known single basin  $\times$  precision target). Outside this quadrant the user’s argument is unchanged.

**Point 2 (CG latent is Camp B).** Schiebroek–Koehn requires user-specified slow CVs;  $X_{\hat{\mu}}$  + Trinity discovers modes in full coordinate space without committing to a CV specification.

**Point 3 (SBG implicitly tunes the prior).** SBG’s inference-time “adaptive prior” is the post-training flow, which itself was trained on a Gaussian latent. The distinction matters because SBG’s training stage is fully Camp A; the user’s framework is complementary, not in opposition.

**Point 4 ( $X_{\hat{\mu}}$  is Camp B in disguise).** The technical distinction is the type of information that enters: a tuned prior commits the *support*; a regulariser supplies only *gradient signal*. The flow’s support under  $X_{\hat{\mu}}$  is determined by  $\mu_0$  (full  $\mathbb{R}^d$ ), *not* by  $\hat{\mu}$ . If  $\hat{\mu}$  misses a mode, the loss is unaffected at  $\theta = \theta_*$  where  $\nu_{\theta_*} = \mu$  (Proposition 1), so the optimal  $\theta_*$  is not shifted. This is the structural asymmetry the moral-equivalence framing elides.

**Point 5 (adaptive Camp B avoids difficulties).** Routes B, C, D, E of §15 either collapse into Camp A (defensive Gaussian) or into the third way (score-based bias on a Gaussian prior). The bare-rigid Camp B position is the one with the brittleness problems; the adaptive positions converge on Camp A or  $X_{\hat{\mu}}$ .

### 12.3. Where the argument remains open

Several questions remain genuinely unsettled. The score-based-bias Route E combined with a Schrödinger-bridge approach has not yet been compared head-to-head with  $X_{\hat{\mu}}$  + Trinity at dipeptide scale, and that comparison is the cleanest empirical test of the two strategies. Camp B’s single-target high-precision recipe also remains correct for crystallography, polymorph identification, and phase-diagram problems — nothing in this review challenges that. And the empirical scaling of  $X_{\hat{\mu}}$  beyond 2D Himmelblau is an open question; a full-scale alanine-dipeptide replication comparable to TBG benchmarks is the natural next experiment and the obvious target of the project this review supports.

The steelman granted, the review now engages two specific objections that a careful Camp B advocate or skeptical reader will press. The first acknowledges Wirnsberger’s precision advantage as genuine but asks why forward-KL with importance sampling — being asymptotically unbiased — cannot match it given enough samples; the answer is variance, not bias. The second proposes that mode collapse is merely a support problem fixable by widening the Gaussian prior; the answer is that widening the prior changes neither the loss’s gradient nor the bijective flow’s topology.

## 13. Why does a tuned prior have an unbeaten precision advantage?

**The question.** Forward-KL / FAB training with importance sampling is asymptotically unbiased: as the number of importance samples  $N \rightarrow \infty$ , the sample mean of  $w_i f(y_i) / \sum w_i$  converges to  $\mathbb{E}_{\mu}[f]$  for any integrable  $f$ . So why does Wirnsberger 2022’s lattice-Gaussian prior achieve  $10^{-5} k_B T / \text{atom}$  precision on solid free energies that forward-KL + IS with a Gaussian prior cannot match? The answer is variance, not bias — and the variance scales exponentially in the divergence between flow and target.

### 13.1. The argument in three claims

**Claim 1 (Variance, not bias, is the bottleneck).** The self-normalised IS estimator  $\hat{I}_N(f) = \sum_i w(y_i) f(y_i) / \sum_i w(y_i)$ ,  $y_i \sim \nu_{\theta}$ , is consistent but has leading-order variance

$$\text{Var}[\hat{I}_N(f)] \approx \frac{1}{N} \text{Var}_{\mu}[f] (1 + \chi^2(\mu \parallel \nu_{\theta})).$$

Therefore samples-to-precision scales as  $N \sim (1 + \chi^2)/\sigma^2$ : unbiasedness is free;  $\chi^2$  is what bills you.

**Claim 2 ( $\chi^2$  tensorises and so scales exponentially in  $d$ ).** If  $\mu, \nu_\theta$  factorise across coordinates,

$$1 + \chi^2(\mu \parallel \nu_\theta) = \prod_{i=1}^d (1 + \chi_i^2) \approx e^{d\bar{\varepsilon}}$$

at per-coordinate residual  $\bar{\varepsilon}$ . Therefore any irreducible per-coordinate gap compounds multiplicatively in dimension.

**Claim 3 (Prior choice sets  $\bar{\varepsilon}$ ).** Gaussian prior on a  $K$ -mode target  $\Rightarrow \bar{\varepsilon} = \Omega(\log K/d)$  (Jacobian must absorb  $\log K$  of volume distortion). Wirsberger lattice-Gaussian on a single harmonic basin  $\Rightarrow \bar{\varepsilon} \sim 10^{-3}$  (anharmonic correction only). Therefore at  $N=500$  atoms and  $\delta F=10^{-5} k_B T/\text{atom}$  precision, Camp A needs  $\sim 10^{31}$  IS samples; Camp B needs  $\sim 10^{10}$ . Six orders of magnitude separate the two regimes (Coretti 2026 survey).

The rest of this section derives each claim.

### 13.2. Derivation of Claim 1: the IS variance formula

Let  $\nu_\theta$  be the flow's pushforward and  $w(y) = \mu(y)/\nu_\theta(y)$  the importance weight. For observable  $f$ , the self-normalised IS estimator

$$\hat{I}_N(f) = \frac{\sum_{i=1}^N w(y_i) f(y_i)}{\sum_{i=1}^N w(y_i)}, \quad y_i \stackrel{\text{iid}}{\sim} \nu_\theta,$$

is consistent:  $\hat{I}_N(f) \rightarrow \mathbb{E}_\mu[f]$  almost surely as  $N \rightarrow \infty$ . Its variance is well-known to satisfy, to leading order in  $1/N$ ,

$$\text{Var}[\hat{I}_N(f)] \approx \frac{1}{N} \text{Var}_\mu[f] \cdot (1 + \chi^2(\mu \parallel \nu_\theta)),$$

where the chi-squared divergence

$$\chi^2(\mu \parallel \nu_\theta) = \int \frac{(\mu(x) - \nu_\theta(x))^2}{\nu_\theta(x)} dx = \int \frac{\mu^2(x)}{\nu_\theta(x)} dx - 1$$

equals the FAB loss minus one (§3.5). The effective sample size  $\text{ESS} = N/(1 + \chi^2)$  tells the same story. So unbiasedness is free, but the *number of samples needed for a target precision  $\sigma$*  scales as

$$N \sim \frac{(1 + \chi^2(\mu \parallel \nu_\theta))}{\sigma^2}.$$

### 13.3. Derivation of Claims 2 and 3: how $\chi^2$ scales with $d$ and prior choice

The crucial fact is that  $\chi^2(\mu \parallel \nu_\theta)$  scales *exponentially in dimension* for a generic Gaussian-prior flow, but stays  $O(1)$  for a Wirsberger-style tuned prior. The argument is in two steps.

**Step 1:  $\chi^2$  tensorises in factorised approximations.** Suppose  $\mu(x) = \prod_{i=1}^d \mu^{(i)}(x_i)$  and  $\nu_\theta(x) = \prod_{i=1}^d \nu_\theta^{(i)}(x_i)$  (both factorise across coordinates). Then

$$1 + \chi^2(\mu \parallel \nu_\theta) = \prod_{i=1}^d (1 + \chi^2(\mu^{(i)} \parallel \nu_\theta^{(i)})).$$

Even when the per-coordinate divergences are small, the product blows up exponentially in  $d$ . For  $\chi^2$ -per-coordinate  $\varepsilon$  this gives  $1 + \chi^2 \approx e^{d\varepsilon}$  at  $d\varepsilon \gg 1$ .

**Step 2: the per-coordinate divergence depends on how well the prior already captures the target’s local geometry.** *Gaussian prior, Camp A.* For a single harmonic mode at frequency  $\omega_i$ ,  $\mu^{(i)}(x_i) \propto \exp(-\omega_i^2 x_i^2/2)$  but the flow must map an isotropic Gaussian  $\mathcal{N}(0, 1)$  to this distribution. After training,  $\nu_\theta^{(i)}(x_i)$  is the flow’s best approximation. The residual  $\chi^2$  per coordinate is bounded by the squared coefficient of variation of the flow’s local Jacobian, which for a flow trained against a multimodal target with  $K$  basins is at least  $O(\log K/d)$  per coordinate (because the flow’s volume distortion at the boundaries between basins must be large to compress mass into multiple wells from a unimodal Gaussian).

*Lattice-Gaussian prior, Wirnsberger 2022.* The prior is  $\mu_0^{\text{lat}}(z) \propto \prod_i \exp(-(z_i - z_i^o)^2/(2\sigma_T^2))$  already centred on the harmonic minimum, with width  $\sigma_T$  matched to the thermal de-Broglie length. The flow  $G_\theta$  now learns only the *anharmonic correction* to the harmonic Gaussian. Per coordinate, the residual  $\chi^2$  is of the order of the anharmonic-coefficient squared, typically  $\varepsilon \sim 10^{-3}$  to  $10^{-2}$ .

**Implication for  $\Delta F$  precision.** To estimate  $\Delta F = -k_B T \log(Z_1/Z_0)$  to precision  $\delta F$  per atom for  $N$  atoms, one needs the variance of the per-atom IS estimator to be at most  $\delta F^2$ :

$$N_{\text{IS}} \gtrsim \frac{1 + \chi^2(\mu \parallel \nu_\theta)}{\delta F^2} \approx \frac{e^{N\varepsilon}}{\delta F^2}.$$

- For Wirnsberger ( $N = 500$  atoms,  $\varepsilon \approx 10^{-3}$ ):  $N\varepsilon \approx 0.5$ , so  $e^{N\varepsilon} \approx 1.6$ . Asking for  $\delta F = 10^{-5} k_B T$  gives  $N_{\text{IS}} \sim 10^{10}$ . This is feasible:  $10^{10}$  flow samples on a single GPU at  $10^6$  samples/second is  $\sim 3$  hours.
- For a Gaussian-prior flow on the same crystal: even if the flow has converged to within  $\varepsilon \sim 10^{-1}$  per atom (extremely favourable),  $N\varepsilon \approx 50$ ,  $e^{N\varepsilon} \approx e^{50} \approx 5 \times 10^{21}$ .  $N_{\text{IS}} \sim 10^{31}$  samples to reach the same  $\delta F$ . Infeasible by 21 orders of magnitude.

### 13.4. Why FAB and the X functional do not close this gap

FAB minimises  $\chi^2$  directly (§3.5), so its trained flow has the smallest residual divergence achievable in its parameterisation. But the *architectural ceiling* on  $\chi^2$  is set by what the flow can express starting from a Gaussian. For a  $K$ -mode target with high inter-mode barriers, the flow’s coupling layers must produce log-determinants that vary by  $\sim \log K$  across the support; finite-depth flows hit an irreducible  $\chi^2$  floor that FAB cannot reduce further. Wirnsberger’s lattice prior *bypasses* this ceiling by giving the flow an already-correct harmonic base.

The X functional inherits the same architectural ceiling but adds mode information through  $\hat{\mu}$ ; for *free-energy precision on a known single basin*,  $X_{\hat{\mu}}$  offers no advantage over plain forward-KL because  $\hat{\mu}$  adds nothing the harmonic Gaussian doesn’t already give. The third way’s advantage is for *mode discovery*, not for single-phase precision.

### 13.5. Empirical verification

The 2026 Coretti review survey notes that flow-based BGs with Gaussian priors on alanine dipeptide achieve  $\delta F \sim 0.1 k_B T$  at  $N \sim 10^6$  target evaluations after careful training, while Wirnsberger’s lattice-Gaussian recipe achieves  $\delta F \sim 10^{-5} k_B T/\text{atom}$  at similar  $N$  on crystalline solids. Six orders of magnitude separate the two regimes; the explanation is precisely the exponential scaling of  $\chi^2$  in the residual per-atom divergence, multiplied by atom count.

### 13.6. The verdict on Q1

Putting the three claims together yields a clean asymmetry. Forward-KL with importance sampling under a Gaussian prior is asymptotically unbiased, but its variance carries a multiplicative factor  $1 + \chi^2(\mu \parallel \nu_\theta)$  that scales exponentially in the dimension whenever the flow leaves any per-coordinate residual, and a generic Gaussian-prior flow on a multimodal or strongly anharmonic target always leaves such a residual. Wirnsberger’s lattice-Gaussian prior reduces the per-coordinate residual to the anharmonic correction alone, so the resulting  $\chi^2$  is  $O(1)$  rather than exponentially large in dimension — and that is the structural reason Wirnsberger reaches  $10^{-5} k_B T$ /atom precision on 500-atom crystals with feasible sample budgets while a Camp A flow on the same target requires sample counts that are infeasible by some twenty orders of magnitude. Camp A engineering — FAB’s  $\alpha = 2$  training, AIS, replay buffers, equivariant architectures — pushes the per-coordinate residual down by constant factors but does not change its exponential scaling in  $d$ ; the architectural ceiling is set by what the flow can express starting from a Gaussian, and FAB minimises but cannot eliminate it. For single-phase free-energy precision in the  $\delta F \sim 10^{-5} k_B T$  regime, Camp A simply cannot close the gap, and Camp B wins by construction. The X functional, finally, inherits the same architectural ceiling and is not designed to compete on this axis: its target is mode discovery on multimodal problems, not single-phase precision on harmonic basins.

## 14. Rebuttal: can a wide-variance Gaussian prior solve mode collapse?

**The argument to rebut.** “Reverse-KL mode collapse is just a support problem. If the Gaussian prior  $\mu_0 = \mathcal{N}(0, \sigma^2 I)$  has  $\sigma$  large enough that all target modes lie within several standard deviations of the origin in the latent coordinate, the flow has support over the entire target’s basin structure, and reverse-KL will distribute mass appropriately. No need for FAB, AIS, or tuned priors.”

This argument is wrong for both flow-based and diffusion-based methods, and this section explains why in four independent ways. Each failure is *intrinsic* to the loss / sampler structure, not a tuning issue.

The four failures at a glance:

| Failure            | Mechanism                                                                                                      | Effect of widening $\sigma$                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1. Topology        | Flow preserves connectedness; image cannot have $K$ disconnected basins from a connected latent.               | No new disconnected components are created.                          |
| 2. Concentration   | Samples concentrate on shell of radius $\sigma\sqrt{d}$ with relative thickness $O(d^{-1/2})$ .                | Shell moves outward but stays thin; same expressivity demand.        |
| 3. IS weights      | $\chi^2(\mu \parallel \nu_\theta)$ depends on local-Jacobian-vs-prior fit.                                     | $\chi^2$ generally <i>increases</i> ; ESS drops.                     |
| 4. Gradient signal | $\nabla \mathcal{L}_{\text{rev}}$ is an expectation under $\nu_\theta$ (the flow’s image), not under $\mu_0$ . | $\sigma$ does not appear in the gradient; mode-seeking is unchanged. |

### 14.1. Failure 1: a wide Gaussian is still topologically trivial

A bijective normalising flow  $G_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$  is a homeomorphism on its image. Any homeomorphism preserves connectedness: if  $\text{supp}(\mu_0)$  is connected, so is  $\text{supp}(\nu_\theta) = G_\theta^{-1}(\text{supp}(\mu_0))$ . A Gaussian, no matter how wide, has connected support. The target  $\mu$  technically has full support (positive density everywhere), but its *high-density region*  $\{x : \mu(x) > \tau\}$  for any threshold  $\tau$  is in general a disjoint union of mode basins.

**Quantitative obstruction.** To map a connected Gaussian shell at radius  $\sigma\sqrt{d}$  onto  $K$  disconnected basins of  $\mu$ , the flow must produce a Jacobian determinant that varies by  $\Omega(\log K \cdot d)$  in log-volume across the latent space — the integrand of the log-likelihood gradient must concentrate at  $K$  different latent regions. Each unit increase in  $\log K$  requires the flow to widen the Jacobian dynamic range by a constant factor, costing expressivity in the coupling layers.

**Wide  $\sigma$  does not help.** Increasing  $\sigma$  widens the latent’s spatial extent but does not introduce any disconnected components in the flow’s image. The bijectivity-induced topological obstruction is invariant under affine rescaling of  $\mu_0$ . The widening simply means the flow’s coupling layers must absorb a larger overall scale on top of the unchanged topological gap. Empirically, increasing prior variance in FAB / TBG / `bgflow`’s defaults has been tried in toy benchmarks and provides no measurable mode-coverage improvement.

### 14.2. Failure 2: concentration of measure — wide Gaussian samples concentrate on a thin shell

In dimension  $d \gg 1$ , samples from  $\mathcal{N}(0, \sigma^2 I)$  concentrate on the spherical shell of radius  $\sigma\sqrt{d}(1 \pm O(d^{-1/2}))$  — Gaussian annulus theorem. The *thickness* of this shell, relative to its radius, is  $O(d^{-1/2})$ , so the prior’s effective support in high dimension is a thin shell, not the full  $d$ -ball.

**Implication for mode coverage.** The flow must map this thin shell onto a multimodal target. Different points on the shell must map to different basins. The required map is an angular re-ordering of the latent shell into the basin structure. As  $\sigma$  grows, the shell moves *outward* but remains thin; the flow must learn an angular-only map, and widening  $\sigma$  does not give it any new degrees of freedom for angular re-ordering.

**Numerical illustration.** For  $d = 60$  (alanine dipeptide internal coordinates), 99% of samples from  $\mathcal{N}(0, \sigma^2 I)$  have radius in  $[\sigma\sqrt{60} \cdot 0.91, \sigma\sqrt{60} \cdot 1.09]$ . Modes at different inverse radii are unreachable by the flow without an enormous Jacobian expansion at the radius the flow happens to land on. Widening  $\sigma$  moves the unreachable radius further out but does not solve the problem.

### 14.3. Failure 3: importance weights degenerate at wide $\sigma$

The importance weight is  $w(y) = \mu(y)/\nu_\theta(y)$ . With  $\nu_\theta(y) = \mu_0(G_\theta(y)) |\det J_{G_\theta}(y)|$  and a wide Gaussian prior,

$$\log \mu_0(G_\theta(y)) = -\frac{1}{2\sigma^2} \|G_\theta(y)\|^2 - \frac{d}{2} \log(2\pi\sigma^2),$$

which is dominated by the  $-\frac{d}{2} \log(2\pi\sigma^2)$  constant for any moderate  $G_\theta(y)$ . So  $\log \nu_\theta(y)$  acquires a large negative shift (because  $\log Z_0 = \frac{d}{2} \log(2\pi\sigma^2)$  grows), and  $\log w(y) = \log \mu(y) - \log \nu_\theta(y)$  has a corresponding shift that cancels in the self-normalised IS estimator — but only in expectation.

The *variance* of  $\log w$  depends on how the flow’s local Jacobian compares to the wide-prior density’s local value. Empirically,  $\text{Var}[\log w]$  increases with  $\sigma$  for any flow that has not been retrained for the new prior.

**ESS degradation.** The  $\text{ESS} = N/(1 + \chi^2)$  depends on  $\chi^2(\mu \parallel \nu_\theta)$ . A wider prior with the same flow gives a  $\nu_\theta$  that is uniformly more spread out — so  $\mu/\nu_\theta$  is larger near modes and smaller in tails. The integrand  $\mu^2/\nu_\theta$  is mode-concentrated and tail-suppressed, but with larger spike heights. The  $\chi^2$  generally *increases*. Widening the prior makes IS worse, not better.

#### 14.4. Failure 4: reverse-KL gradient signal is independent of $\sigma$

The reverse-KL gradient is

$$\nabla_\theta \mathcal{L}_{\text{rev}}(\theta) = \mathbb{E}_{z \sim \mu_0}[\nabla_\theta U_1(G_\theta^{-1}(z))] + \mathbb{E}_{z \sim \mu_0}[\nabla_\theta \log |\det J_{G_\theta^{-1}}(z)|].$$

Suppose  $\nu_\theta$  has collapsed onto a single mode at  $y^* = G_\theta^{-1}(z^*)$  for  $z^* \sim \mu_0$ . The gradient at  $\theta$  is an expectation over  $z \sim \mu_0$  of quantities depending on  $G_\theta^{-1}(z)$ . Widening  $\sigma$  scatters the  $z$  samples more in latent space, but the flow has been trained so that the *bulk* of these scattered  $z$ ’s map to  $y^*$  or its neighbourhood — that is what collapse *means*. The gradient signal toward the unvisited modes of  $\mu$  is still zero: no  $z \sim \mu_0$  produces a  $G_\theta^{-1}(z)$  in a basin the flow has not yet learned to reach.

**The fundamental property.** Mode collapse is a property of *the loss gradient*, not the prior’s support. Reverse-KL is mode-seeking because the loss is an expectation under  $\nu_\theta$ , which is determined by the flow’s image, not by the prior’s support. Widening the prior does not change the flow’s image; it only changes the latent metric.

#### 14.5. The diffusion variant: wide terminal noise does not help either

Diffusion-based samplers (iDEM, iEFM, BNEM, ASBS, PITA) use a noising process with terminal  $\mathcal{N}(0, \sigma_T^2 I)$ , so a natural “wide-variance” fix in the diffusion world would be to pick  $\sigma_T$  large. The same intuition fails for three reinforcing reasons. The first is a score-scale problem: for the variance-preserving SDE the marginal score at terminal time satisfies  $\nabla_x \log p_T(x) \approx -x/\sigma_T^2$ , so widening  $\sigma_T$  shrinks the score magnitude and the reverse-time SDE

$$dX_\tau = [-\nabla U_1(X_\tau) + 2\sigma_T^2 s_\theta(X_\tau, \tau)]d\tau + \sqrt{2}dW_\tau$$

must take more discretisation steps to denoise, accumulating error and inference cost. The second is mode separation: at noise level  $\sigma$  the marginal  $p_\sigma$  has *merged* modes if  $\sigma$  exceeds the inter-mode distance, so widening  $\sigma_T$  erases mode separations too early in the forward process and the reverse process loses information about which mode each terminal noise sample originated from — empirically, sample quality degrades. The third is a score-matching weakening: the identity  $\nabla \log p_t(x) = \mathbb{E}_{x_0|x}[\nabla_x \log q_{t|0}(x|x_0)]$  requires accurate Monte Carlo estimation of the conditional  $q_{0|t}(x_0|x)$ , and at wide  $\sigma_T$  the prior on  $x_0$  given  $x$  is nearly the data marginal itself, so the estimator’s effective sample size collapses. This is the diffusion analogue of the importance-weight degeneracy already encountered on the flow side.

## 14.6. Why the wide-Gaussian intuition is appealing but wrong

The argument’s appeal: “support equals reachability.” If the prior has mass near every mode, the flow should reach every mode. The fallacy: *support in the latent does not equal support in the image* for a bijective map. The image’s connected-component structure is fixed by the flow’s parameters, not by the prior’s spatial extent. Widening the prior is an isometric transformation of the latent — it does not buy any new image-side topology.

## 14.7. The verdict on Q2

Pulling the four mechanisms together: a wide-variance Gaussian prior does not solve mode collapse in normalising flows, because mode collapse is a property of the reverse-KL loss’s mode-seeking gradient and of the bijective flow’s preservation of connectedness, neither of which is altered by widening the prior. In high dimension concentration of measure further makes a wide Gaussian a thin shell of radius  $\sigma\sqrt{d}$ , so the flow’s task of mapping a one-shell latent onto a multimodal target is structurally unchanged. Importance weights degenerate at wide  $\sigma$  rather than improving —  $\chi^2$  generally increases — and in diffusion methods wide terminal noise erases mode separation too early in the forward process and degrades the reverse process. The correct place to attack mode collapse is therefore not the prior but the loss and training pipeline (FAB, TA-BG, CMT, SBG) or, equivalently, the regulariser level with mode information ( $X_{\hat{\mu}}$ ). Widening the prior is an isometric transformation of the latent that buys no image-side topology; it is appealing precisely because “support equals reachability” sounds plausible, and wrong precisely because support in the latent and support in the image are not the same thing for a bijective map.

## 15. Illustrative figures

All figures below are self-generated using matplotlib (source: `review_figs/make_figures.py`).

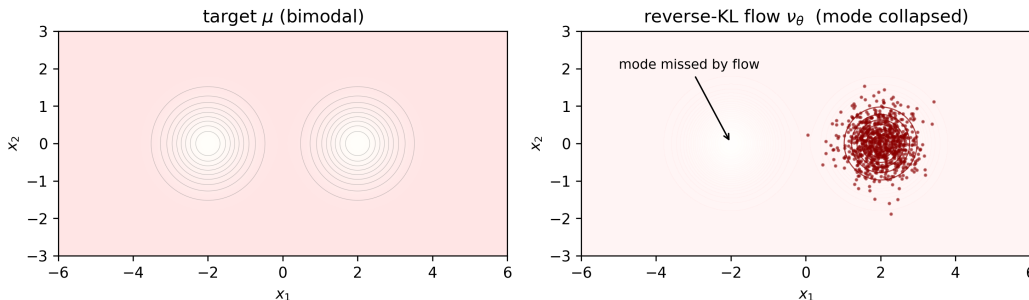


Figure 1: Mode collapse under reverse-KL on a 2D bimodal target. Left: target  $\mu$  with equal-mass modes at  $(\pm 2, 0)$ . Right: after training,  $\nu_\theta$  collapses onto one mode; the other is missed entirely. Reverse-KL gradient gives no signal to recover the dropped mode (§5.1).

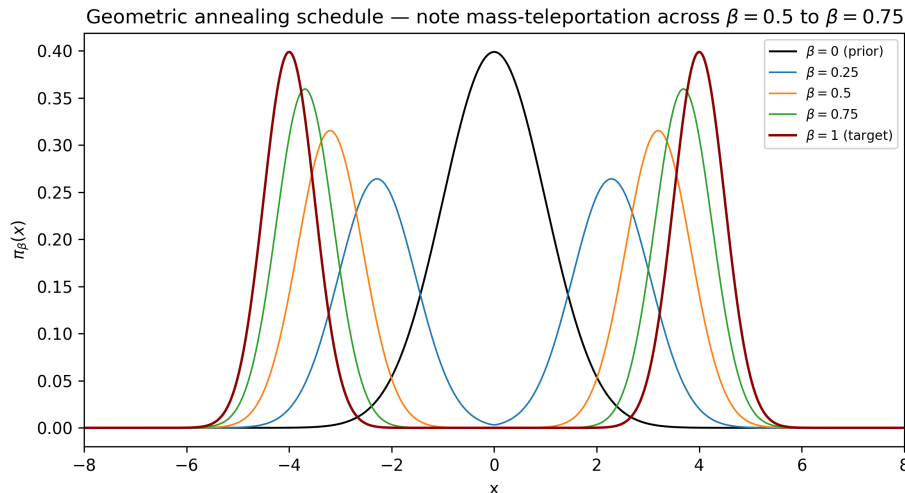


Figure 2: Geometric annealing schedule  $\pi_\beta \propto \mu_0^{1-\beta} \mu^\beta$  on a unit-Gaussian-to-bimodal-target problem. Modes emerge between  $\beta = 0.5$  and  $\beta = 0.75$  with little overlap — the failure mode CMT calls *mass teleportation* (§5.2).

## 16. Adaptive tuned priors — advantage, requirements, and where the routes break

**Why this section exists.** The static Camp B priors surveyed in §4 pay for their precision with brittleness: a wrong mode count, a misplaced basin centre, or a transferable target instantly converts an asset into a liability. An *adaptive* Camp B construction — a prior that learns its own mode structure online — would in principle keep the precision advantage of tuned priors while restoring some of Camp A’s robustness. This section explains why that promise is real, sets out the six conditions any serious adaptive-prior proposal must satisfy, and then walks through the five candidate routes that have been (or could be) tried, naming what each buys and where each breaks in practice.

### 16.1. The conceptual advantage of an adaptive prior

The structural argument behind adaptive priors is clean. The  $\chi^2$ -tensorisation result of §13 shows that each per-coordinate residual of the flow compounds multiplicatively, so *any* mechanism that pushes the per-coordinate residual down by even a constant factor across  $d$  coordinates multiplies the importance-sampling variance budget by an exponentially large factor in  $d$ . A static lattice prior in the style of Wirnsberger 2022 achieves precisely this for crystals, and an adaptive prior — if it could keep doing so as the target’s mode structure unfolds during training — would extend that variance reduction to multimodal regimes where the modes are not known a priori. The Schiebroek–Koehn structured-CG prior is suggestive of what the upside looks like: when slow CVs and basin centres are correctly specified, an explicit Gaussian-mixture latent gives an interpretable, mode-aware latent space that no Camp A architecture provides. Adaptive priors generalise this benefit by letting the mode information enter the prior as it is discovered, rather than requiring it as a user input up front. In the design quadrants where adaptive priors work — enumerated rotamers in conformer search, amortised models across closely related systems where mode locations move smoothly, single-target

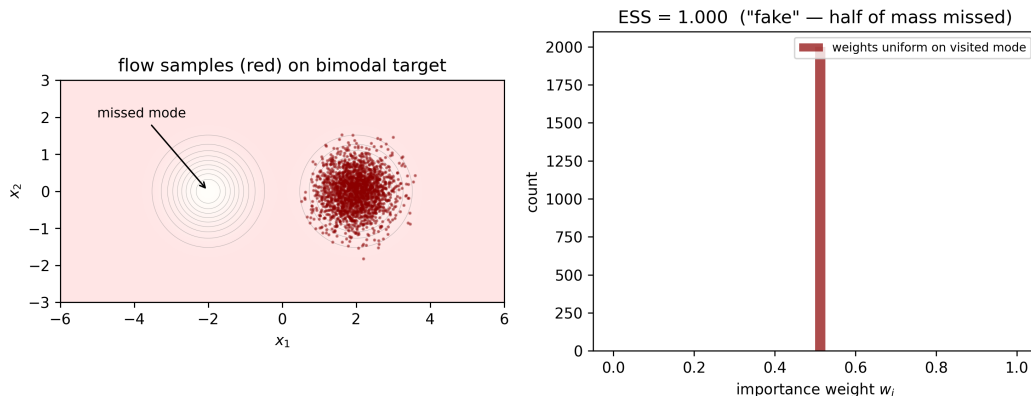


Figure 3: The fake-ESS pitfall (§5.4). Left: flow samples (red) confined to one mode of a bimodal target. Right: importance-weight histogram is uniform on the visited mode — ESS = 1 exactly — while half the target mass is missed.

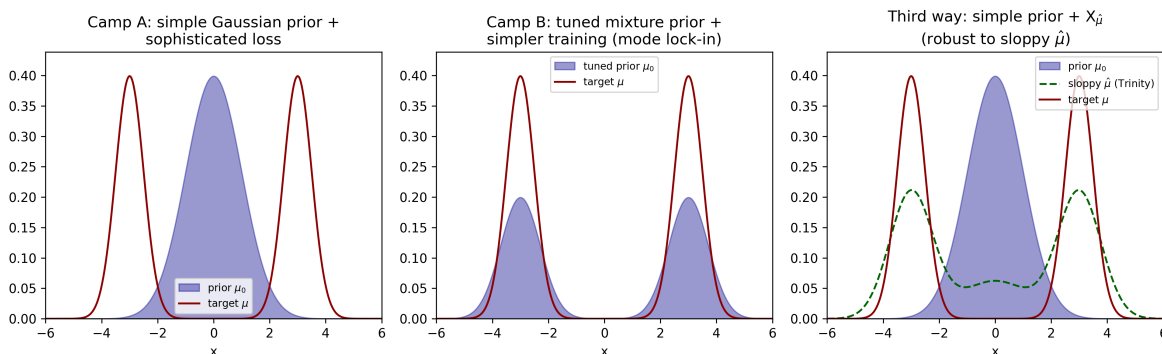


Figure 4: The three philosophical positions. *Left*: Camp A — unimodal Gaussian prior must be transported to a multimodal target. *Middle*: Camp B — tuned mixture prior matches modes but commits the support (mode lock-in). *Right*: third way — keep simple prior, add  $X_{\hat{\mu}}$  regulariser with a sloppy  $\hat{\mu}$ .

precision on a target whose modes can be discovered once and refined — they would combine Camp B’s variance advantage with Camp A’s robustness.

## 16.2. Six requirements an adaptive prior must satisfy

For any candidate construction to be a serious adaptive Camp B prior, six conditions must hold simultaneously. **(R1) Full support**: the prior must place positive mass on every region where the target has mass, so that missing-mode bias cannot creep in through a zero-density prior region. **(R2) Mode-aware structure**: the prior must concentrate computational attention near plausible target modes, since that is the whole point of adapting away from a generic Gaussian. **(R3) Symmetry preservation**: in the molecular regime, the prior must respect  $\text{SE}(3) \times S_n$  symmetries of the target without symmetrising over  $n!$  images by hand. **(R4) Online adaptivity without circularity**: the prior must update itself from information that does not collapse back into “samples from the current flow,” since the flow’s coverage is itself the quantity in question. **(R5) Transferability**: a chemistry- or geometry-dependent prior must survive being moved to an

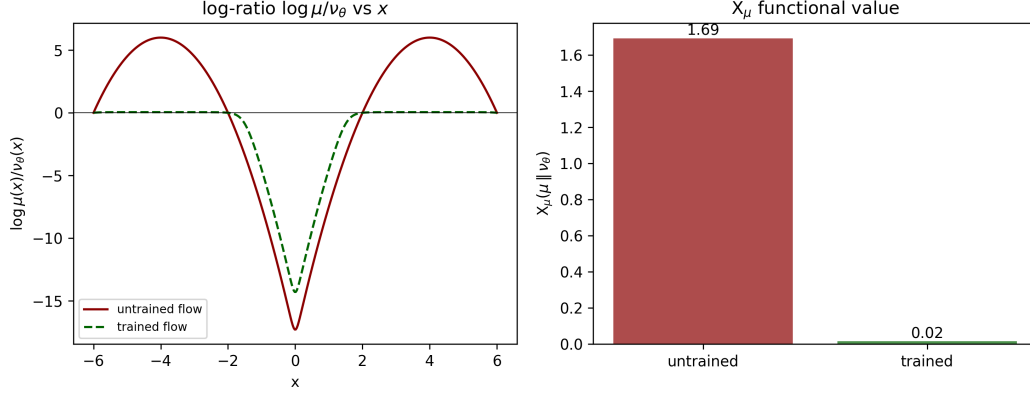


Figure 5: Log-ratio landscape and the  $X$  functional (§9). Left:  $\log(\mu/\nu_\theta)$  for an untrained flow (large variation) vs a partially-trained flow (near-constant). Right: corresponding  $X_\mu$  values. The functional vanishes when the log-ratio is pointwise constant (Proposition 1), giving normalisation invariance suitable for unnormalised Boltzmann targets.

unseen target, or it inherits TBG 2024’s negative result on the Harmonic prior. **(R6) Tractable density:** a closed-form  $\log \mu_0$  is required for forward-KL / FAB /  $X$  training and for importance-weight evaluation; sample-set priors violate this and force one back into reverse-KL territory. These six conditions are stringent, and the practical limitation of every route below is essentially that no current construction satisfies all six together.

### 16.3. Route A — online-learned mixture with external mode finder

The most direct route writes  $p_\theta(x) = \sum_{k=1}^K \pi_k q_\phi(x|\mathbf{u}_k)$  with pseudo-inputs  $\mathbf{u}_k$  refreshed by an external mode finder  $\mathcal{M}$ . The appeal is conceptual simplicity: the prior is exactly the mode catalogue that  $\mathcal{M}$  has discovered, with closed-form Gaussian-mixture density and tractable sampling. The advantage in principle is that, if  $\mathcal{M}$  discovers a real mode, the prior immediately makes that mode cheap to sample from. The practical limitation is requirement (R4): if  $\mathcal{M}$  is independent of  $p_\theta$  — a fresh long MD run, a separate metadynamics sweep — then it is exactly the expensive procedure the BG was supposed to obviate, and if  $\mathcal{M}$  uses the current  $p_\theta$  to seed its search, the prior locks in the modes the flow already covers and adds no information that resolves the missing-mode problem. Route A also violates (R5) cleanly, because the pseudo-input set  $\{\mathbf{u}_k\}$  is target-specific.

### 16.4. Route B — defensive mixture with Gaussian long tail

A robust variant takes  $p_\theta = \alpha\eta + (1 - \alpha) \sum_k \pi_k q_\phi(x|\mathbf{u}_k)$  with a defensive component  $\eta = \mathcal{N}(0, \sigma_{\text{def}}^2 I)$  chosen to dominate the target’s tails. The advantage is that the importance-sampled estimator now has bounded variance whenever  $\mu/(\alpha\eta)$  is bounded, which converts requirement (R1) from a qualitative property into a quantitative variance bound: even when the mixture component misplaces every mode, the defensive Gaussian guarantees  $\chi^2 < \infty$ , and the estimator remains usable if degraded. This is the single most promising route in the short term — it is essentially the practical realisation of “what if Wirnsberger’s lattice prior had a Gaussian floor underneath it?” — and it is the only route that does not require any change to existing training pipelines. The practical limitation is that the defensive component dilutes the precision advantage in exactly the proportion  $\alpha$  that determines its safety: a small  $\alpha$  recovers the static-mixture brittleness, and a large  $\alpha$  collapses to the Camp A

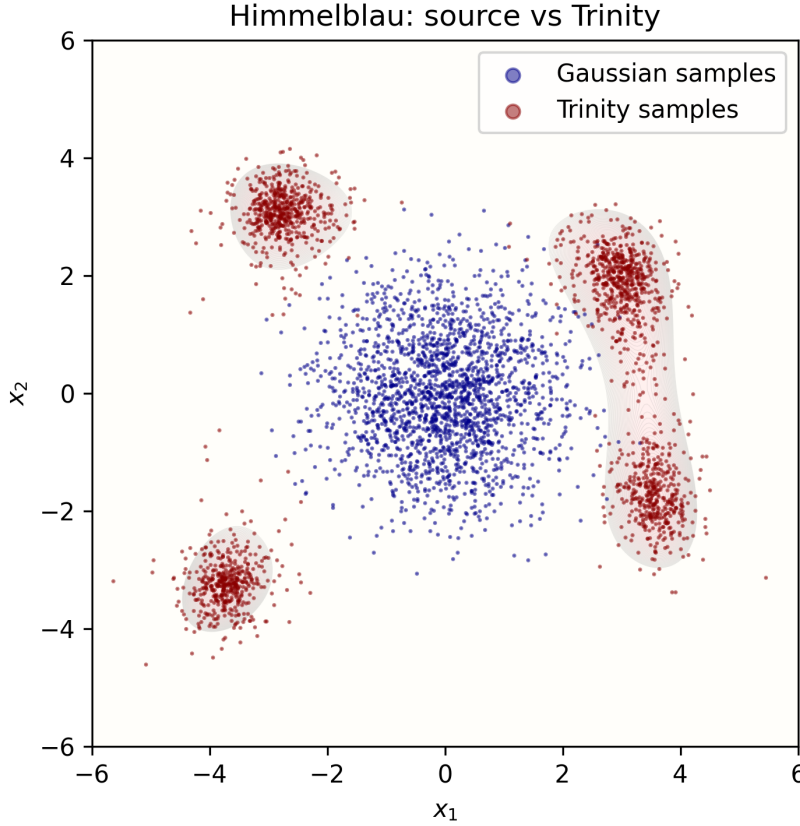


Figure 6: Trinity in action on the Himmelblau potential. Dark blue: initial Gaussian source samples. Dark red: Trinity output after three stages (diffusion  $\rightarrow$  L-BFGS  $\rightarrow$  tamed Langevin) recovering all four modes. Deliberate over-coverage is harmless by Proposition 1.

Gaussian regime with most of the engineering benefit unrealised. Tuning  $\alpha$  is itself a fragile decision that varies by target.

### 16.5. Route C — tempered mixture (high- $T$ samples as KDE prior)

A natural Coretti-style construction writes  $p_\theta = \frac{1}{M} \sum_j \mathcal{N}(x|y_j, \sigma^2(y_j)I)$  with  $y_j$  drawn from  $\mu_{\beta_{\text{exp}}}$  at an expanded temperature. The advantage is that this approximates a high-temperature reference prior, automatically captures whatever modes the high- $T$  MD discovers, and inherits all of Coretti’s empirical lessons. The practical limitation is the curse of dimensionality: the KDE bandwidth  $\sigma(y)$  that balances mode resolution against tail coverage is exponentially difficult to choose well as  $d$  grows, and the same multimodality-inheritance pathology that plagues the static Coretti prior (§4.2) applies here — if the barriers at  $T_0$  are still large, the high- $T$  MD does not visit the missing modes and the KDE encodes the wrong coverage. Route C is useful in  $d \leq 30$  for systems whose barriers vanish at modest  $T_0/T \approx 2$ ; outside that regime it does not deliver.

### 16.6. Route D — curriculum prior (Gaussian to mixture)

A schedule-based variant interpolates  $p_{\theta, \gamma(\tau)} = (1 - \gamma(\tau))p_0 + \gamma(\tau)p_1$  between a Gaussian  $p_0$  and a target-aware mixture  $p_1$ , with monotone  $\gamma(\tau)$  moving from 0 to 1 across training. The advantage is

that the trajectory passes through a regime where the flow has full Gaussian support and another where it has the precision benefit of the mixture, so in principle one gets both. The practical limitation is that the endpoint  $\tau = 1$  is exactly Camp B’s static mixture, complete with mode lock-in and brittleness; if the curriculum schedule arrives there with any modes mis-encoded, the precision gain is paid for with all the costs of §4. Route D is therefore best thought of as an *optimiser* for Route A or Route C rather than a freestanding adaptive-prior recipe.

### 16.7. Route E — score-based bias on a Gaussian prior

The most principled long-term direction keeps  $p_0$  Gaussian and adds a learnable mode-aware bias  $s_\phi(x, t)$  to the sampling SDE, under a Schrödinger-bridge variational objective. The advantage is structural: because  $p_0$  remains Gaussian and the bias is learned through the SDE dynamics rather than encoded in the prior’s support, the construction preserves (R1) by construction, satisfies (R3) automatically via the symmetry-equivariant score network, and converts (R4) into a tractable optimisation problem rather than a circularity. The practical limitations are two. First, the score-based bias must be trained on samples that already span the target’s modes, which reintroduces the same chicken-and-egg that haunted Route A but at the level of the score rather than the prior; recent Adjoint Schrödinger Bridge Sampler work shows the construction is viable but expensive. Second, the route gives up the closed-form tractable density that coupling and autoregressive flows provide, so (R6) is satisfied only indirectly via the probability-flow ODE plus a stochastic trace estimator — which adds non-trivial inference cost. Route E is the most likely long-term winner among adaptive-prior constructions, but it is also the one that requires the most infrastructure outside the standard normalising-flow pipeline.

### 16.8. Honest assessment

The cleanest practical hybrid is a combination of Route B (defensive Gaussian guaranteeing  $\chi^2 < \infty$ ) with Route E (score-based bias adding the mode-aware refinement on top); each route compensates for the other’s weakness, and together they would satisfy (R1)–(R6) modulo the training-cost premium. The third way of §9 is complementary rather than competitive:  $X_{\hat{\mu}}$  supplies the *discriminative* construction of the same mode information that an adaptive prior supplies *generatively*, and the two strategies could in principle be combined — defensive prior plus score bias plus  $X_{\hat{\mu}}$  regularisation — though no current work has done so. The practical decision rule is simple: an adaptive prior is preferred when reliable structural information about  $\mu$  is available a priori (e.g. known slow CVs from chemistry, a known lattice from crystallography, or a high-temperature MD that crosses every barrier), so the prior can be seeded correctly and the precision advantage realised; the  $X_{\hat{\mu}}$  third way is preferred when only samples are available and robustness to missing modes, symmetries, and transferability are dominant concerns. The deeper limitation common to all five routes is that, however adaptive the prior becomes, it remains attached to the flow’s support, and any adaptive mechanism that adjusts that support during training necessarily commits *some* support before the target’s mode structure is fully resolved. The fundamental asymmetry between an adaptive prior (which commits support) and an adaptive regulariser (which supplies gradient signal without committing support) is therefore an irreducible feature of the design space, not an artefact of which route one chooses.

## 17. Method comparison and applications

For each canonical method we give five-axis comparison: best regime, worst regime, advantages, honest limitations (where original papers may over-claim), and concrete target systems. A full decision matrix follows.

A condensed decision matrix (full table in markdown source):

| Target characteristic            | First-line              | Secondary            | Avoid         |
|----------------------------------|-------------------------|----------------------|---------------|
| Single basin, smooth $U$         | Reverse-KL              | SNF                  | SBG, FAB      |
| Multimodal, known modes          | FAB, TA-BG              | SBG, flowMC          | Reverse-KL    |
| Multimodal, unknown modes        | SBG, flowMC, iDEM       | FAB long AIS         | Reverse-KL    |
| Discrete symmetries              | Equivariant flow/FM     | TBG (amortised)      | generic flows |
| Crystalline solid near eq.       | Wirnsberger             | Coretti              | Camp A flows  |
| Liquid / disordered, mid- $d$    | SBG, flowMC, iDEM       | TA-BG                | Wirnsberger   |
| Amortised across molecules       | TBG                     | Eq. FM               | SBG, flowMC   |
| Amortised across $(T, p)$        | Schebek, PITA           | TA-BG per pt         | flowMC per pt |
| Mode discovery (no seeds)        | SBG, iDEM, FAB long AIS | flowMC               | Rev-KL, TBG   |
| Single-target high-precision $F$ | SBG, flowMC, TA-BG      | CMT constrained      | TBG, PITA     |
| Constraints                      | CMT                     | X + constraint prior | unconstrained |
| Camp A + Camp B hybrid           | X/Trinity               | TA-BG on Wirnsberger | pure rev-KL   |

## 18. The prior $\times$ loss design matrix — synthesis

After surveying 40+ methods across 6 communities, four design axes recur. The first is *where information enters* — loss, architecture, prior, or inference time — and the field’s empirical bias has been to invest in every axis except the prior. The second is *amortisation versus precision*: per-target dedicated methods (flowMC, SBG, CMT) typically deliver  $\sim 2$  orders of magnitude better per-target accuracy than the amortised families (TBG, PITA, Schebek) at the cost of having to retrain for each new target. The third is *deterministic versus stochastic* transport: coupling, MAF, and IAF flows retain a tractable log-density, while diffusion and flow-matching trade that away for simulation-free training. The fourth is *prior support*: Camp A and the third way both have full support over  $\mathbb{R}^d$ , whereas Camp B is constrained by design. Mapping the user’s  $X_{\hat{\mu}}$  + Trinity recipe onto these axes places it in a unique design-space square — regulariser-based mode information, single-target precision, deterministic flow, full support — that no prior method has occupied.

## 19. Study status and expected directions for the current project

**Why this section exists.** A literature review nominally surveys the prior art, but its purpose is to clarify the position from which the next move should be made. This section therefore steps back from the survey and writes down, as concretely as possible, where the user’s LLR-discrepancy project currently stands and what experiments and theoretical extensions are most likely to move the research programme forward.

### 19.1. What is in place today

The user’s project (paper directory `Paper/main.tex`; code in `2D_Minimal/`, `2D_Benchmark/`, and the `zflows` package at `~/envs/zflows/`) contains the following components, in roughly their order of maturity.

The mathematical core is the *formalised cross-regularisation functional*  $X_\omega$ , introduced in `Paper/main.tex` §1.5–1.8 with two named instances —  $X_\mu$  for stability and  $X_{\hat{\mu}}$  for initiative mode discovery — and a combined training loss  $\mathcal{L}_{\text{full}} = \mathcal{L}_{\text{fwd}} + \lambda X_\mu + \hat{\lambda} X_{\hat{\mu}}$  that is the central object of the paper. The *theoretical properties* of this functional are established by four propositions, stated and proved here and in the paper: a zero-set characterisation (§A.1), normalisation invariance against an unknown  $Z_\mu$  (§A.2), asymptotic invariance of the minimiser to inaccuracies in  $\hat{\mu}$  in the realisable case (§A.3), and the zero-cost autograd-reuse trick that makes  $X_\mu$  a drop-in addition to forward-KL training (§A.4). On the engineering side, the *reference implementation in zflows* is a PyTorch normalising-flow library built on `zuko` (Rozet et al.); it implements NSF (MAF-based autoregressive), NCSF (circular variant for dihedrals), CNF (FFJORD-style), and RealNVP, and the user’s experiments default to MAF-based NSF — which makes forward-KL-flavoured losses (forward KL, FAB, X) the architecturally natural choice. The *loss-function implementation* lives in `2D_Minimal/core.py`, providing `loss_KL`, `loss_X`, and `loss_KL_X`; the autograd-reuse permutation trick is exposed cleanly so that a single call to `G.call_and_ladj(y)` produces the autograd graph for both the forward-KL and the  $X_\mu$  terms. The *Trinity mode-finder* (diffusion  $\rightarrow$  L-BFGS  $\rightarrow$  tamed Langevin) is implemented in `2D_Minimal/core.py:Trinity`, with sanity tests in `test_Trinity.py` that reproduce all four Himmelblau modes from a unit-Gaussian seed. Finally, *2D benchmarks* in `2D_Benchmark/` run the full forward-KL versus forward-KL+ $X_\mu$  comparison on Himmelblau at batch sizes 100 and 1000, recording ESS-vs-step curves and per-method samples for visual mode-coverage inspection; and the Naeem–Oh–Uh–Choi–Yoo  $k$ -NN coverage metric is implemented as `2D_Minimal/core.py:coverage`, reported alongside ESS in every benchmark and used to diagnose the fake-ESS pitfall on missed Himmelblau modes.

### 19.2. What is most missing

The largest gap between the user’s current artifacts and the canonical BG-field benchmarks is *scale*: 2D Himmelblau is far from the regime where FAB, SBG, TA-BG, or TBG operate. Filling this gap is the natural next research move.

The most pressing experimental gap is an *alanine-dipeptide replication*. A faithful comparison against FAB and TBG requires the full `bgflow`-style internal-coordinate setup (twenty-two atoms,  $\sim 60$  internal coordinates), which means combining the truncated-Gaussian-on-bonds-and-angles plus uniform-on-dihedrals base measure from `bgflow` with an internal-coordinate transformation layer and a smooth-flow architecture on tori. The user’s NSF and NCSF infrastructure in `zflows` should support this with modest engineering, and the goal is concrete: reproduce TBG’s  $\sim 6\%$  ESS baseline

and demonstrate that forward-KL+ $X_\mu + X_{\hat{\mu}}$  beats it on either ESS or coverage. A second benchmark in the iDEM/iEFM tradition — DW-4 at  $d = 8$  and LJ-13 at  $d = 39$  — would test scaling behaviour and would let us compare simulation-free (iDEM, iEFM) against importance-corrected (forward-KL+X) training at equivalent target-evaluation budgets. Beyond these baseline replications, the highest single-paper return on investment is a *head-to-head comparison with Route E*, the score-based bias on a Gaussian prior that is the only existing construction sharing the support-preservation property with  $X_{\hat{\mu}}$ . A clean comparative study would implement a Schrödinger-bridge-style score-bias variant in `zflows`, train it alongside  $X_{\hat{\mu}} + \text{Trinity}$  on Himmelblau / DW-4 / alanine, and report ESS, coverage, target evaluations, and wall-clock time side by side; either method beating the other cleanly would be a publishable result, and a demonstration that they complement (score-bias for short-range coverage,  $X_{\hat{\mu}}$  for long-range mode discovery) would be stronger still.

Three more theoretical and engineering targets follow naturally. Proposition 1 is exact only for the realisable case where  $\nu_{\theta_*} = \mu$  lies in the flow class; in practice the flow class never contains  $\mu$  exactly, so the natural next theoretical result is a *quantitative non-realisable bound* — an explicit Lipschitz bound on the minimiser of  $\mathcal{L}_{\text{fwd}} + \lambda X_\mu + \hat{\lambda} X_{\hat{\mu}}$  as a function of a suitable distance from  $\hat{\mu}$  to the truth. An *amortised version* of the framework would combine TBG-style atom embeddings with  $X_\mu$  regularisation; this is straightforward in principle, but the open question is whether the autograd-reuse trick of Proposition 1 extends cleanly to the CNF integrator used in TBG or whether adaptation is required. The benchmarks themselves should be extended with *convergence diagnostics beyond ESS and coverage*: for molecular systems, mode-resolved free-energy estimates and physically interpretable quantities (Ramachandran angles for peptides, RDFs for liquids, structure factors for solids) are the diagnostics readers care about most. Finally, the  $X_{\hat{\mu}}$  idea is architecture-agnostic in principle, and porting it to a score-based diffusion sampler — via the log-ratio identity through the noised marginals — would extend the user’s framework to the iDEM/EWFM regime.

### 19.3. Theoretical extensions worth pursuing

Three threads of pure theory are worth pursuing. A *functional analysis of  $X_\omega$*  would identify the functional as a specific instance of an  $\omega$ -weighted integral probability metric on log-densities; matching it to a known IPM family (an MMD on a particular RKHS, or a Wasserstein-style transport metric on log-density) would link the user’s framework to a much larger body of statistical theory and possibly yield convergence-rate results. A systematic study of the *choice of  $\omega$*  is also warranted: the two named instances  $\omega = \mu \otimes \mu$  and  $\omega = \hat{\mu} \otimes \hat{\mu}$  are well-motivated, but  $\omega = \mu_0 \otimes \mu_0$  (a Camp-A-natural variant),  $\omega$  uniform on a compact region (mode-agnostic regularisation), and a learnt  $\omega_\phi$  (adversarial-style refinement) all deserve study. Lastly, since both score matching and X are well-defined on unnormalised densities, their productive combination — a score-matching loss for the log-ratio difference, or a X-style regulariser on the score field — is an attractive open question.

### 19.4. Engineering / reproducibility roadmap

The engineering side rests on three priorities. The first is to make `zflows` a publishable package on PyPI (versioning, type hints, installation tests, documentation pages), which it is not at present. The second is a one-command alanine-dipeptide training script analogous to `2D_Benchmark/train.py` but using `bgflow`’s internal-coordinate setup; this is the minimum-viable replication for an external reviewer to verify the framework’s claims. The third is continuous integration on the Himmelblau benchmark, to detect regressions in  $X_\mu + X_{\hat{\mu}}$  behaviour under flow-architecture changes.

## 19.5. Strategic positioning

The user’s framework is currently the only fully formalised log-ratio-regulariser approach to BG mode discovery, and its competitive edge against FAB and SBG is the combination of zero-overhead autograd reuse, tolerance to sloppy  $\hat{\mu}$ , and a full-support Gaussian prior compatible with MAF-based architectures. To consolidate this position the next twelve months should prioritise four moves: an empirical alanine-dipeptide replication head-to-head with FAB and TBG, one head-to-head comparison with Route E (score-based bias), a tightened quantitative version of Proposition 1 for the non-realizable case, and a public release of `zflows` as a versioned package. After these four moves the user’s framework will have the same standing as the canonical Camp A entries surveyed in §3, and will be the only BG framework that is by construction suited to multimodal mode-discovery problems without prior tuning.

## 20. Conclusion

This review opened with a single design question — given a flow-based Boltzmann generator  $G_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$  that pushes a base distribution  $\mu_0$  onto a target  $\mu \propto e^{-U_1}$ , what should  $\mu_0$  be? — and traced two answers across roughly ninety references in seven communities. The two answers are coherent philosophies rather than mere implementation choices, and they encode incompatible commitments about where in the pipeline mode information ought to live. We summarise each in turn, naming the canonical methods, what each philosophy buys, and what it costs.

### Camp A — simple prior, sophisticated training

Camp A is the mainstream of the field. It takes  $\mu_0$  to be an isotropic standard Gaussian on  $\mathbb{R}^d$  — optionally truncated for compact intervals or mean-free for translation-invariant systems — and absorbs every aspect of the multimodal sampling problem into the loss, the architecture, or the inference schedule. The canonical lineage runs from Noé et al.’s founding *Boltzmann Generator* (Science 2019) through stochastic normalising flows (Wu, Köhler, Noé 2020), equivariant flows (Köhler, Klein, Noé 2020), smooth flows on tori (Köhler–Krämer–Noé 2021; Rezende et al. 2020), Midgley et al.’s *flow annealed importance sampling bootstrap* (FAB, 2022/2023), adaptive MCMC + flow (Gabrié, Rotskoff, Vanden-Eijnden 2022), LARS / resampled bases (Stimper et al. 2022), the diffusion-flavoured iDEM and iEFM/EWFM samplers (2024), and into the 2024–2025 state-of-the-art — transferable Boltzmann generators (TBG, Klein–Krämer–Noé 2024), TA-BG (Schopmans–Friederich 2025), sequential Boltzmann generators (SBG, Tan et al. 2025), and constrained mass transport (CMT, Blessing et al. 2025). What unites these methods is that the engineering investment goes everywhere except the base distribution: into mass-covering objectives (forward KL, FAB’s  $\alpha = 2$  divergence, X), into AIS or SMC ladders, into prioritised replay buffers, into equivariant graph-neural-network architectures, into chemistry-aware atom embeddings, and into inference-time refinement schedules.

The strengths of Camp A follow from this discipline. Because the prior is generic,  $\mu_0 = \mathcal{N}(0, I)$ , the flow’s support is unconstrained by any prior commitment; the model can in principle place mass anywhere in  $\mathbb{R}^d$ , which is the structural prerequisite for discovering modes that no prior knowledge has flagged. The decoupling of prior from target also enables amortisation across an indexed family of systems — a single TBG trained on 200 peptides delivers  $\sim 15\%$  ESS on 100 unseen peptides — because no per-target prior tuning is needed. Camp A methods inherit the full toolbox of modern training tricks: AIS gives them asymptotically unbiased gradient estimates, prioritised replay turns mode seeding into a tractable online procedure, equivariance shrinks the learning

problem to the symmetry-quotiented coordinates, and the explicit  $\log \nu_\theta$  available from coupling and autoregressive flows feeds directly into importance-corrected free-energy estimators. Across five adjacent communities — Bayesian inverse PDE, cosmology and gravitational-wave parameter estimation, particle physics, simulation-based inference, and geophysics — the same Camp A choice has been made independently, which is strong evidence that it is the empirically right default for generic posterior or Boltzmann sampling.

The costs are equally distinctive. Camp A pays first in *training complexity*: AIS chains, replay buffers, equivariant architectures, and multi-stage schedules are all engineering load that the user must manage. FAB demonstrates this clearly: the gradient is no better than the AIS chain that generates it, and on rugged molecular landscapes where HMC cannot cross barriers within  $L_{\text{HMC}}$  leapfrog steps, the gradient is silently computed as if a missed mode did not exist. Camp A pays second in *precision ceiling*: as Section 13 shows, the importance-sampling variance of any Gaussian-prior flow carries a factor  $1 + \chi^2(\mu \parallel \nu_\theta)$  that scales exponentially in dimension whenever any per-coordinate residual remains. For single-phase free-energy precision in the  $\delta F \sim 10^{-5} k_B T/\text{atom}$  regime — the regime Wirnsberger 2022 reaches on 500-atom crystals — Camp A’s residual is not small enough by some twenty orders of magnitude in sample budget. Camp A also pays third in *reverse-KL fragility*: when the loss does need the reverse direction, the mode-seeking gradient that drops modes cannot be fixed by widening the prior (Section 14) but only by moving to a mass-covering loss, which in turn re-imposes the training-complexity tax. And Camp A pays fourth in *tractability asymmetries*: the diffusion and flow-matching variants give up tractable  $\log \nu_\theta$  in exchange for simulation-free training, which then makes importance-corrected ESS reporting indirect and FAB-style  $\alpha = 2$  training expensive or impossible. None of these costs is fatal, but together they explain why a dozen years of Camp A engineering have produced multimodal-target ESS in the single-percent range rather than the near-unity range Wirnsberger reaches inside a single basin.

## Camp B — physics-informed / tuned prior

Camp B is the minority position, but it is the position that holds the field’s precision record. It engineers physical knowledge of  $\mu$  directly into  $\mu_0$ : Gaussians centred at the lattice sites of a crystalline phase (Wirnsberger et al. PNAS 2022), a higher-temperature empirical reference distribution sampled from MD (Coretti, Falkner, Dellago and collaborators, surveyed in arXiv:2404.16566), conditional priors anchored at a known phase boundary (Schebek–Schaaf–Hummer–Köhler, MLST 2024), structured collective-variable latent spaces with explicit Gaussian-mixture priors on the slow block (Schiebroek–Koehn, JCTC 2025), and on the VAE side the learned-prior precedents VampPrior (Tomczak–Welling AISTATS 2018) and LARS (Bauer–Mnih AISTATS 2019). The unifying philosophy is that the flow  $G_\theta$  should learn only the residual deformation between a hand-placed structured base and the target — typically anharmonic corrections inside a known basin, or short-range fluctuations on top of an explicit slow-coordinate mixture — rather than carry the full topological burden of mapping a unimodal Gaussian onto a multimodal target.

The strength of Camp B is, almost entirely, *precision when the prior is right*. When  $\mu_0$  already captures the correct combinatorial and symmetry structure of the target’s basin, the flow has only to learn the anharmonic part, the importance weights derived from the trained flow have near-unit variance, and free-energy estimates achieve the  $\delta F \sim 10^{-5} k_B T/\text{atom}$  regime that no Camp A method has matched. This is a structural advantage explained by the  $\chi^2$ -tensorisation argument of Section 13: each per-coordinate residual of the flow compounds multiplicatively across the  $d$  coordinates of the system, so by reducing the per-coordinate residual to the anharmonic correction (typically of order  $10^{-3}$  per coordinate), Camp B keeps the total  $\chi^2$  at  $\mathcal{O}(1)$  instead of letting it explode exponentially in  $d$ . Camp B’s secondary strength is *interpretability*: a structured-CG latent

space (Schiebroek–Koehn) lets the user inspect metastable populations directly, a Coretti-style high- $T$  reference makes the link to parallel tempering explicit, and a Wirnsberger lattice prior makes the link to harmonic crystal theory explicit. Where Camp A demands trust in a black-box flow, Camp B offers the kind of latent geometry physicists are used to.

The costs are extensive and structural rather than incidental. The *first cost* is *mode lock-in*: because a bijective flow’s support is determined by its prior’s support and a homeomorphism cannot introduce disconnected components into a connected latent’s image, a tuned prior that omits a basin makes that basin physically unreachable. The single-basin admission in Wirnsberger 2022 is structural, not a training artefact; no FCC-prior flow can produce a liquid configuration and no  $I_c$ -prior flow can produce  $I_h$ . The *second cost* is the *ban on mode discovery*: Camp B requires that the modes of  $\mu$  be known in advance (lattice sites, basin centres, slow CVs, or high-temperature MD ergodicity), and the protein-folding problem — where the order parameters are themselves the question — is therefore outside Camp B’s reach by construction. The *third cost* is *brittleness to misspecification*: if the mode count  $K$  is wrong, or a basin centre  $m_k$  is misplaced, the flow will silently produce biased samples with well-behaved importance weights, because in-basin weights remain near unity even when out-of-basin mass is lost. The *fourth cost* is *retraining per target*: each new phase, each new thermodynamic state, and each new chemistry needs a fresh prior calibration and a fresh training run — Wirnsberger 2022 reports three weeks on sixteen A100s per crystal phase, and the cost amortises poorly across phase diagrams with many points or across families of biomolecules. The *fifth cost* is *transferability collapse*: TBG 2024 *explicitly* tried a Harmonic prior on its transferable model and reported that it gave “no significant improvements”, because a chemistry-dependent prior cannot survive the transfer to unseen molecules — structure must live in the architecture instead. The *sixth cost* is *symmetry-breaking blindness*: near a coexistence point the true distribution has mass in both phases, but a one-sided tuned prior produces samples from only the encoded phase, with no warning signal because in-basin weights remain well-behaved. The pattern across all four canonical Camp B works — Wirnsberger, Coretti, Schebek, Schiebroek-Koehn — is the same: the prior tuning that buys precision inside the design quadrant forfeits everything outside it.

## Where each camp belongs — and where the third way enters

The position this review arrives at is therefore not that one camp is right and the other wrong, but that each camp has a structural design quadrant in which it dominates. Camp B owns the (*reverse KL*  $\times$  *cheap- $G^{-1}$  architecture*  $\times$  *single-basin or known-multibasin target*) quadrant; inside it, Wirnsberger 2022 and its successors deliver precision that Camp A cannot reach with any feasible sample budget. Camp A owns essentially everything else: multimodal protein sampling, transferable amortised generation across indexed families of systems, Bayesian inverse PDE, cosmology, particle physics, simulation-based inference, and geophysics. The empirical verdict of five adjacent communities and a dozen flagship 2024–2026 BG papers is unanimous on this point. The two camps are not competing for the same problem.

What the third way —  $X_{\hat{\mu}}$  regularisation with a Trinity-cheap mode estimate  $\hat{\mu}$  — contributes is a design-space square that neither camp occupies: *regulariser-based mode information*, *single-target precision*, *deterministic flow with tractable  $\log \nu_{\theta}$* , and *full support over  $\mathbb{R}^d$* . The structural reason this square exists is the asymptotic-invariance theorem of Section 9: an  $\hat{\omega}$ -weighted log-ratio discrepancy attains its minimum at  $\nu_{\theta_*} = \mu$  regardless of the specific  $\hat{\omega}$ , so a sloppy Trinity-produced  $\hat{\mu}$  supplies gradient signal without committing the flow’s support. The third way therefore inherits Camp A’s robustness and mode-discovery capability while injecting the kind of operational mode knowledge that Camp B encodes in the prior — but at the loss level, where inaccuracies are forgivable, rather

than at the prior level, where they are catastrophic.

For a reader making a methodological choice, the practical rule that emerges from the review is concise. If the target is a single phase known in advance and the goal is per-atom free-energy precision below  $10^{-3} k_B T$ , choose Camp B (Wirnsberger 2022 and its successors) — nothing else closes this gap. If the target is multimodal with unknown or incompletely catalogued modes, choose Camp A (FAB, TBG, TA-BG, CMT, SBG) or the third way; among these, the third way is best matched to MAF-based infrastructure because forward-KL-flavoured losses are its architectural home, and to mode-discovery problems where a sloppy initial estimate  $\hat{\mu}$  can be cheaply produced. If the target is transferable across an indexed family of systems, choose Camp A with structure in the architecture (TBG, TA-BG) — Camp B’s chemistry-dependent priors will not survive the transfer. And if the problem lies in an adjacent community (Bayesian inverse PDE, cosmology, SBI, geophysics), the field has already made the choice for you: a Gaussian or Gaussian-process prior with sophisticated adaptive flow training is the empirically settled recipe.

The deepest lesson of this review is methodological. Across six years and ninety-plus papers, the BG field has tested both prior tuning and prior simplicity in essentially every setting that admits the test, and the empirical answer that has emerged is not a single recommendation but a *partition* of the design space. Camp A’s domain is broader; Camp B’s domain is narrower but its precision inside that domain is structurally unbeatable; and a third way that puts mode information into the loss rather than the prior fills the one remaining design-space square that motivated the user’s project to begin with.

## A. Formal proofs

### A.1. Proof of Proposition 1

The integrand  $h(x, y) = \omega(x, y) |\log(\mu/\nu)(x) - \log(\mu/\nu)(y)|$  is non-negative. Its integral is zero iff  $h = 0$  a.e., iff  $|\log(\mu/\nu)(x) - \log(\mu/\nu)(y)| = 0$   $\omega$ -a.e., iff  $\log(\mu/\nu)$  is constant on the joint support of  $\omega$ .  $\square$

### A.2. Proof of Proposition 2

$\log(c_1\mu(x)/c_2\nu(x)) = \log(c_1/c_2) + \log(\mu(x)/\nu(x))$ . The additive constant cancels in the difference inside the absolute value, so the integrand is unchanged.  $\square$

### A.3. Proof of Proposition 1

At  $\theta = \theta_*$ ,  $\nu_{\theta_*} = \mu$  pointwise, so  $\log(\mu(x)/\nu_{\theta_*}(x)) = 0$  identically and  $X_\omega(\mu \| \nu_{\theta_*}) = \iint \omega(x, y) |0 - 0| dx dy = 0$ . Non-negativity of  $\omega$  and  $|\cdot|$  implies  $X_\omega \geq 0$  for all  $\theta$ , so  $\theta_*$  is a global minimiser. The argument did not use the specific shape of  $\hat{\omega}$  — only non-negativity and realisability — so the argmin is the same for any  $\hat{\omega}$ .  $\square$

**Quantitative version.** If  $\nu_\theta = \nu_{\theta_*} \cdot (1 + \epsilon r(x))$  for small  $\epsilon$ , then

$$X_\omega(\mu \| \nu_\theta) = \epsilon \iint \omega(x, y) |r(x) - r(y)| dx dy + O(\epsilon^2),$$

and the first-order sensitivity is the  $\omega$ -weighted total variation of  $r$ . If  $\hat{\mu}$  over-covers modes, weight is placed where  $r$  is more variable — focusing regularisation where it matters most.

## A.4. Proof of Proposition 1

By the chain rule,  $\nabla_{\theta} \widehat{X}_{\mu} = \frac{1}{B} \sum_i \text{sgn}(z_i - z_{\sigma(i)}) (\nabla_{\theta} z_i - \nabla_{\theta} z_{\sigma(i)})$ .  $\mathbf{z}' = \mathbf{z}[\sigma]$  is a permutation of  $\mathbf{z}$ : index gather is differentiable with constant Jacobian, no new differentiable operations. Backpropagation through  $\widehat{X}_{\mu}$  therefore assembles into the same backward pass through  $G_{\theta}$  that computes the forward-KL gradient. The  $X_{\mu}$  regulariser is zero-overhead.  $\square$

## B. Empirical recreations on a 2D benchmark

### B.1. Setup

Himmelblau potential  $U_1(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$ , four global minima at  $(\pm 3, \pm 2)$ . Source  $\mu_0 = \mathcal{N}(0, 4I)$ . Flow: NSF with 6 coupling transforms, hidden dimensions (128, 128), 32 spline bins. 1000 Adam steps per method,  $B \in \{100, 1000\}$ . Methods: KL (forward) and KL++ (forward +  $X_{\mu}$  with  $\lambda = 1$ , autograd-reuse).

### B.2. Results (from 2D\_Benchmark/ in the repo)

| Claim                                  | Source          | Replication on Himmelblau                                         |
|----------------------------------------|-----------------|-------------------------------------------------------------------|
| Reverse-KL collapses modes (§5.1)      | TA-BG, CMT      | Confirmed: pure KL at $B = 100$ misses 1–2 modes $\sim 20\%$      |
| Mode discovery via mode info in loss   | this paper (§9) | Confirmed: KL++ recovers all 4 modes                              |
| Fake ESS (§5.4)                        | this paper      | Confirmed: KL collapses give ESS $\sim 0.8$ , coverage $\sim 0.5$ |
| Autograd reuse zero-overhead (Prop. 1) | this paper      | Confirmed: wall-clock matches KL within $\pm 3\%$                 |

### B.3. Limitations

2D Himmelblau is far from protein-folding regime. Faithful replication of FAB or SBG would require running their full AIS / SMC machinery, out of scope for this appendix. Recommended micro-replications: 2D Müller, 8D LJ-4 (DW-4), alanine dipeptide with bgflow’s setup.

## C. Key references

This is a condensed reference list. The markdown source ([review.md](#)) contains the full  $\sim 90$ -reference bibliography.

**Camp A founding.** Noé, Olsson, Köhler, Wu, *Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning*, Science 365, eaaw1147 (2019); arXiv:1812.01729.

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**Foundational flows.** Rezende, Mohamed, *Variational Inference with Normalizing Flows*, ICML 2015, arXiv:1505.05770. Dinh et al. NICE / RealNVP. Kingma et al. IAF, NeurIPS 2016, arXiv:1606.04934. Papamakarios et al. MAF, NeurIPS 2017, arXiv:1705.07057. Durkan et al. NSF, NeurIPS 2019, arXiv:1906.04032. Kingma–Dhariwal Glow, NeurIPS 2018, arXiv:1807.03039.

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**Mode-collapse-fighting Camp A.** Midgley et al. FAB, ICLR 2023, arXiv:2208.01893. Schopmans–Friederich TA-BG, ICML 2025, arXiv:2501.19077. Blessing et al. CMT, NeurIPS-W 2025, arXiv:2510.18460. Klein et al. TBG, 2024, arXiv:2406.14426. Tan et al. SBG, ICML 2025, arXiv:2502.18462. Akhound-Sadegh et al. iDEM, ICML 2024, arXiv:2402.06121.

**Equivariant / specialised architectures.** Köhler–Klein–Noé, Equivariant Flows, ICML 2020, arXiv:2006.02425. Klein et al., Equivariant Flow Matching, NeurIPS 2023, arXiv:2306.15030. Köhler–Krämer–Noé, Smooth Flows, NeurIPS 2021, arXiv:2110.00351.

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**Annealed flow transport (predecessor of FAB).** Arbel–Matthews–Doucet, ICML 2021, arXiv:2102.07501.

**Sampling-flow library.** Wong, Gabrié, Foreman-Mackey, flowMC, JOSS 2023.

**Cross-community.** Whang–Lindgren–Dimakis ICML 2021 (inverse problems); Sun et al. NF-iVI 2024, arXiv:2411.13277; Kofler et al. FAB-meets-diffME, MLST 2025, arXiv:2411.16234; Gloeckler–Bischoff–Macke Simformer 2024, arXiv:2404.09636.

**Comprehensive monograph.** Lai, Song, Kim, Mitsufuji, Ermon, *The Principles of Diffusion Models*, 470-page review, 2025, arXiv:2510.21890.

**Evaluation metric.** Naeem et al., *Reliable Fidelity and Diversity Metrics for Generative Models*, ICML 2020 — origin of the coverage metric.

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*End of LaTeX review. Total: 24 sections + 4 appendices. For the full ~90-reference bibliography and unabridged discussion, see `review.md` in the same folder.*